

Charged-particle multiplicity distributions in pp and p-Pb collisions with ALICE



ALICE

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on behalf of the ALICE collaboration

7th International Conference on
New Frontiers in Physics



KØBENHAVNS
UNIVERSITET

Overview



- ➔ Introduction
 - Heavy-ion collisions
 - p-Pb collisions
 - Detectors
- ➔ Multiplicity Distributions
- ➔ Analysis
 - Recipe to get $P(N_{ch})$
- ➔ Previous Results
- ➔ p-Pb collisions - MC studies
- ➔ Summary



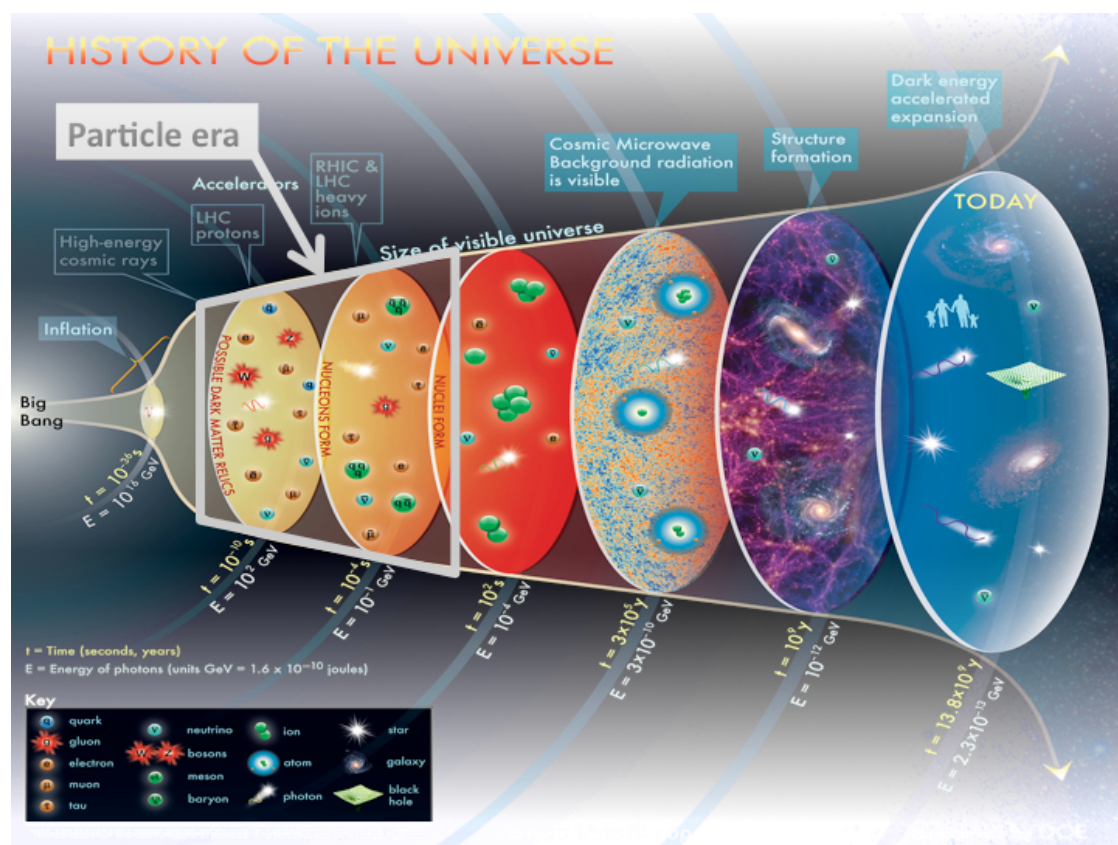
ALICE

Introduction

Heavy-ion Collisions



➔ Heavy-ion colliders recreate the conditions that existed after the Big Bang.

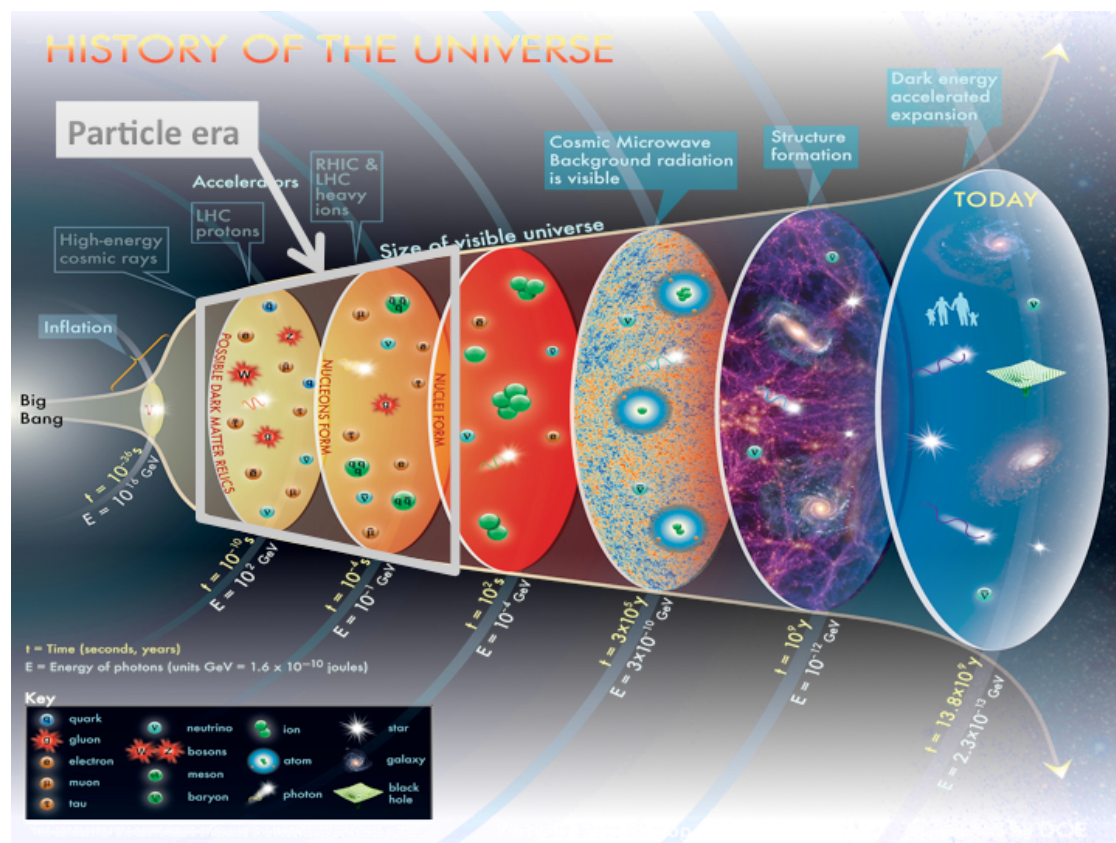


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- Quark-Gluon Plasma (QGP) → deconfined state of QCD matter

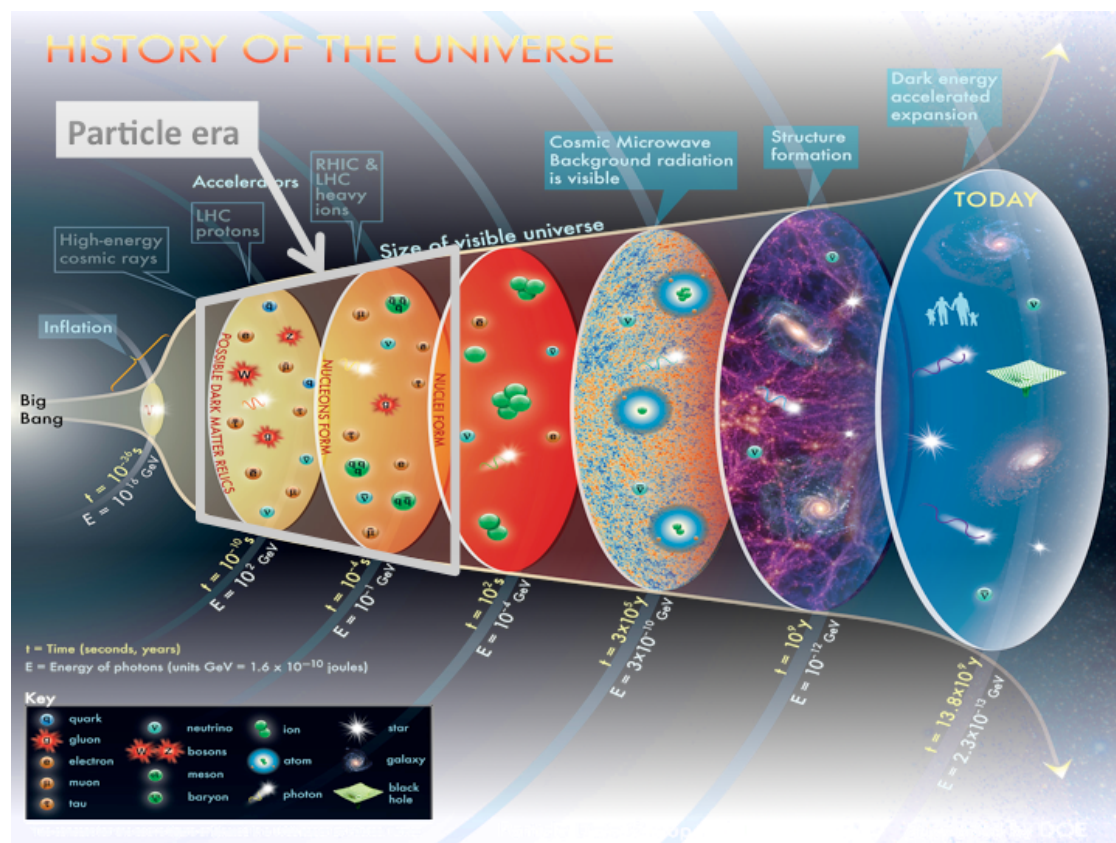


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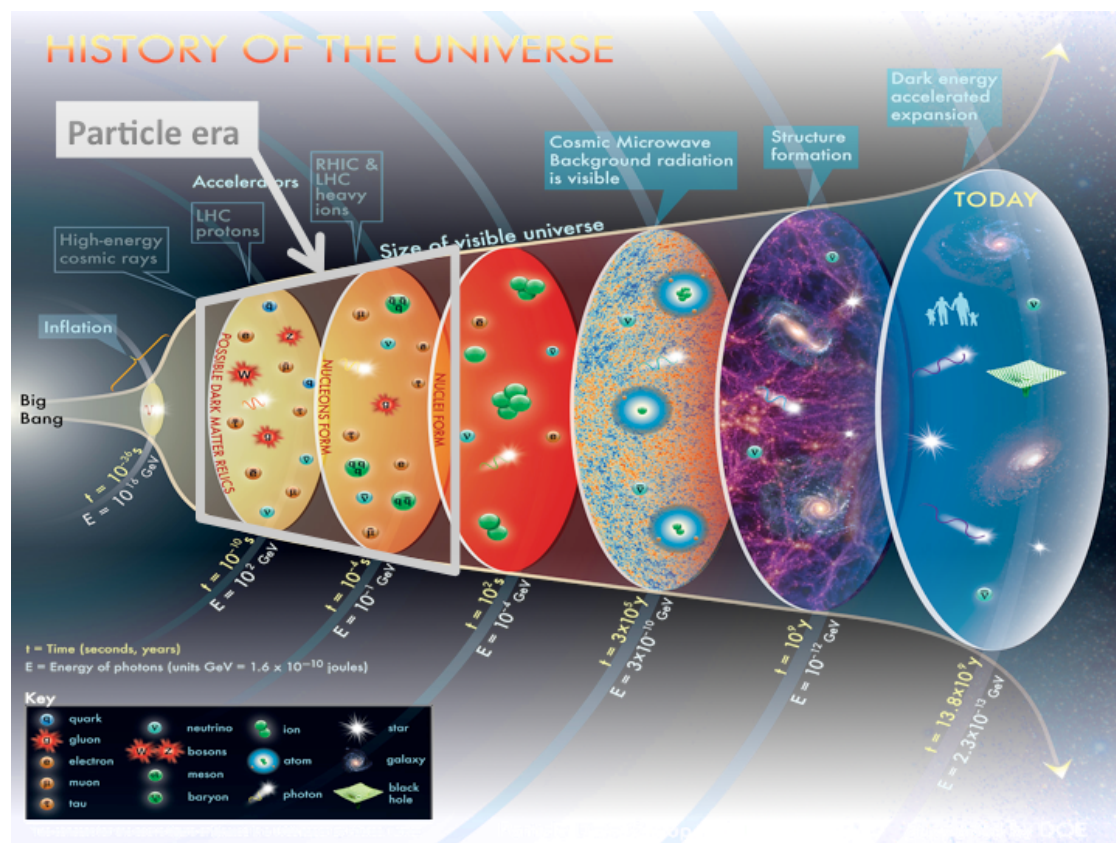


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- Propagation of jets through the QGP

Interaction between hard jets and the colored medium which leads to parton energy loss.

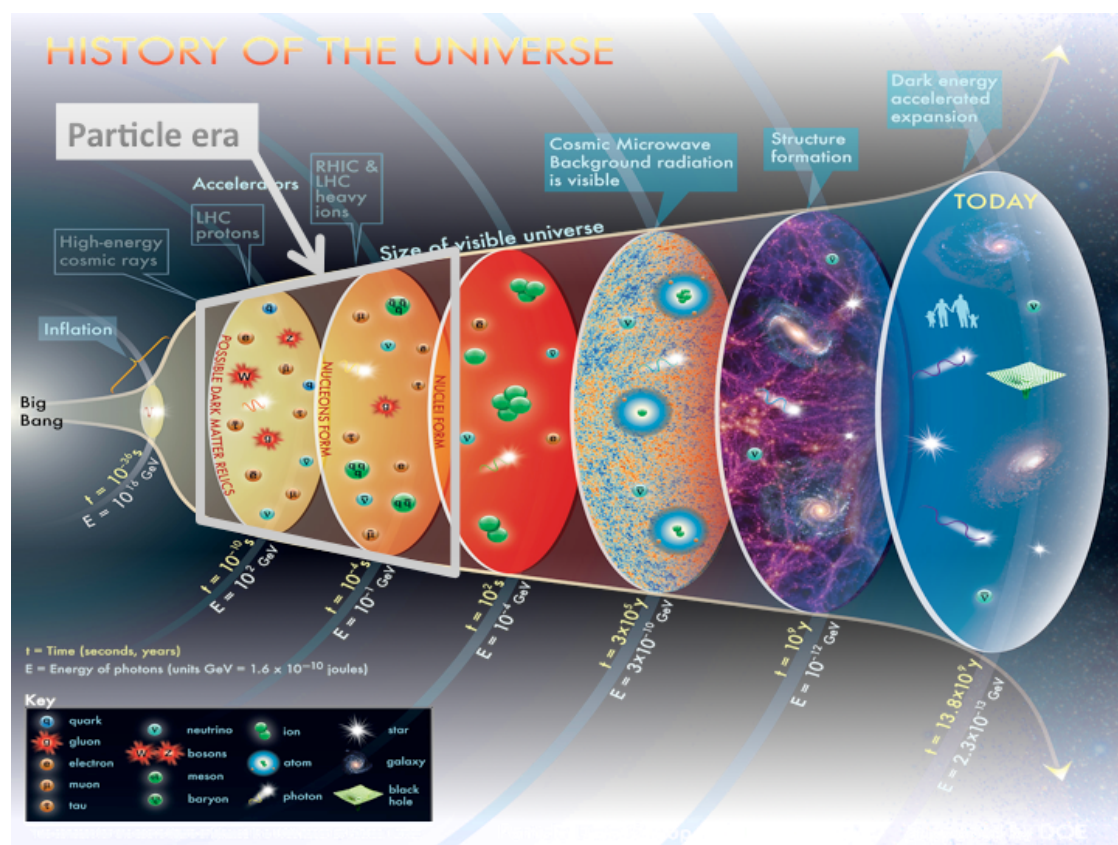
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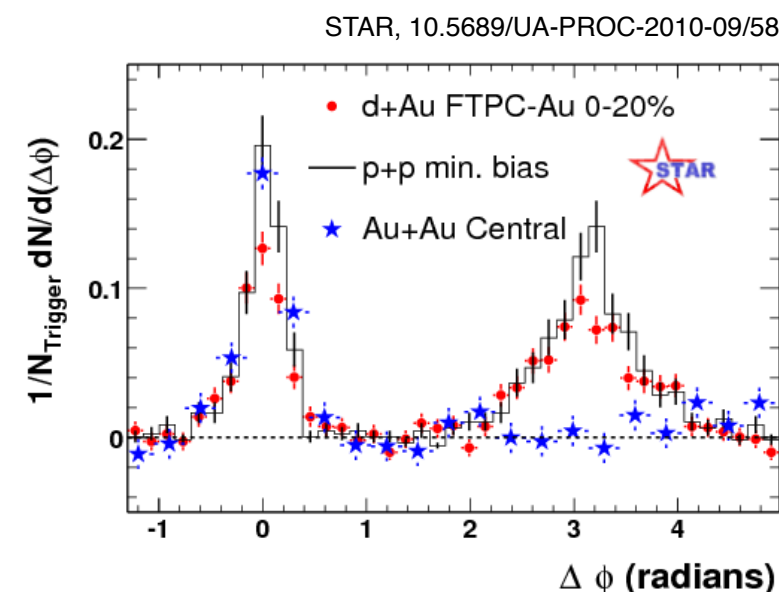
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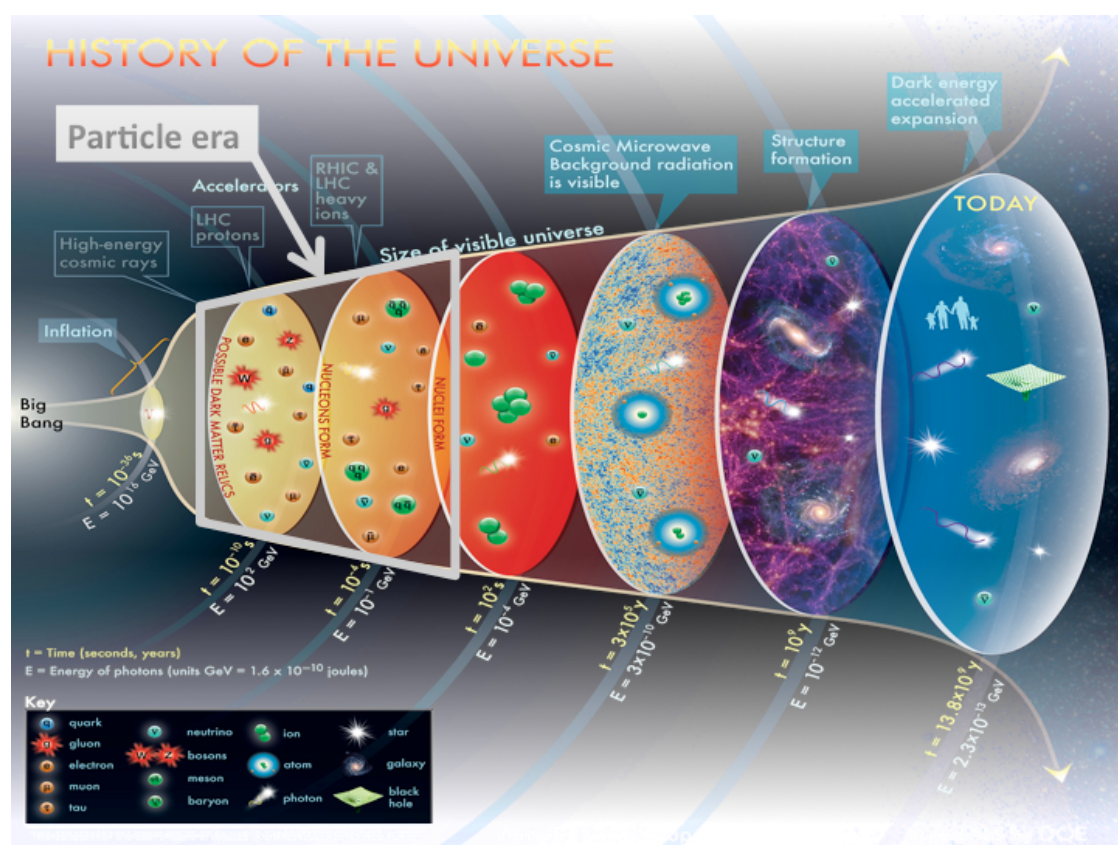
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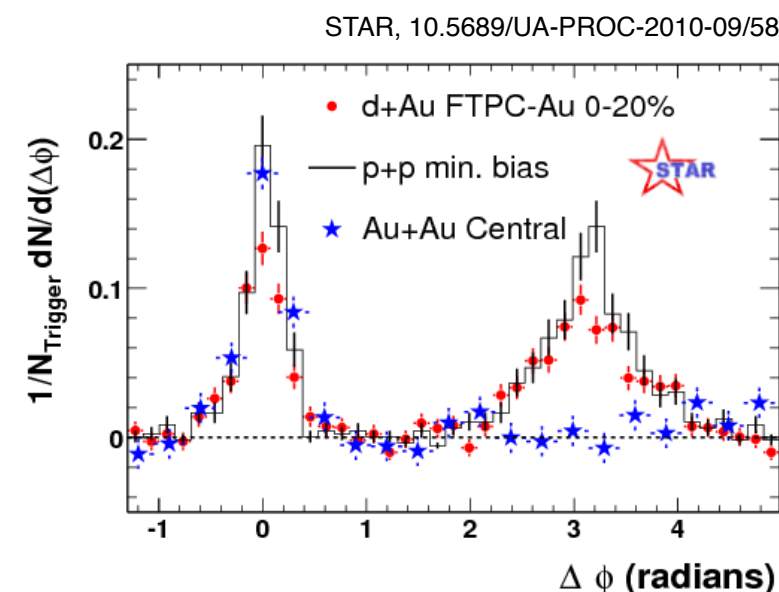
Interaction between hard jets and the colored medium which leads to parton energy loss.

Jet quenching

- QGP undergoes a collective expansion

Provide experimental information on the EoS and the transport properties (viscosity) of the created QGP.

Flow



p-Pb Collisions

- Understanding the QGP

- ↳ Baseline measurements

- ↳ pp collisions

- ↳ p-Pb collisions



p-Pb Collisions



- Understanding the QGP

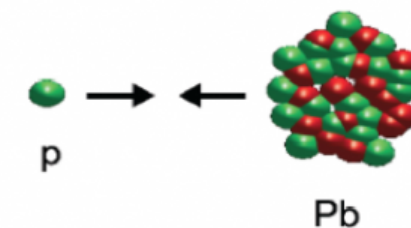
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Hot nuclear matter effects
vs.
Cold nuclear matter effects



p-Pb Collisions



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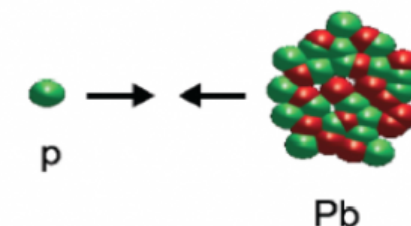
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A system interesting by itself!

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p-Pb Collisions



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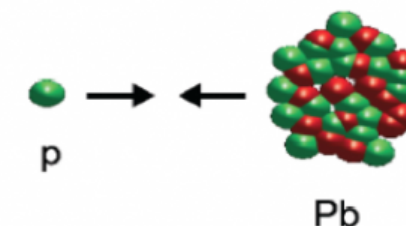
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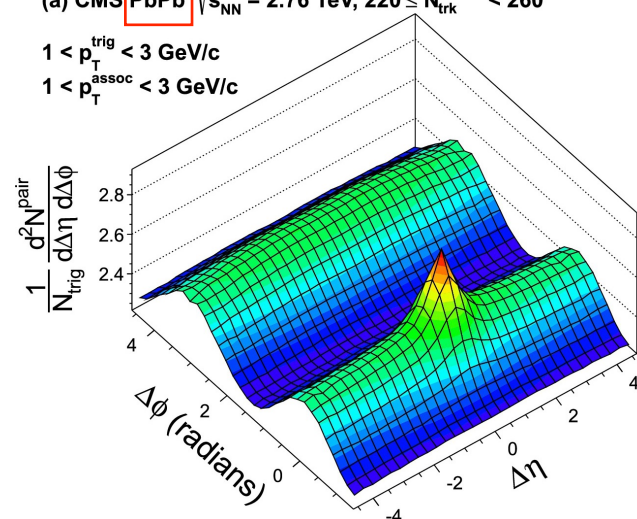
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CMS, 10.1016/j.nuclphysa.2014.09.049

(a) CMS **PbPb** $\sqrt{s_{NN}} = 2.76$ TeV, $220 \leq N_{trk}^{offline} < 260$

$1 < p_T^{trig} < 3$ GeV/c

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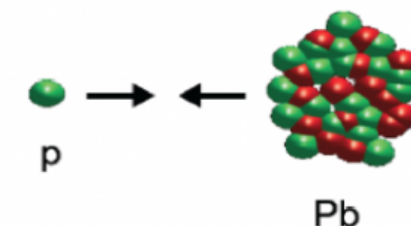
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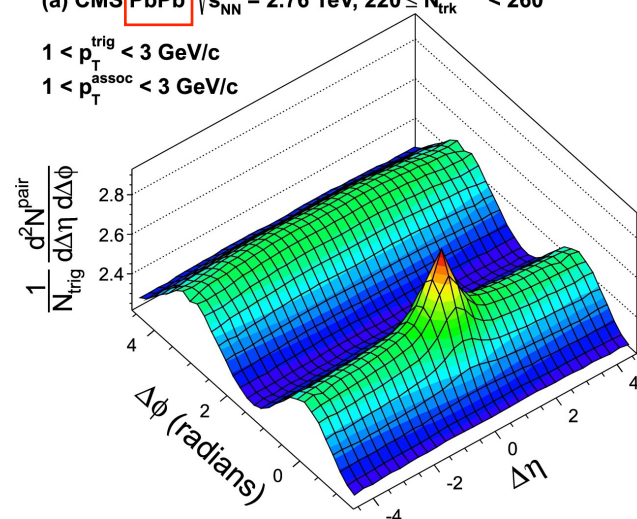
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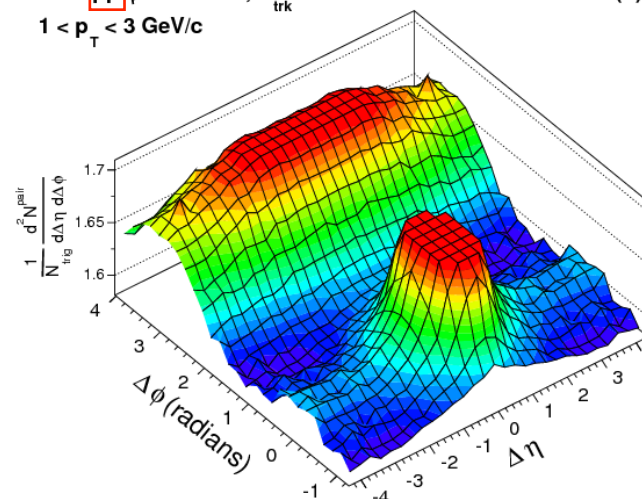


CMS, 10.1103/PhysRevLett.116.172302

CMS pp $\sqrt{s} = 13$ TeV, $N_{trk}^{offline} \geq 105$

$1 < p_T < 3$ GeV/c

(b)



- Understanding the QGP

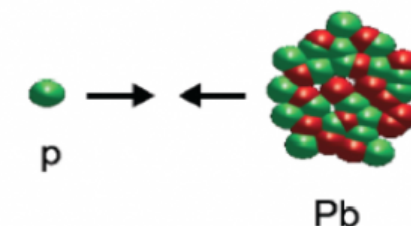
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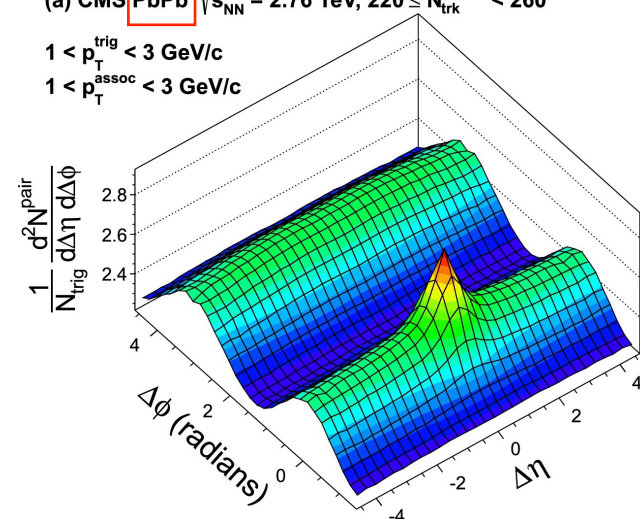
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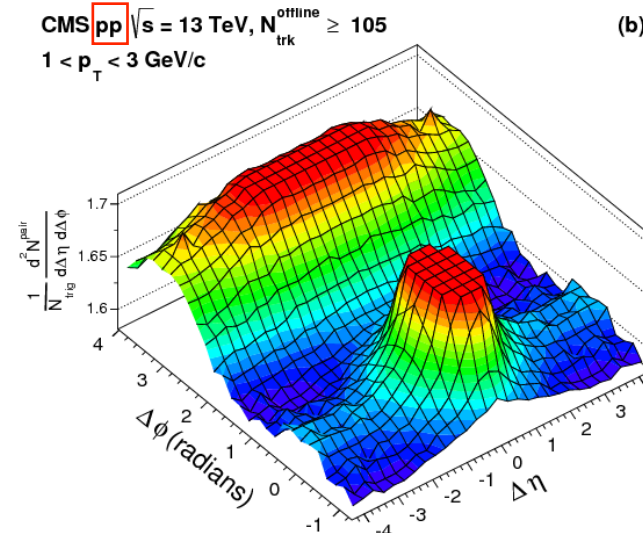
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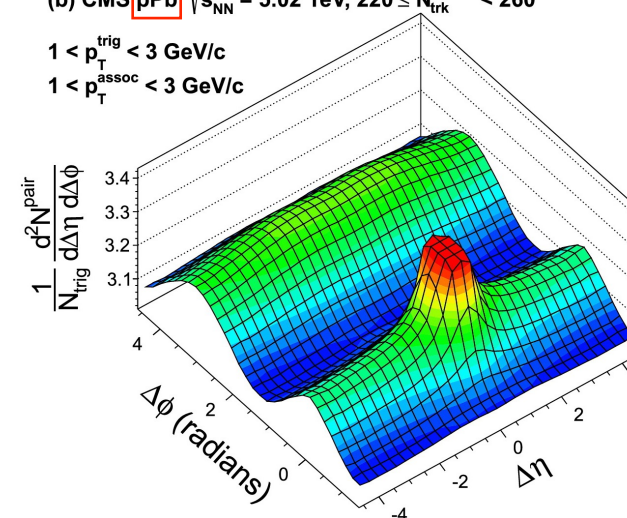
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(b) CMS **pPb** $\sqrt{s_{NN}} = 5.02$ TeV, $220 \leq N_{trk}^{offline} < 260$

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 $1 < p_T^{assoc} < 3$ GeV/c



Multiplicity Distributions



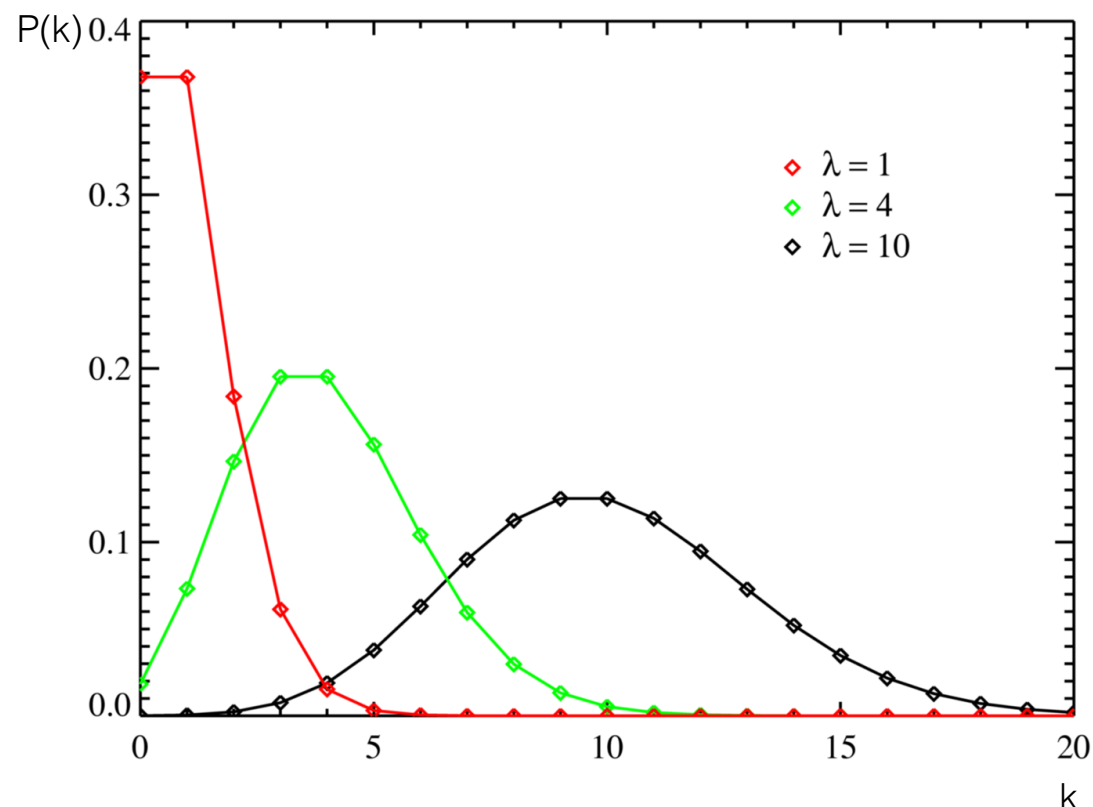
Total multiplicities of events in intervals of pseudorapidity

- ↳ The probability $P(N_{ch})$ for producing N_{ch} particles in the final state.
- ↳ Give insights to the particle production mechanism and provide constraints to models.



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Poisson distribution

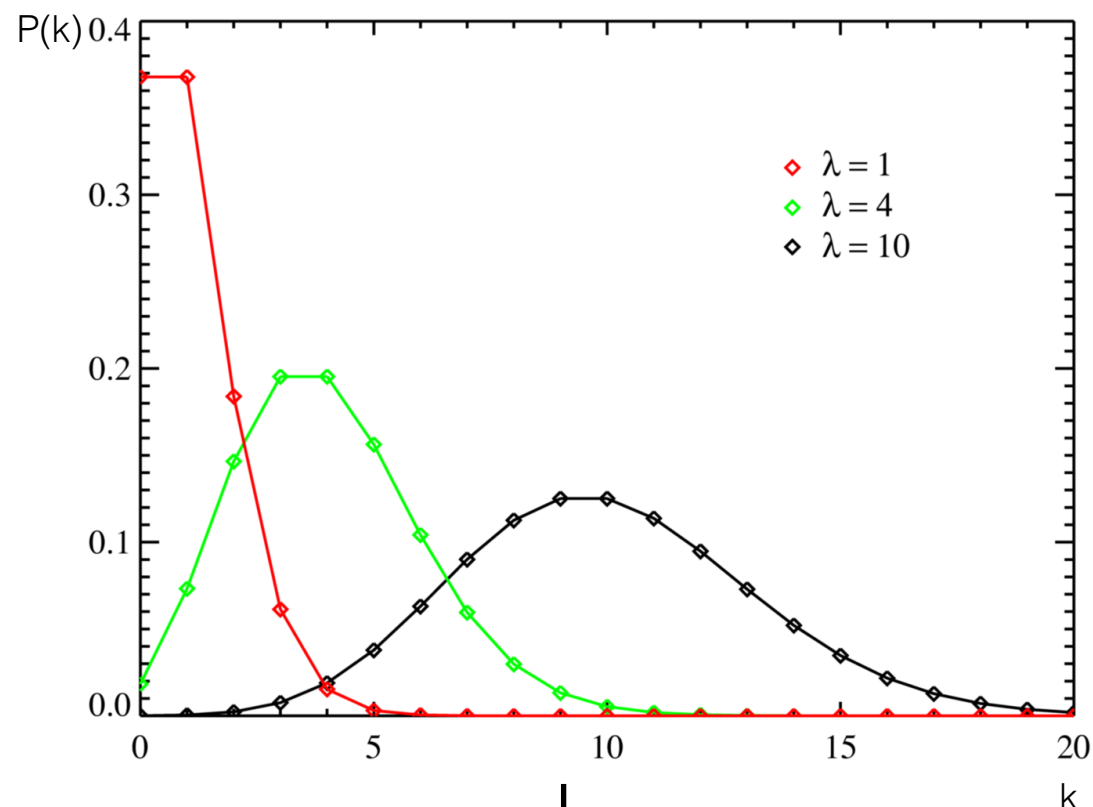
$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$



independent events

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Multiplicity distribution

Poisson distribution

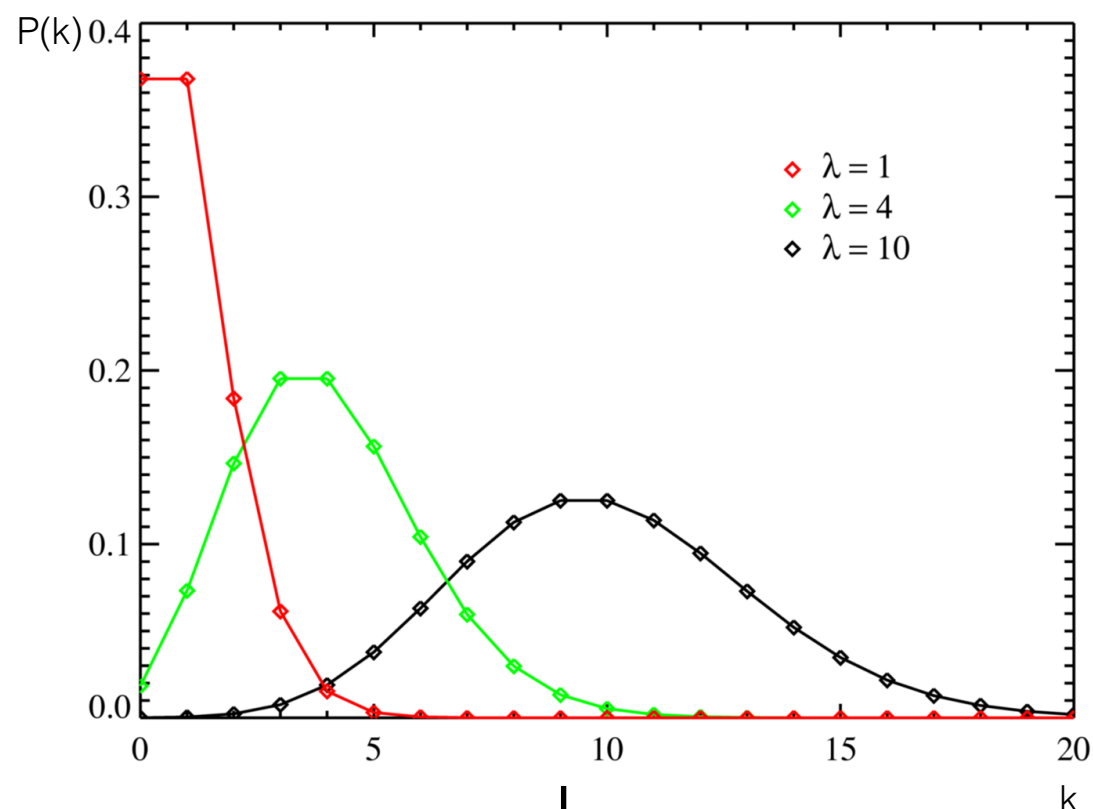
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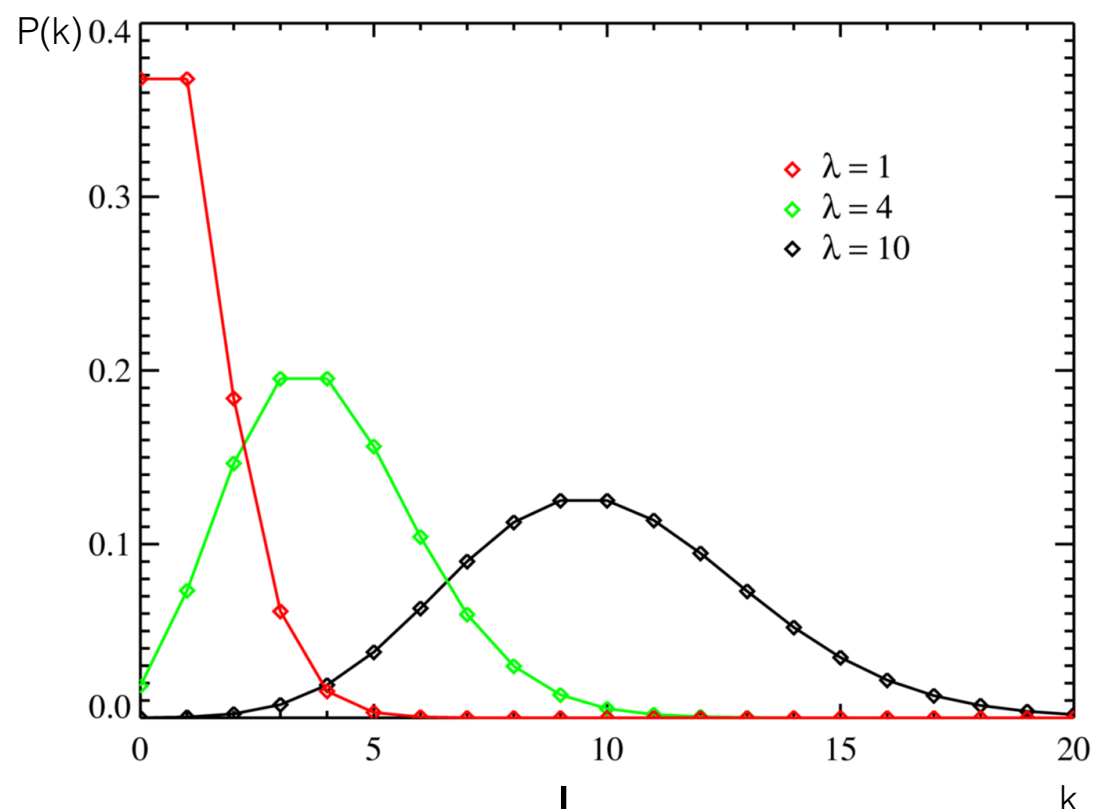
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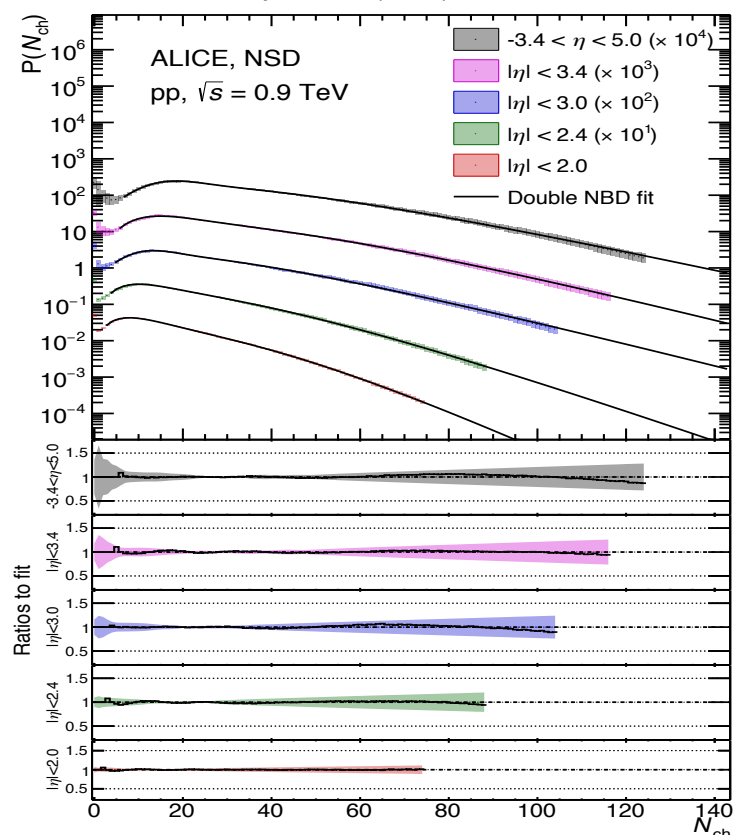
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Final state

↓

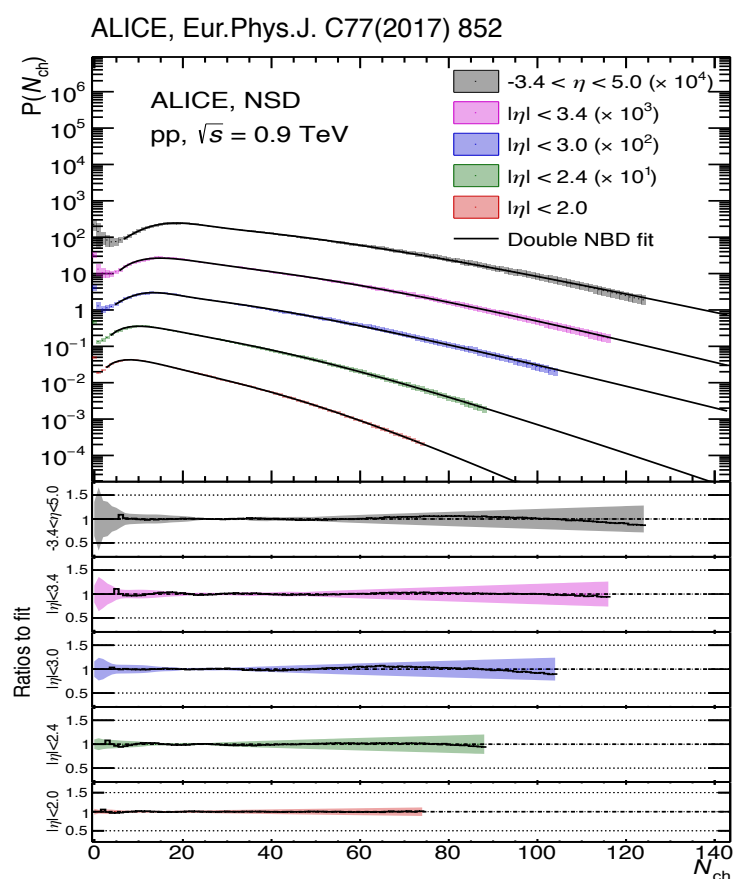
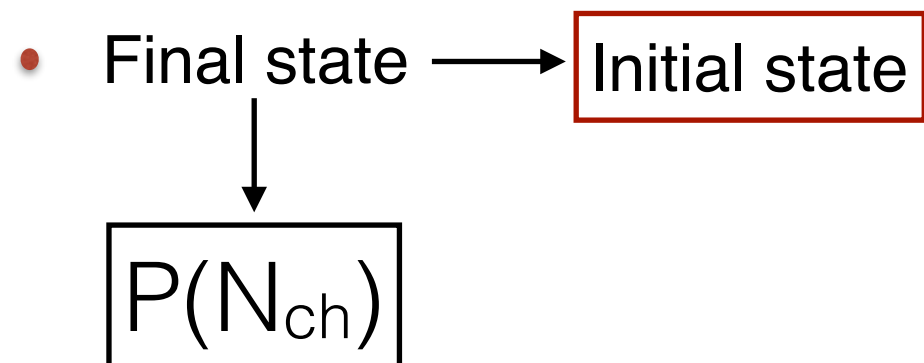
$$P(N_{ch})$$

ALICE, Eur.Phys.J. C77(2017) 852



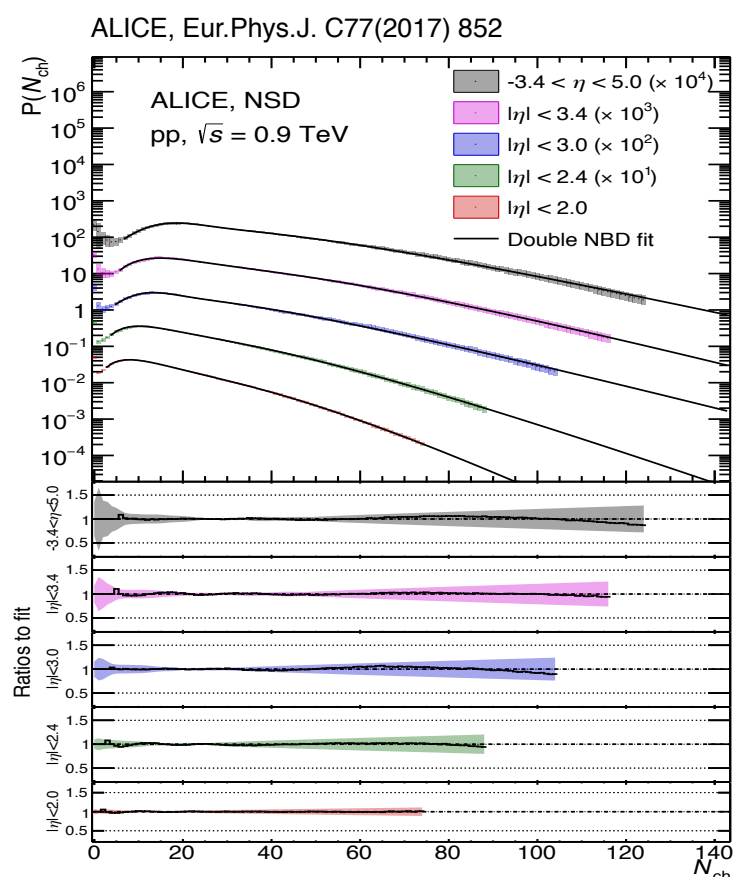
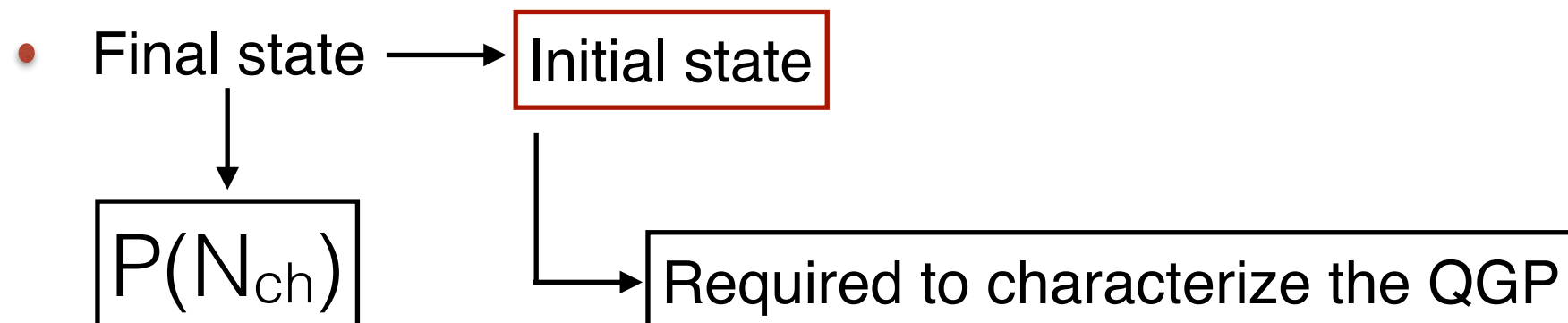
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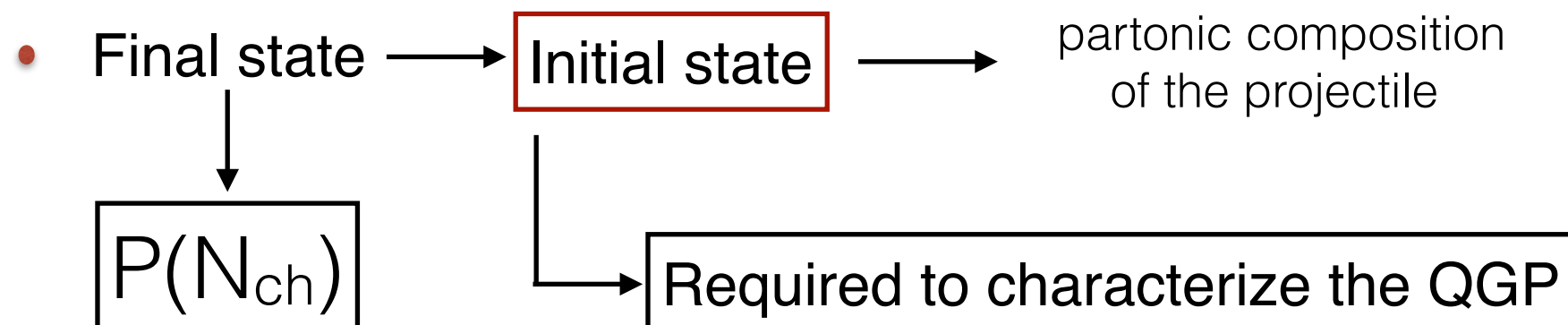
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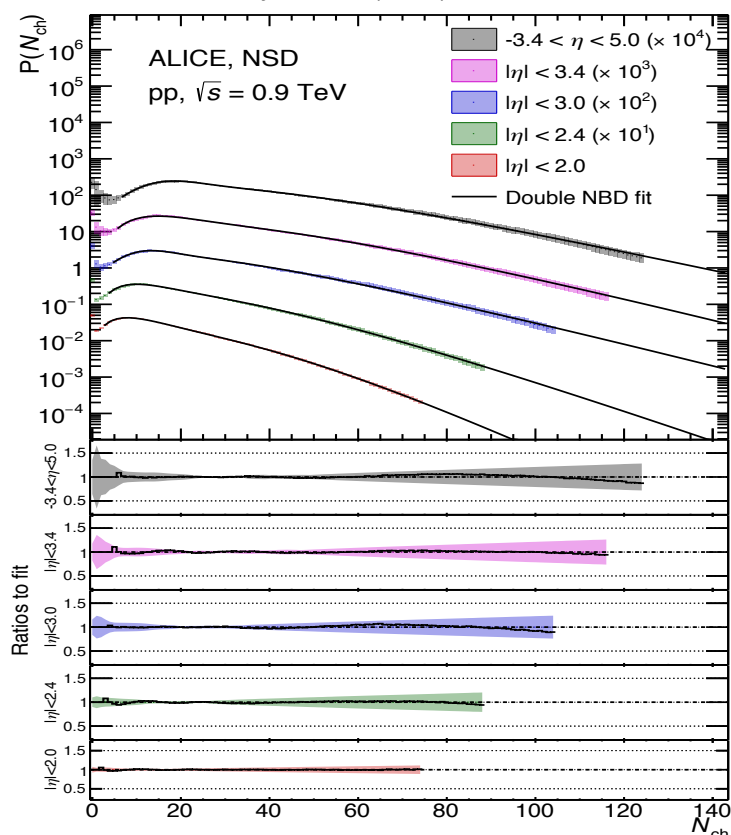


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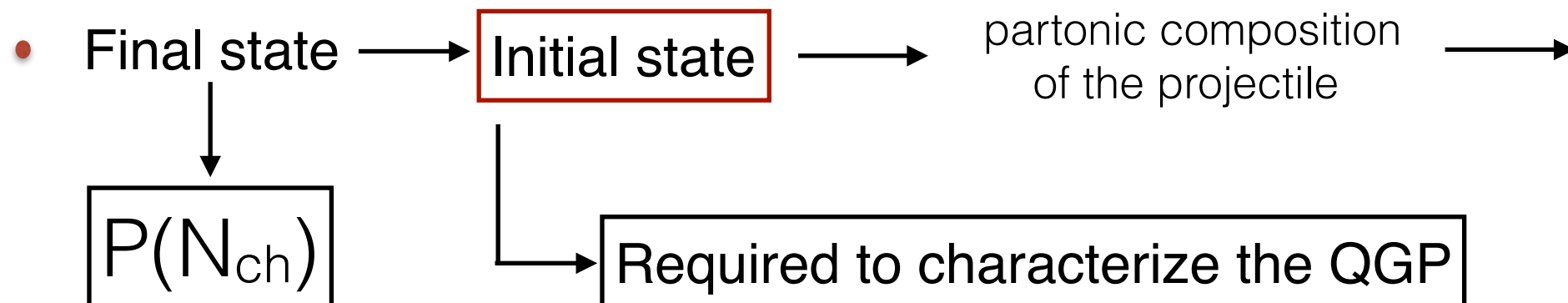


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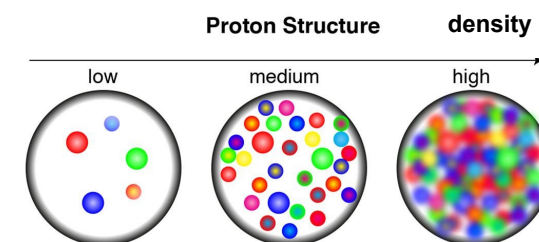


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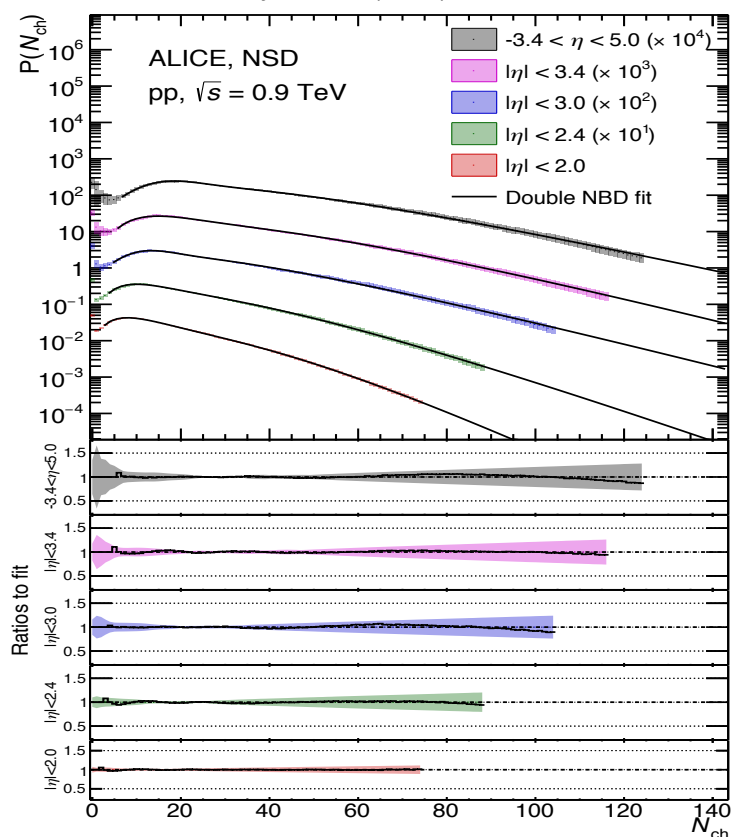
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Gluon saturation

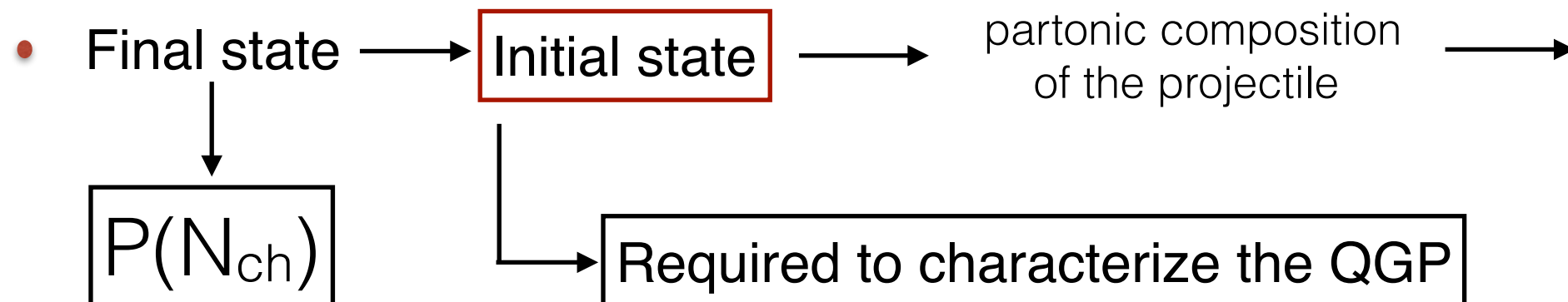


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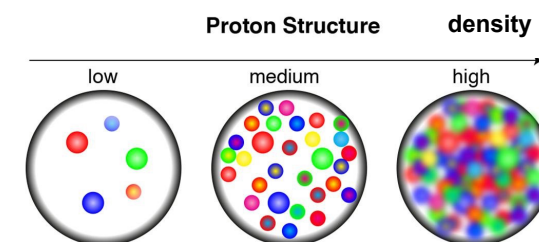


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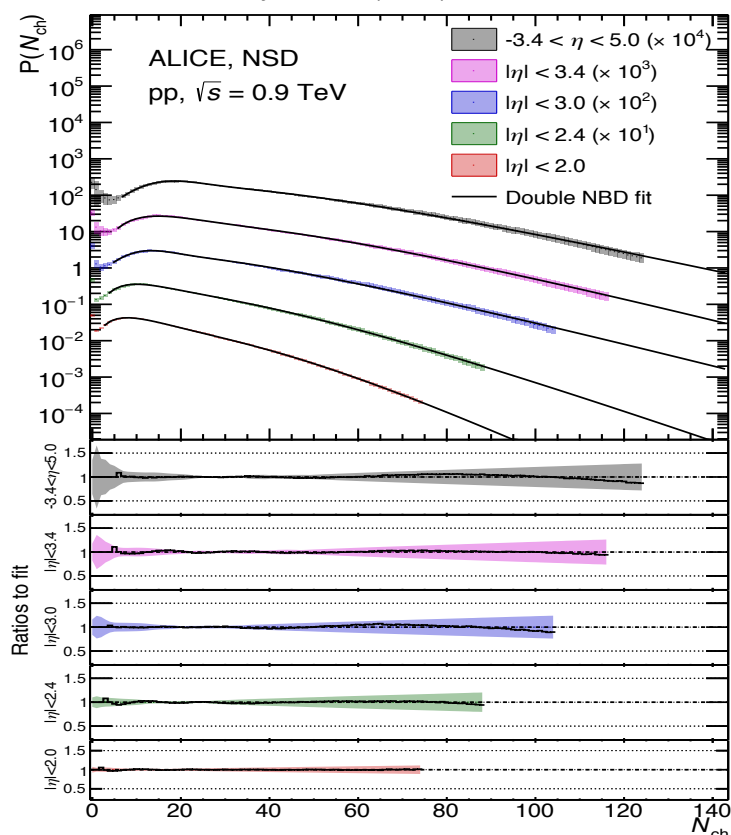
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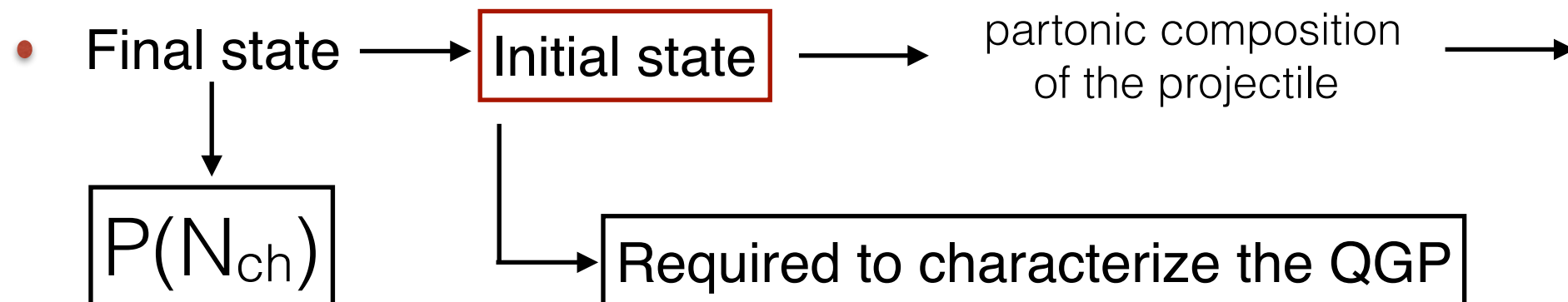
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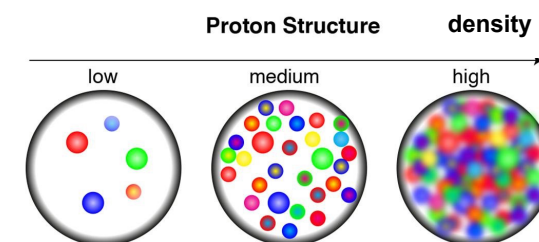
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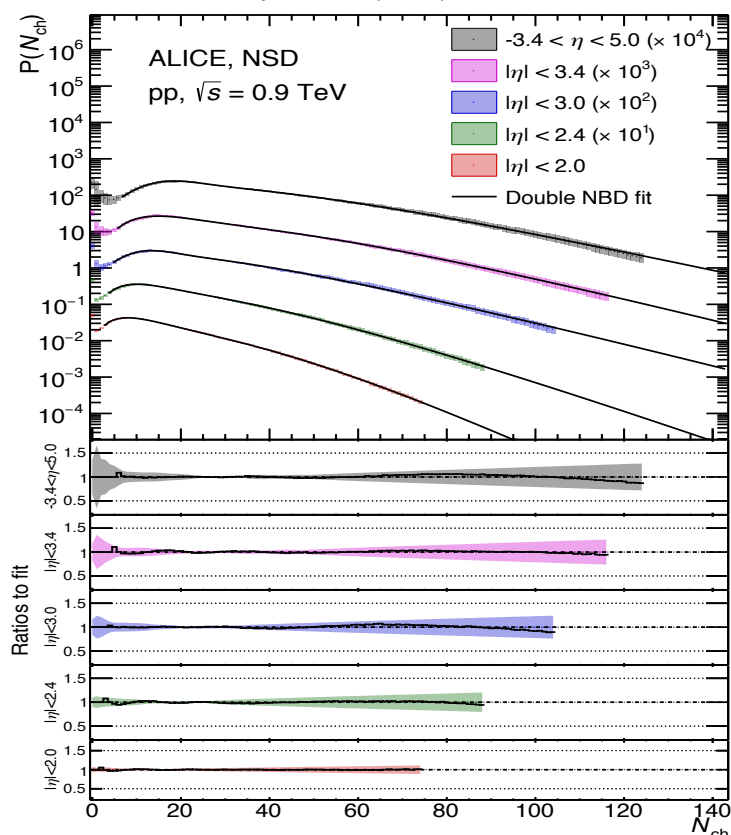
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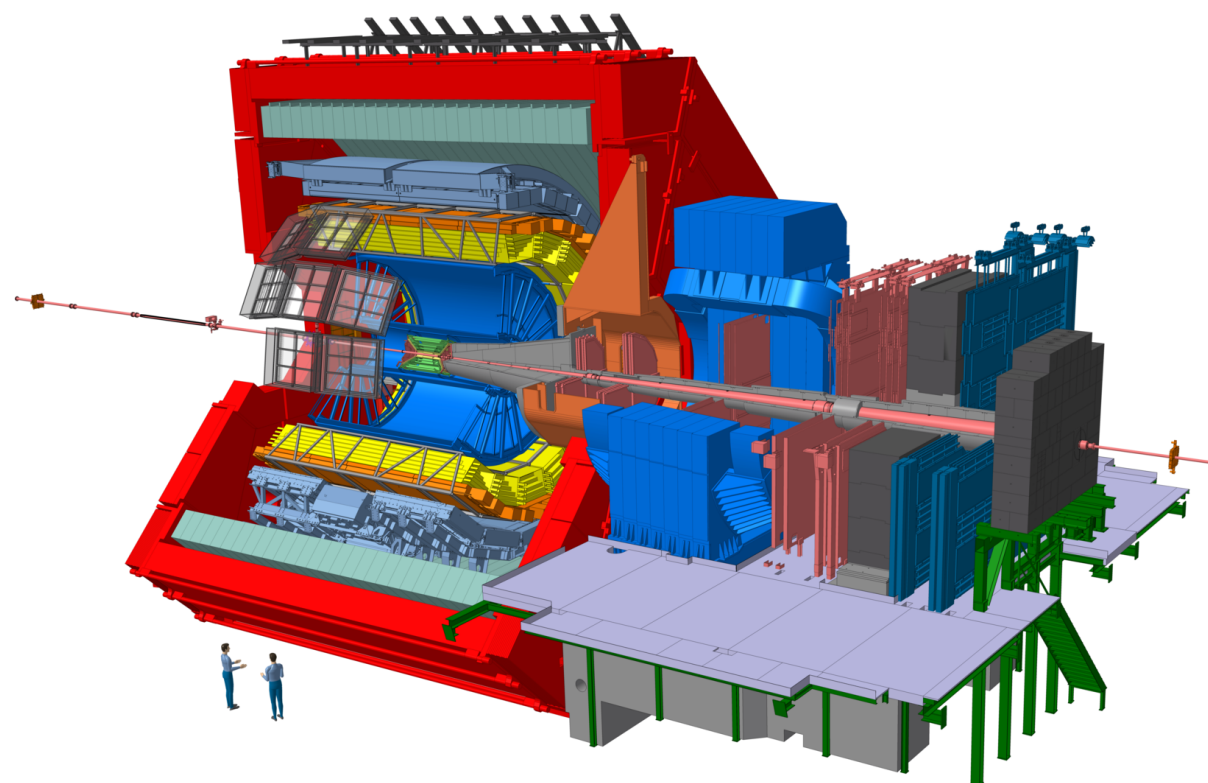


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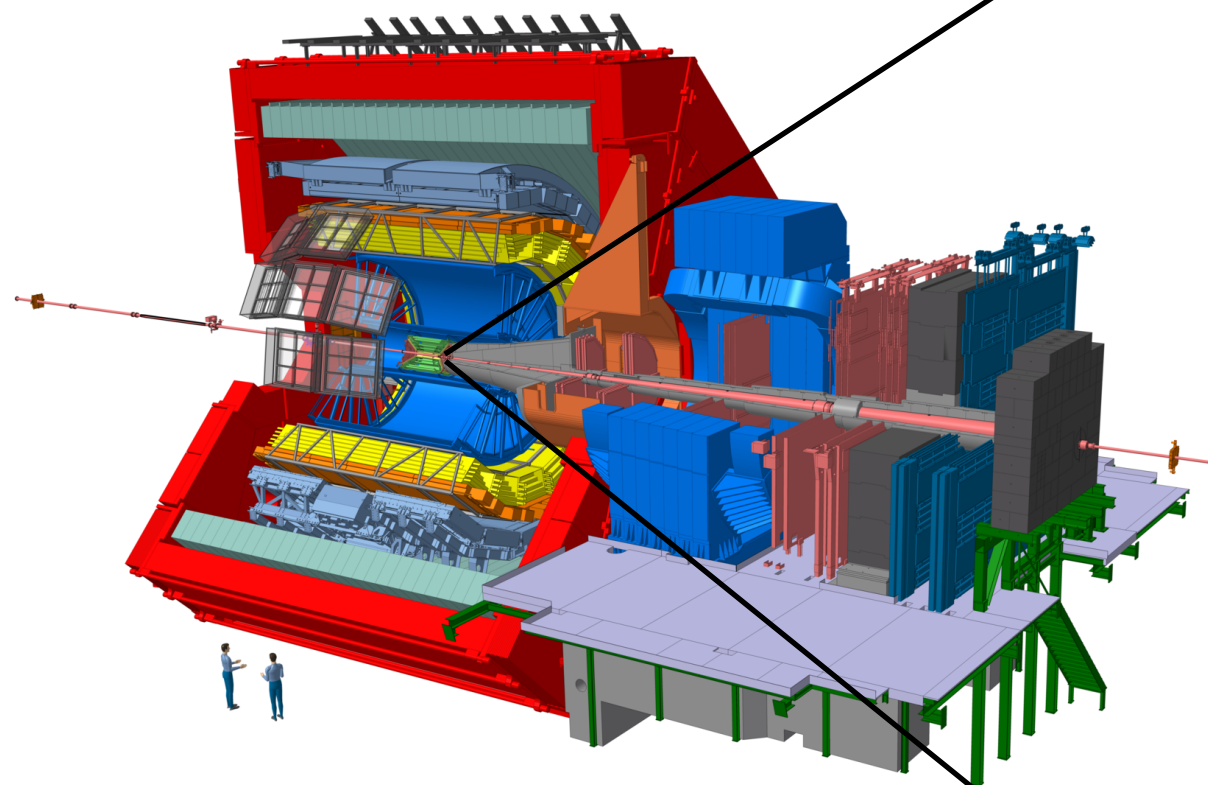
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↳ Presence of collective effects in p-Pb

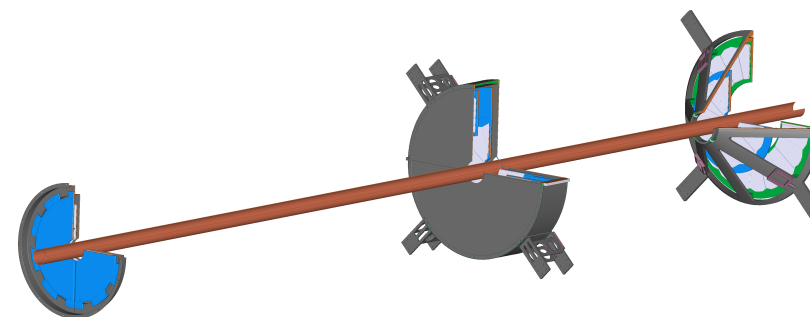


Optimized to study
heavy-ion collisions





Forward Multiplicity Detector

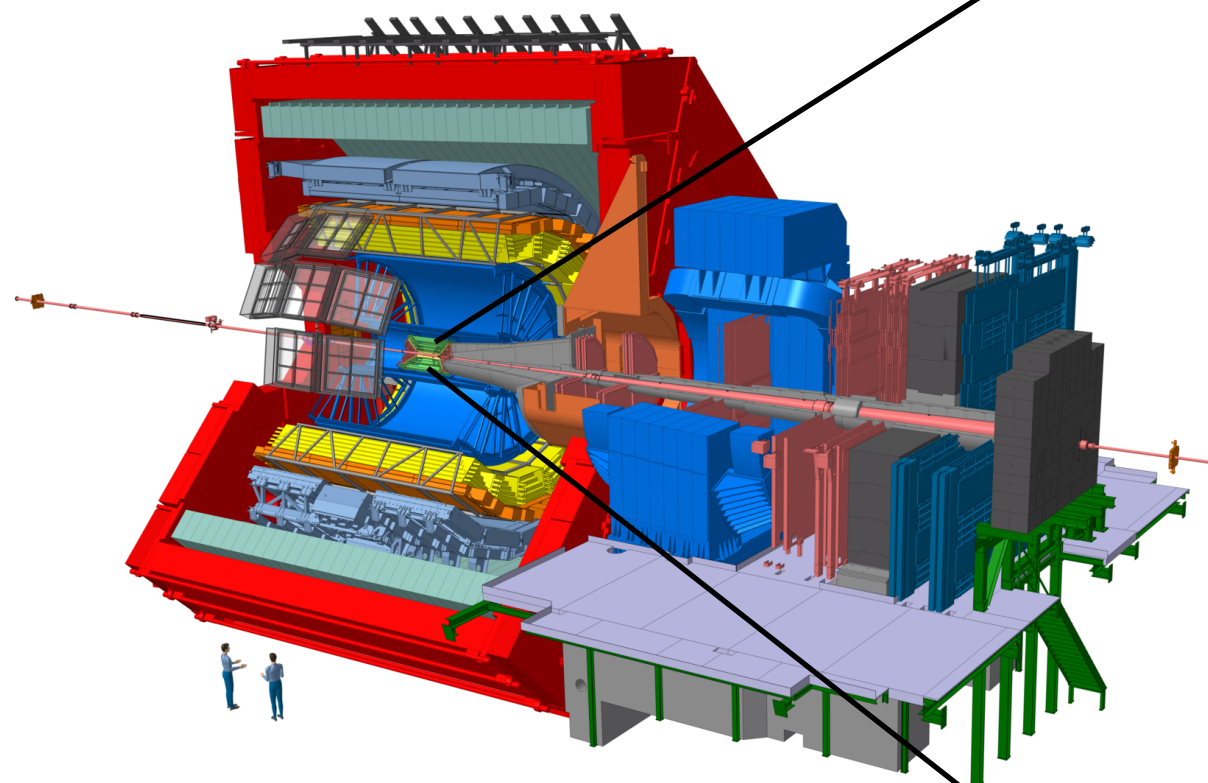


- Measure the multiplicity distribution of charged particles.
- Measurements in the forward regions $-3.4 \leq \eta \leq -1.7$ and $1.7 \leq \eta \leq 5.1$.

$$\eta = -\ln \left[\tan\left(\frac{\theta}{2}\right) \right]$$

$\theta \rightarrow$ The angle between the particle momentum $\mathbf{p}$ and the beam axis.

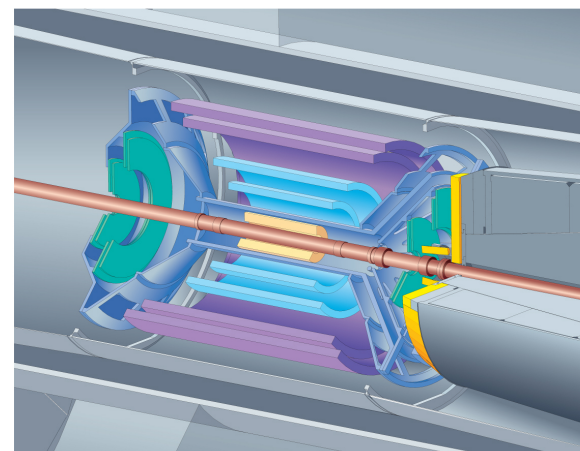




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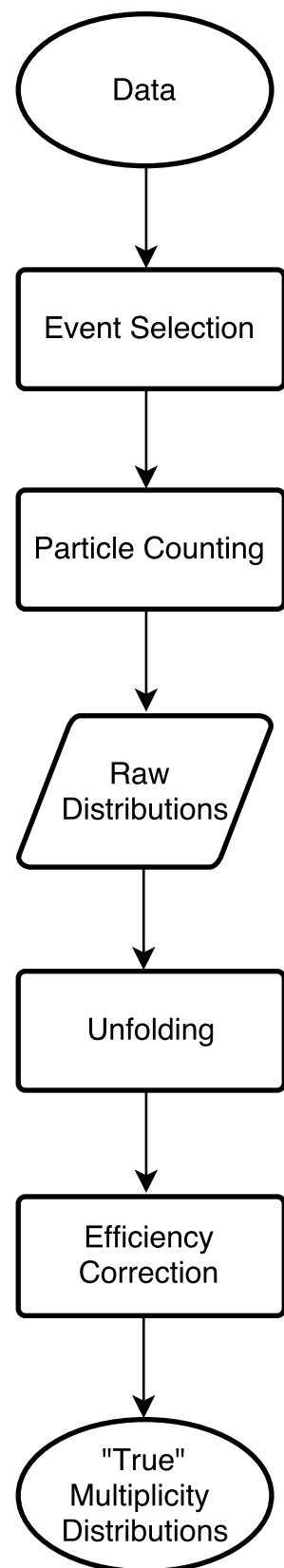
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Inner Tracking System

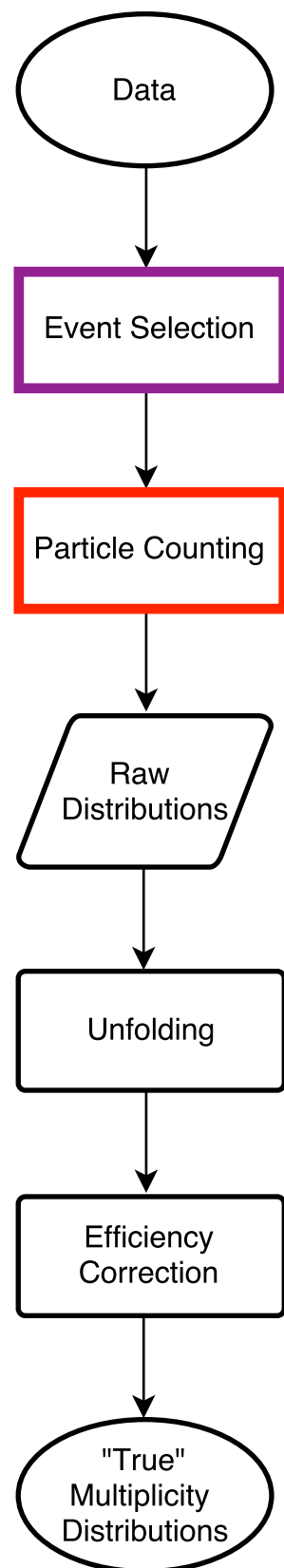


- SPD - Silicon Pixel Detector
- Coverage of $|\eta| < 2.0$
- FMD + SPD - continuous coverage for the measurements

Recipe to get $P(N_{ch})$



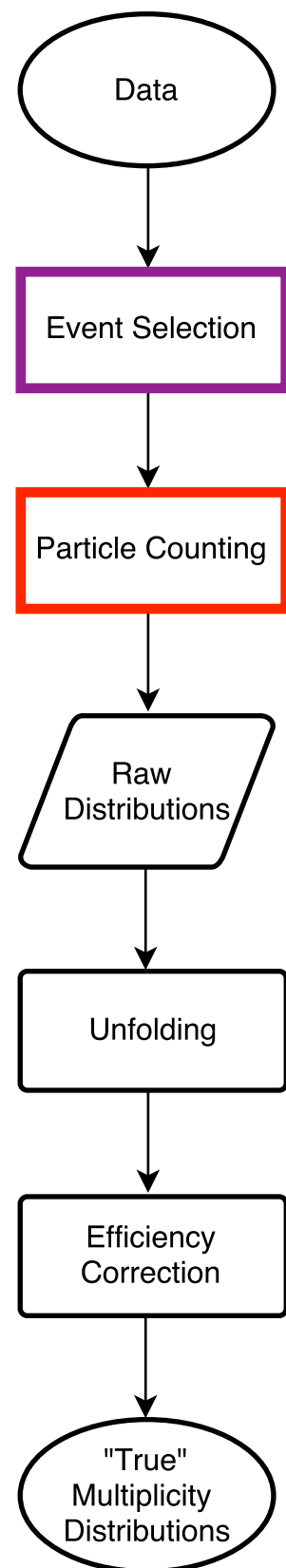
Event Selection and Particle Counting



Selection of trigger and vertex

Particle counting by FMD and SPD

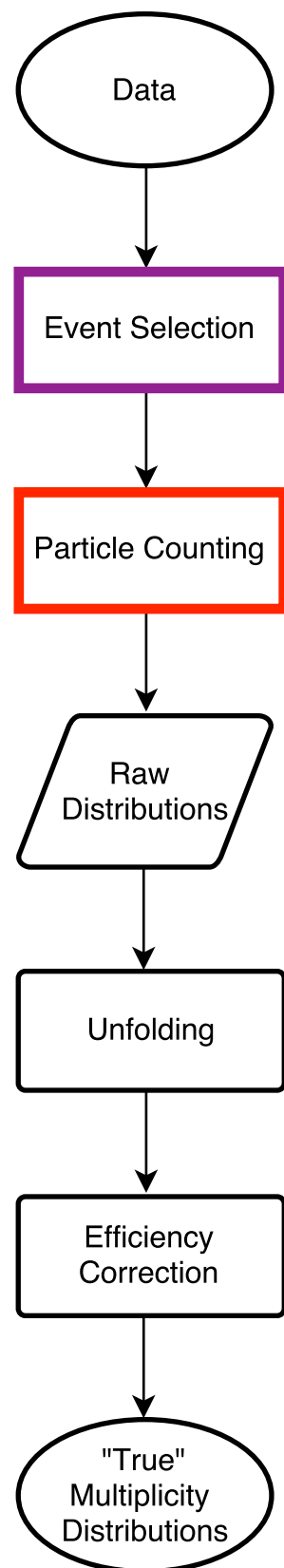




Selection of trigger and vertex

- ↳ The V0 and SPD detectors are used to select events
- ↳ $|v_z| < 4\text{cm}$ → Remove acceptance gaps

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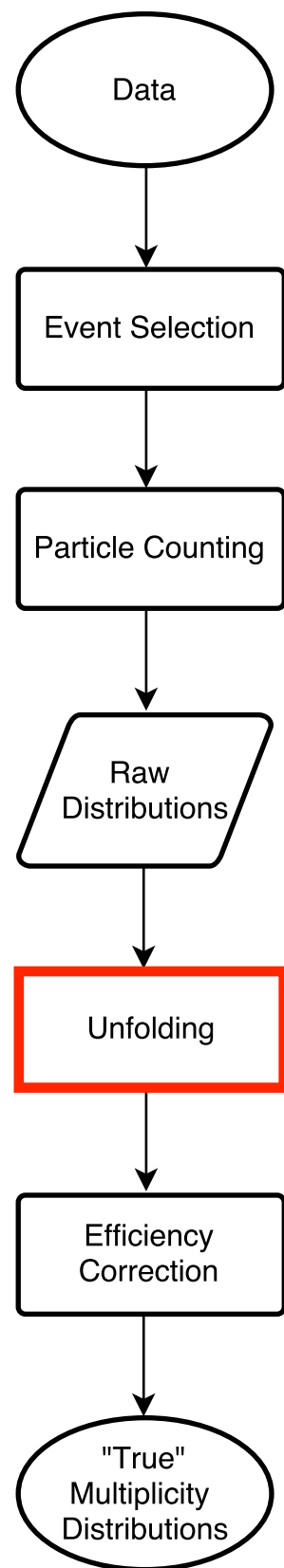


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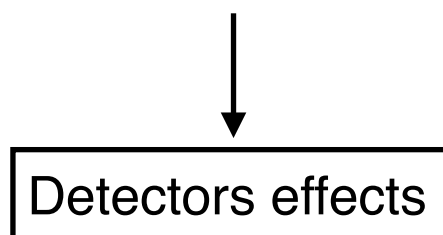
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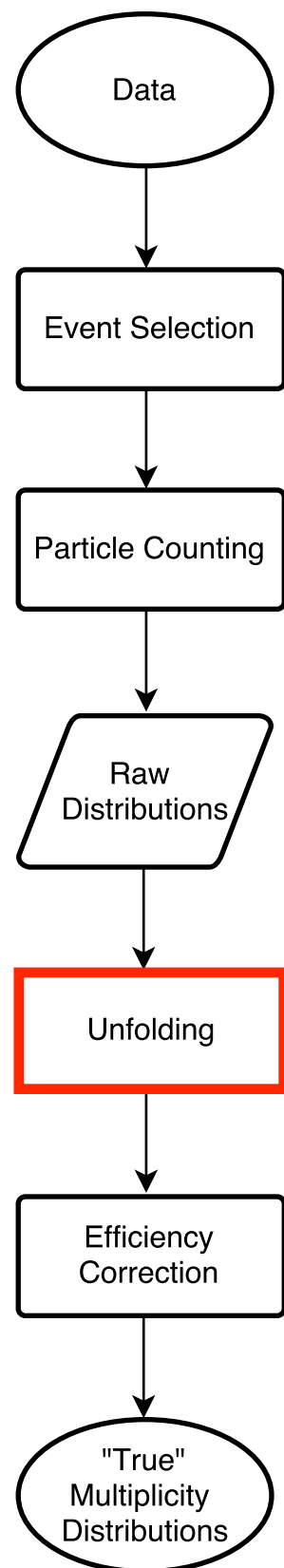
Particle counting by FMD and SPD

- ↳ Occasionally, particles hit FMD and deposit energy in more than one strip
- ↳ Shared signals are merged into single signals

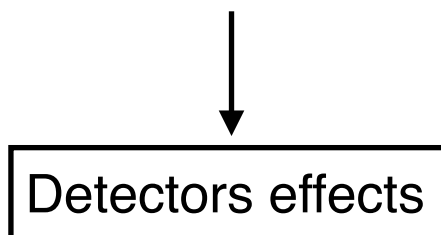


The observed distribution is a distortion of the “true” distribution

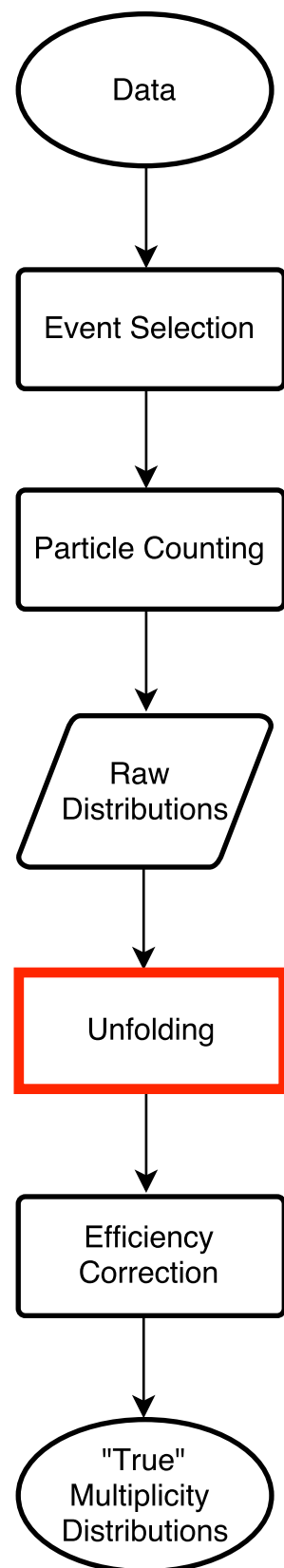




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Unfolding: Corrects the raw distribution for the effects of secondary particles, acceptance gaps, and resolution.



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Detectors effects

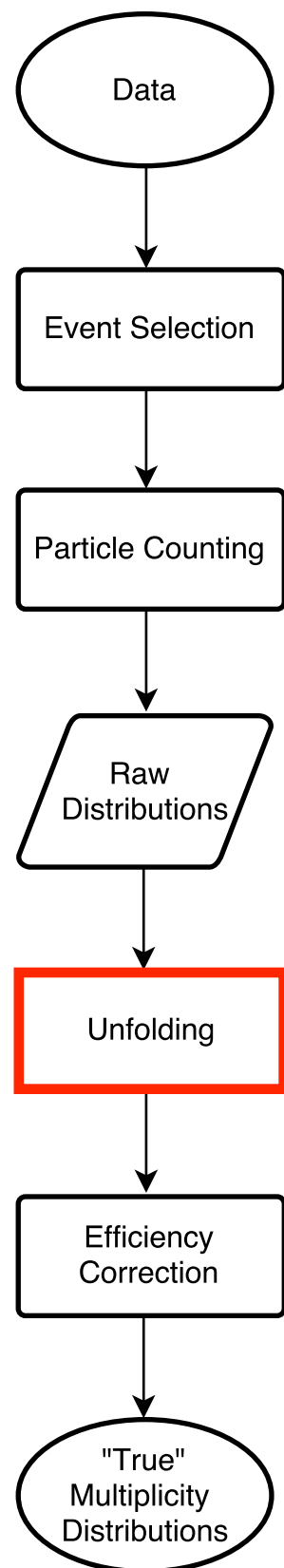
Unfolding: Corrects the raw distribution for the effects of secondary particles, acceptance gaps, and resolution.

Mathematical definition:

$$f_{meas}(x) = \int R(x|y) f_{true}(y) dy$$

For binned data $\rightarrow x_i = R_{ij} y_j$





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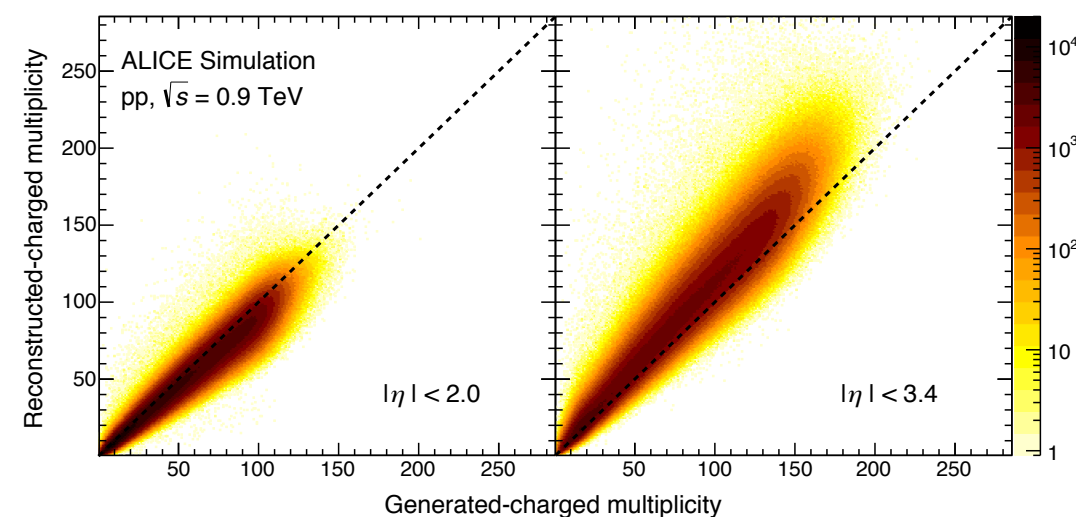
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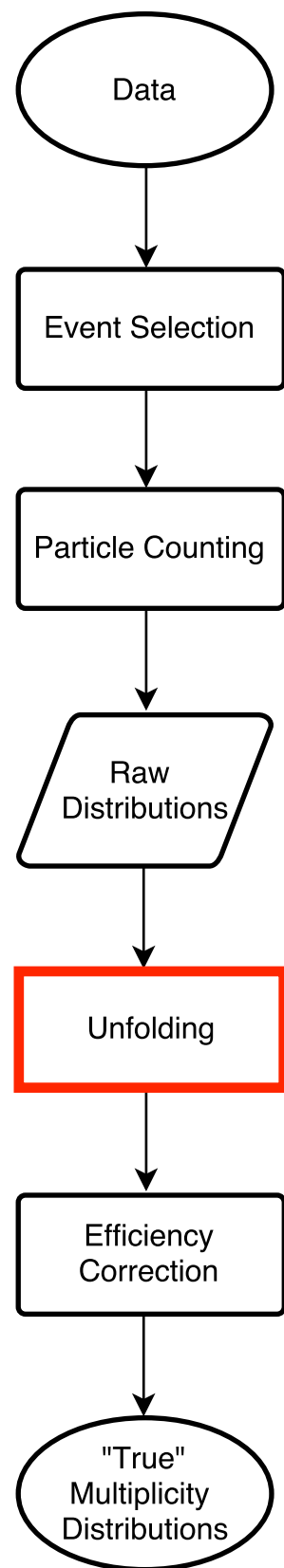
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ALICE, arXiv:1708.01435





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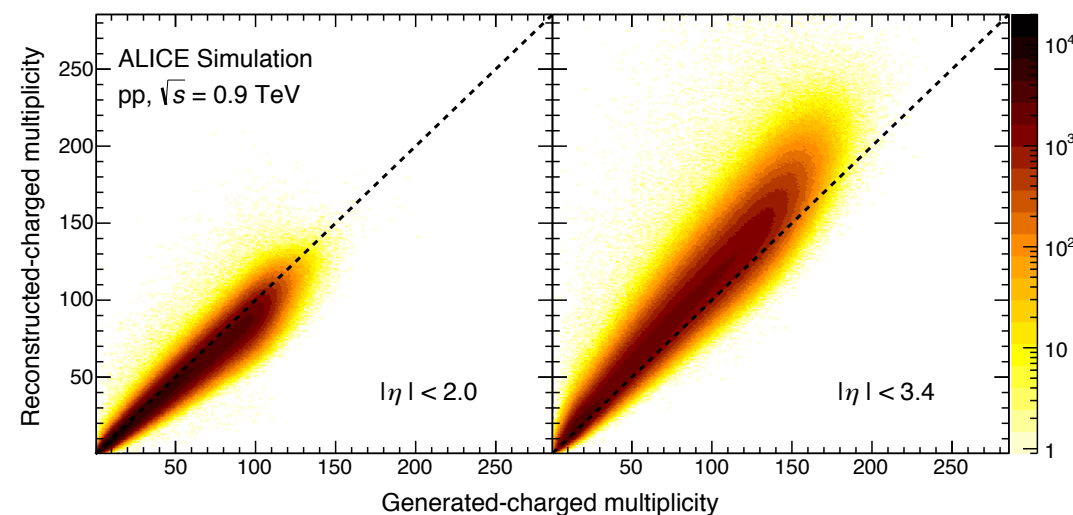
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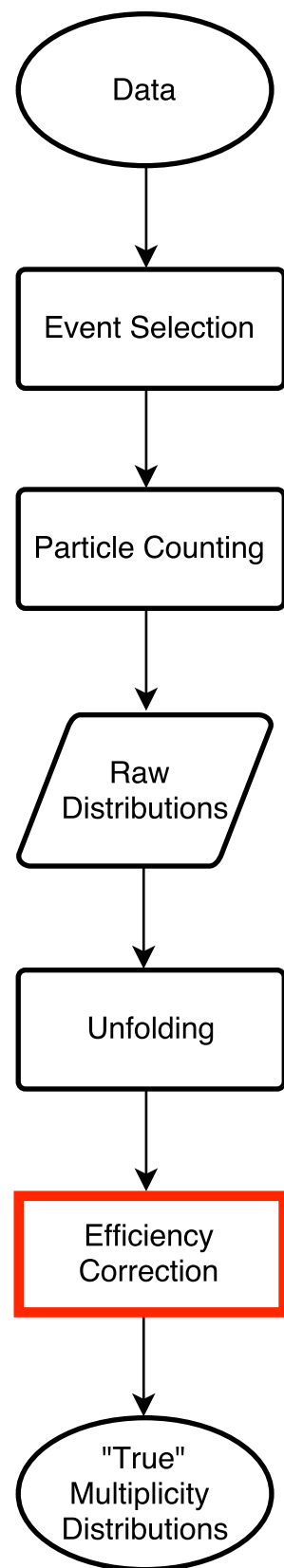
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Bayesian Unfolding

ALICE, arXiv:1708.01435



Efficiency Correction



Low Multiplicities

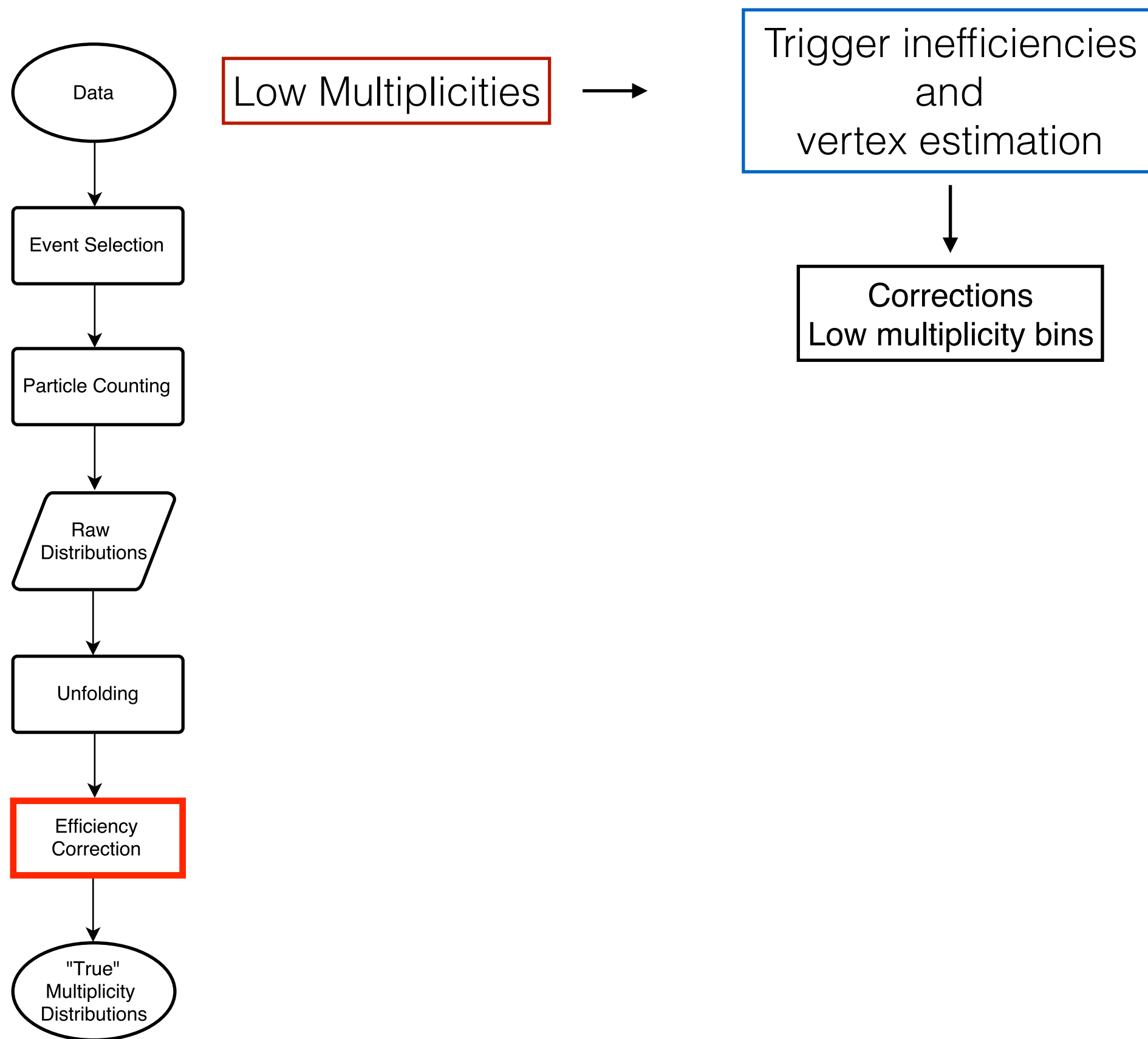


Trigger inefficiencies
and
vertex estimation

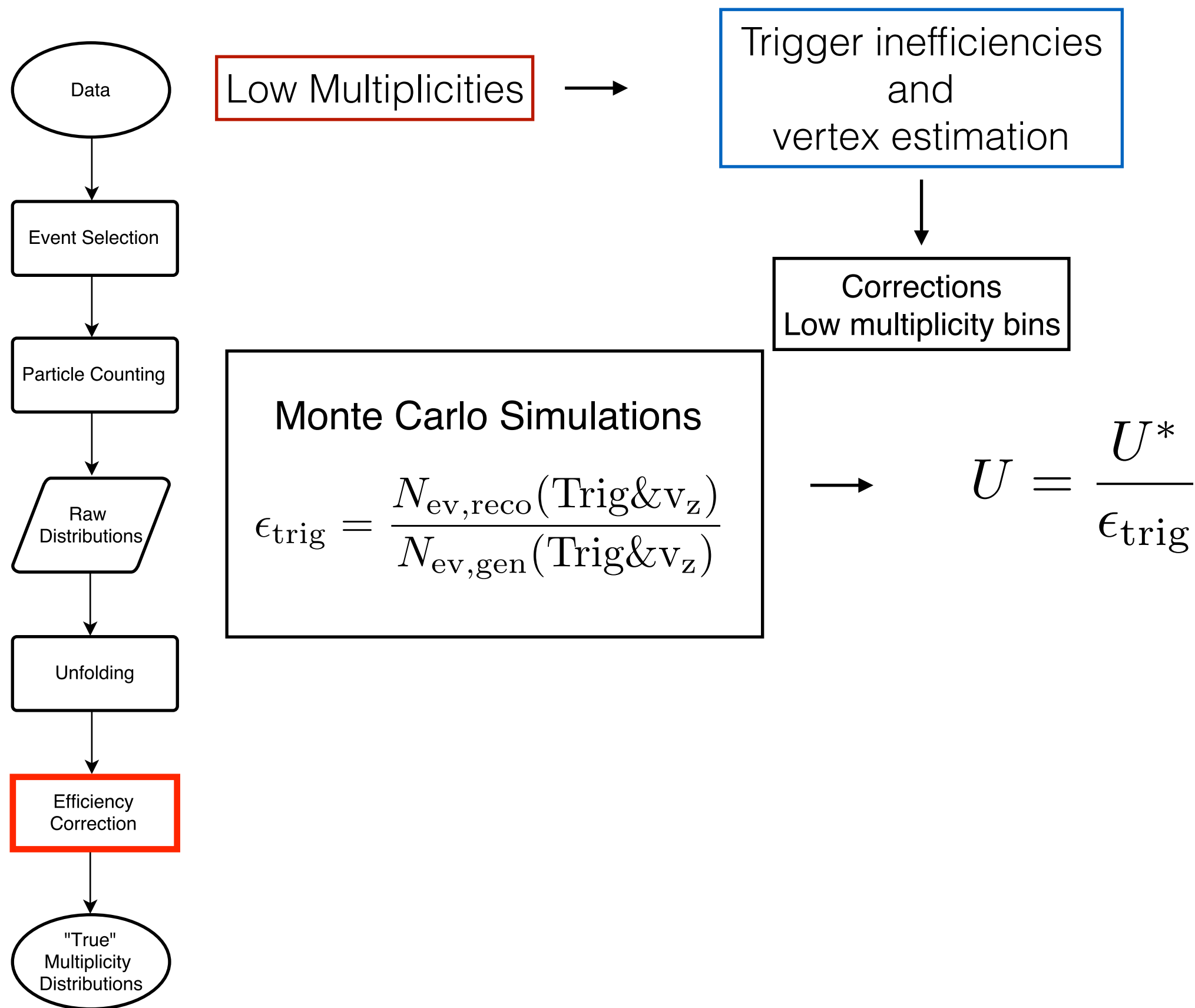


ALICE

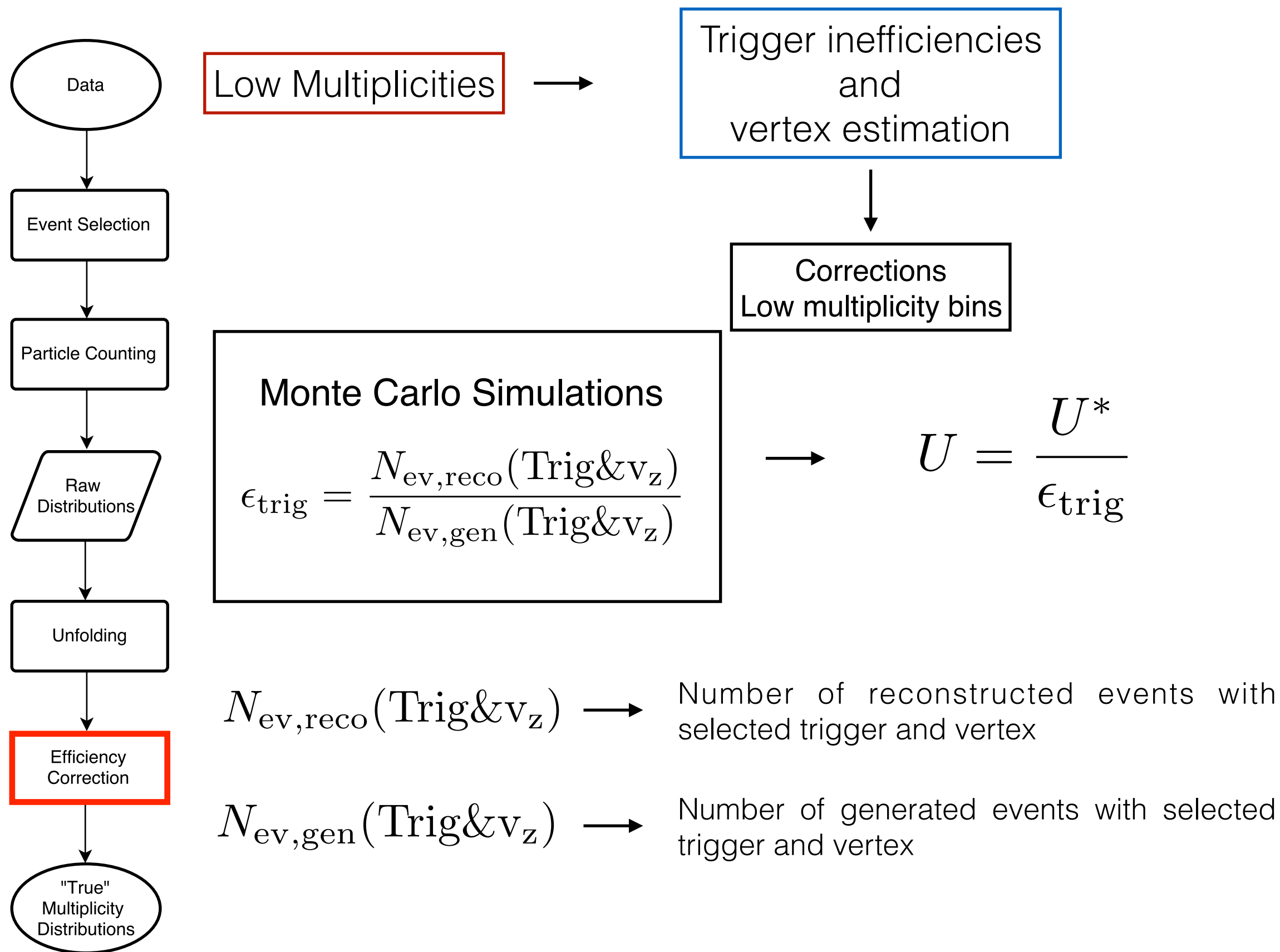
Efficiency Correction

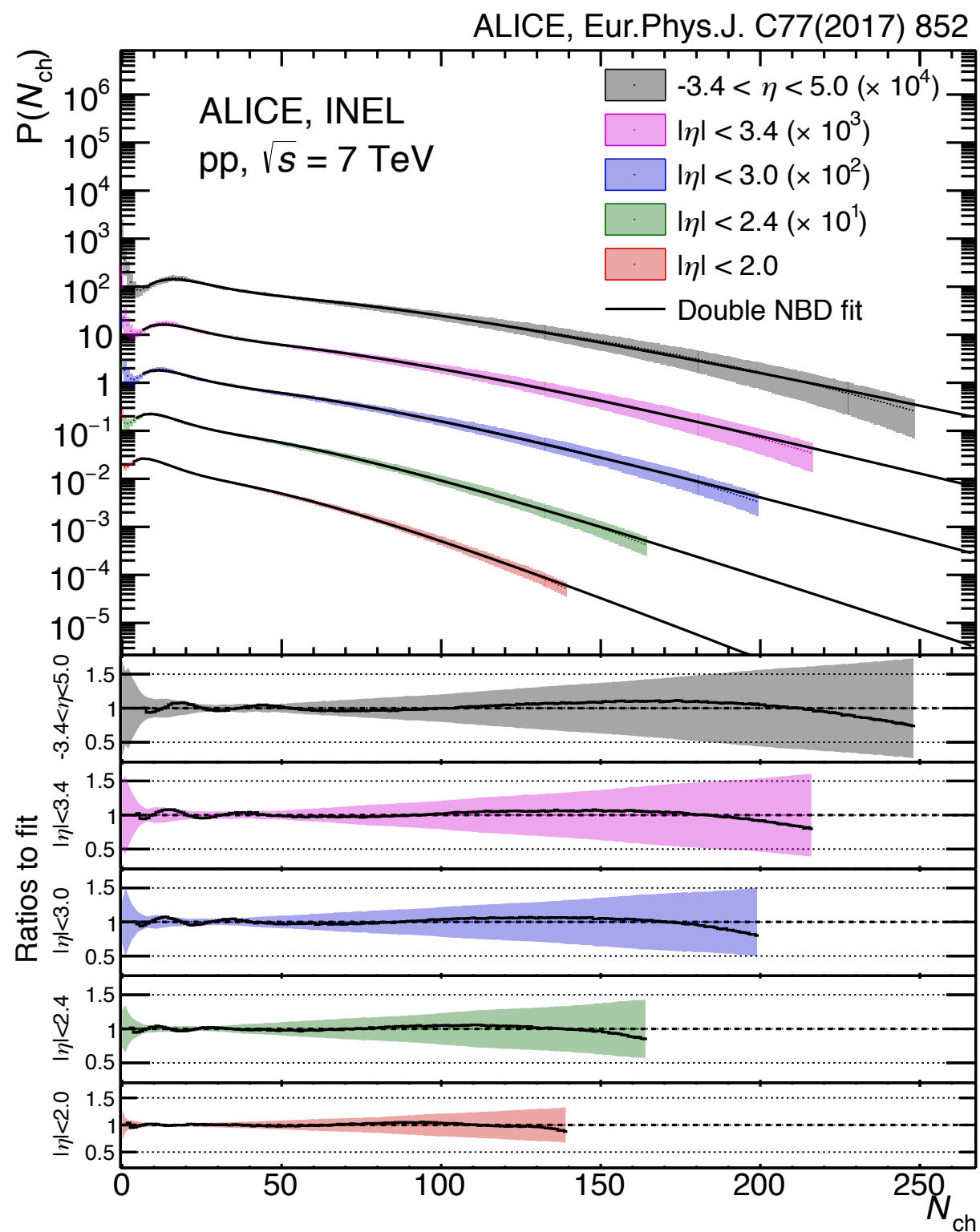


Efficiency Correction

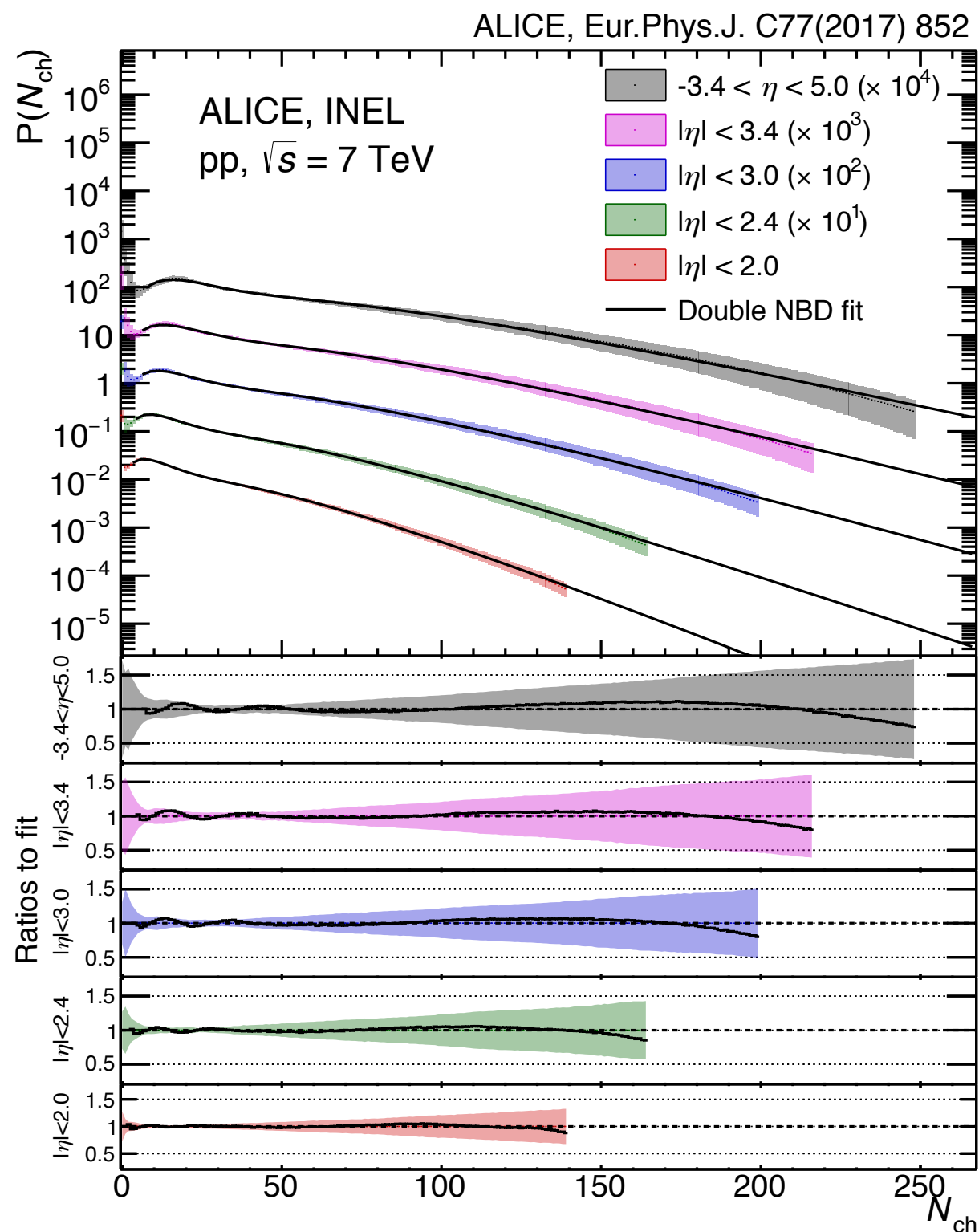


Efficiency Correction

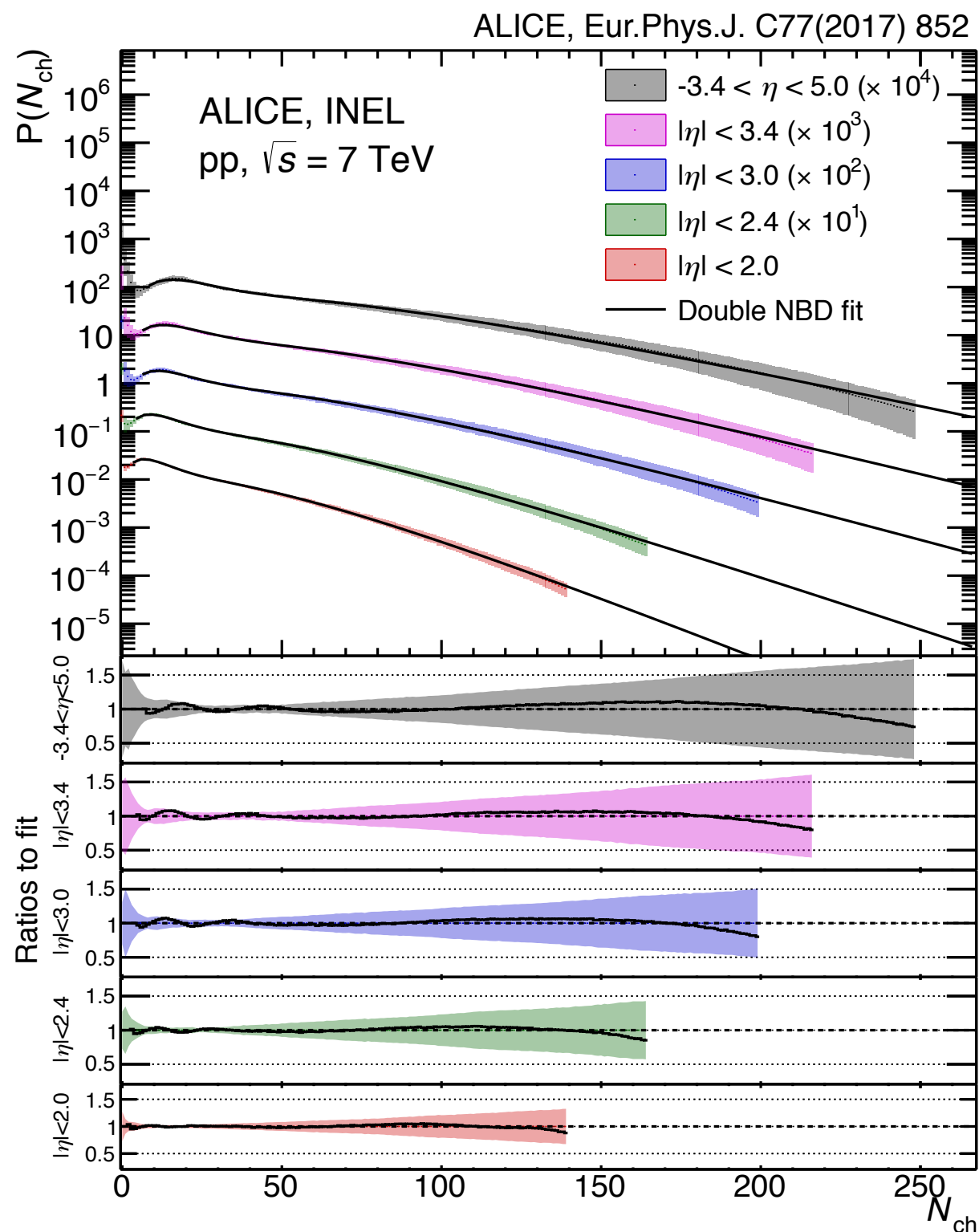




- INEL trigger: It includes all the inelastic events.



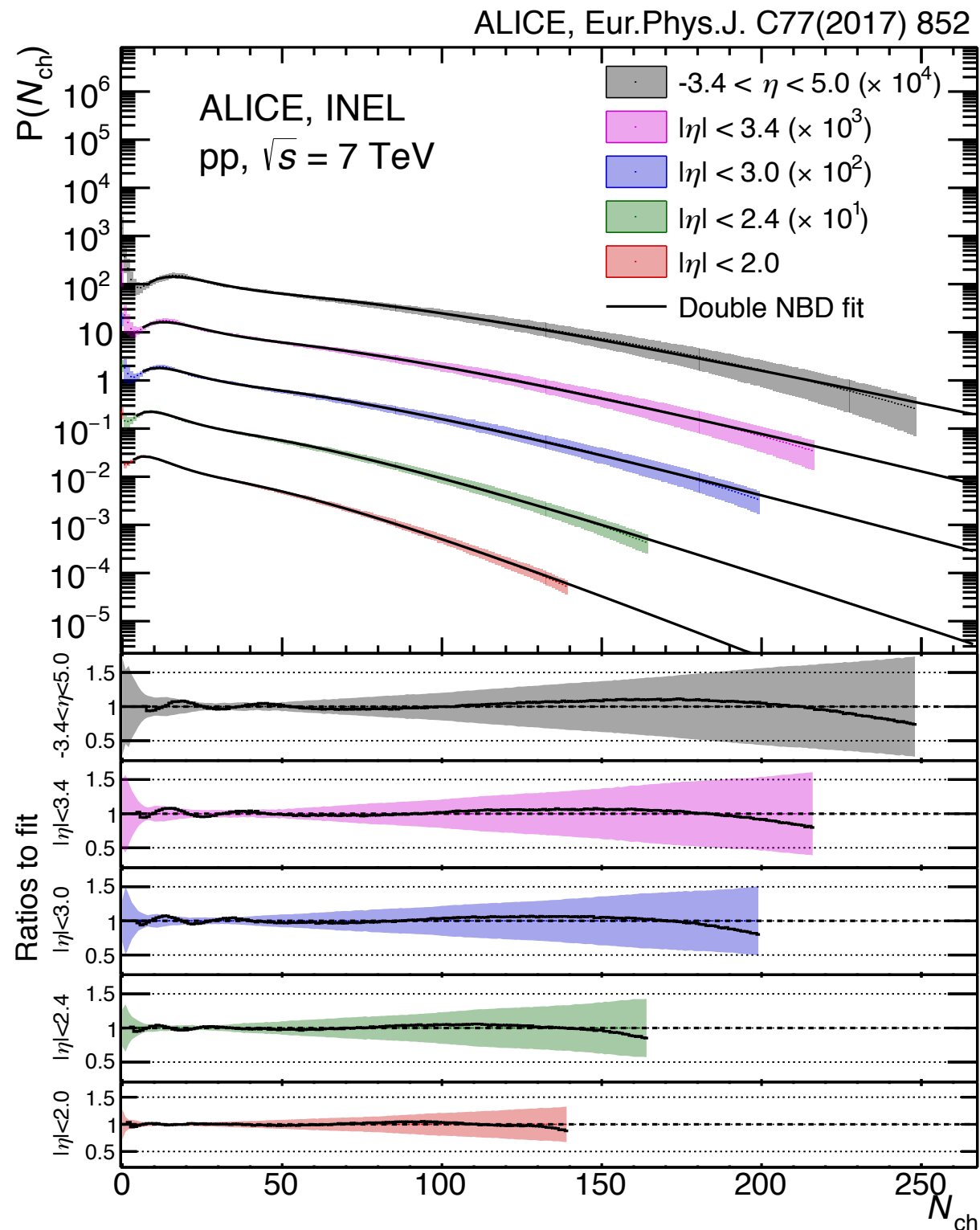
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- Sum of two NBDs fit to the charged-particle multiplicity distributions.



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$$P(n) = \lambda[\alpha P_{\text{NBD}}(n, \langle n \rangle_1, k_1) + (1 - \alpha) P_{\text{NBD}}(n, \langle n \rangle_2, k_2)]$$



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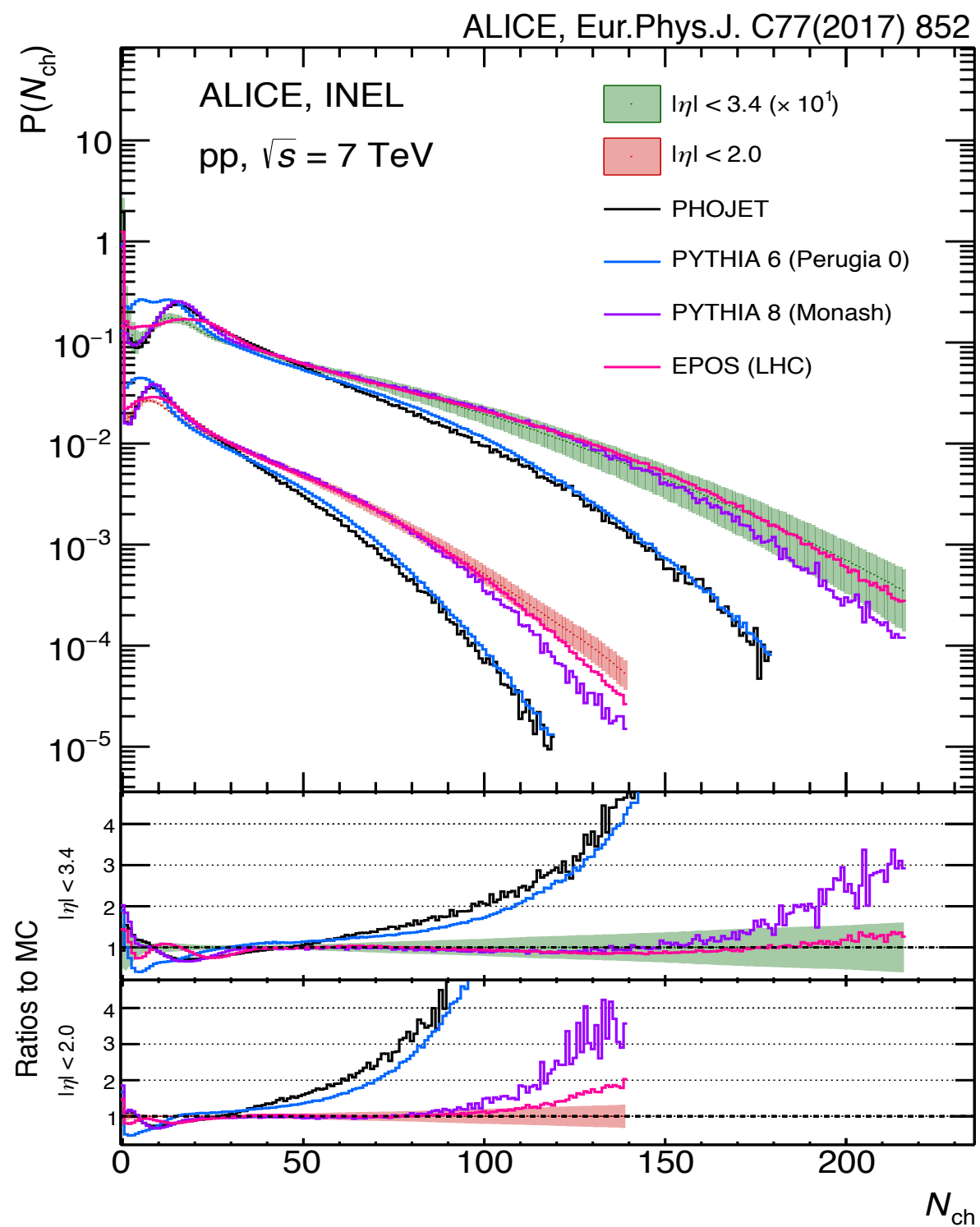
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Phys.Rev. D59 (1999) 094020

- According to the clan model

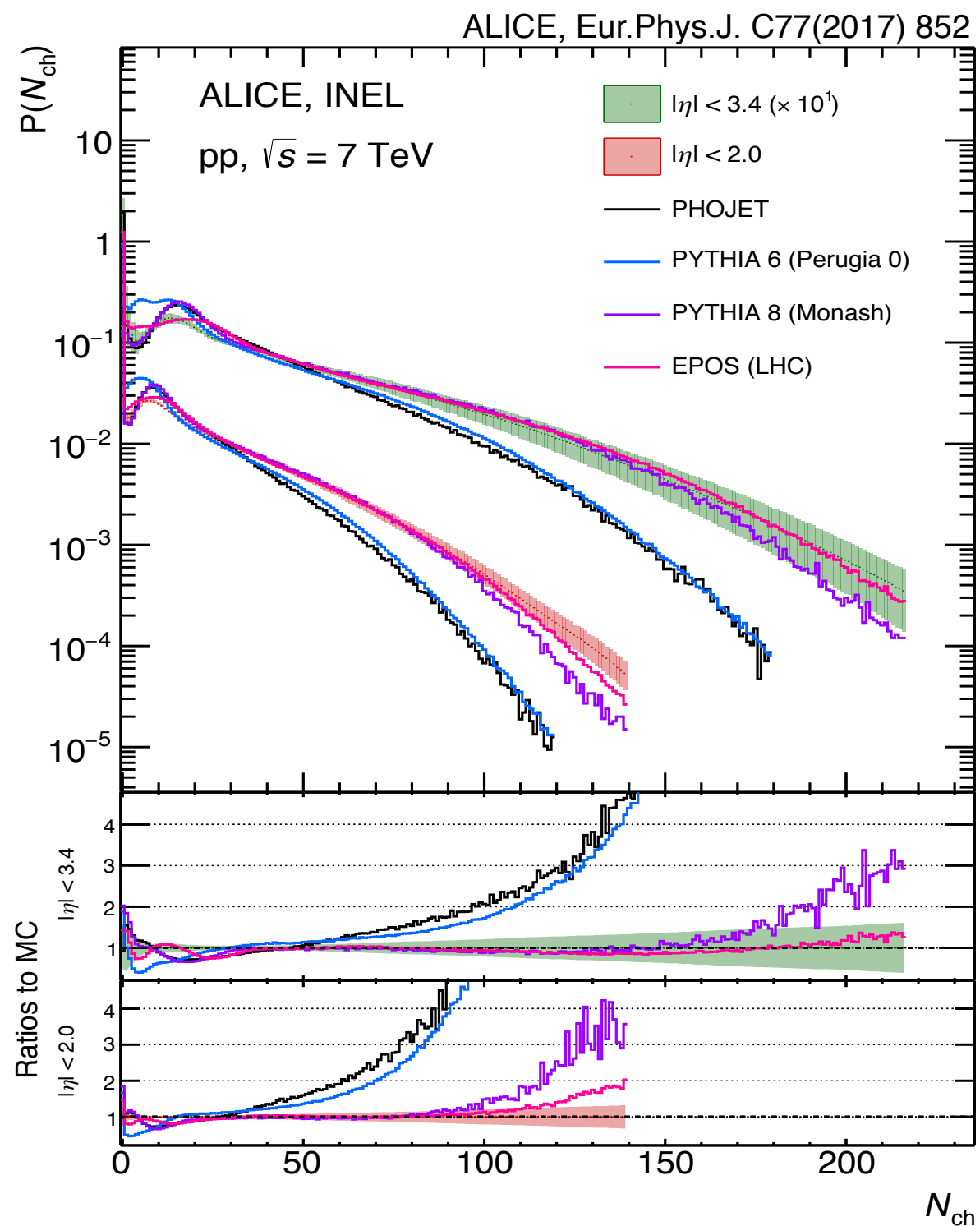
$P_{\text{NBD}_1} \longrightarrow$ Soft component
 $\searrow$ Small momentum transfer
 $\searrow$ Diffraction

$P_{\text{NBD}_2} \longrightarrow$ Semi-hard component
 $\searrow$ Large momentum transfer



- Comparison with 4 MC models





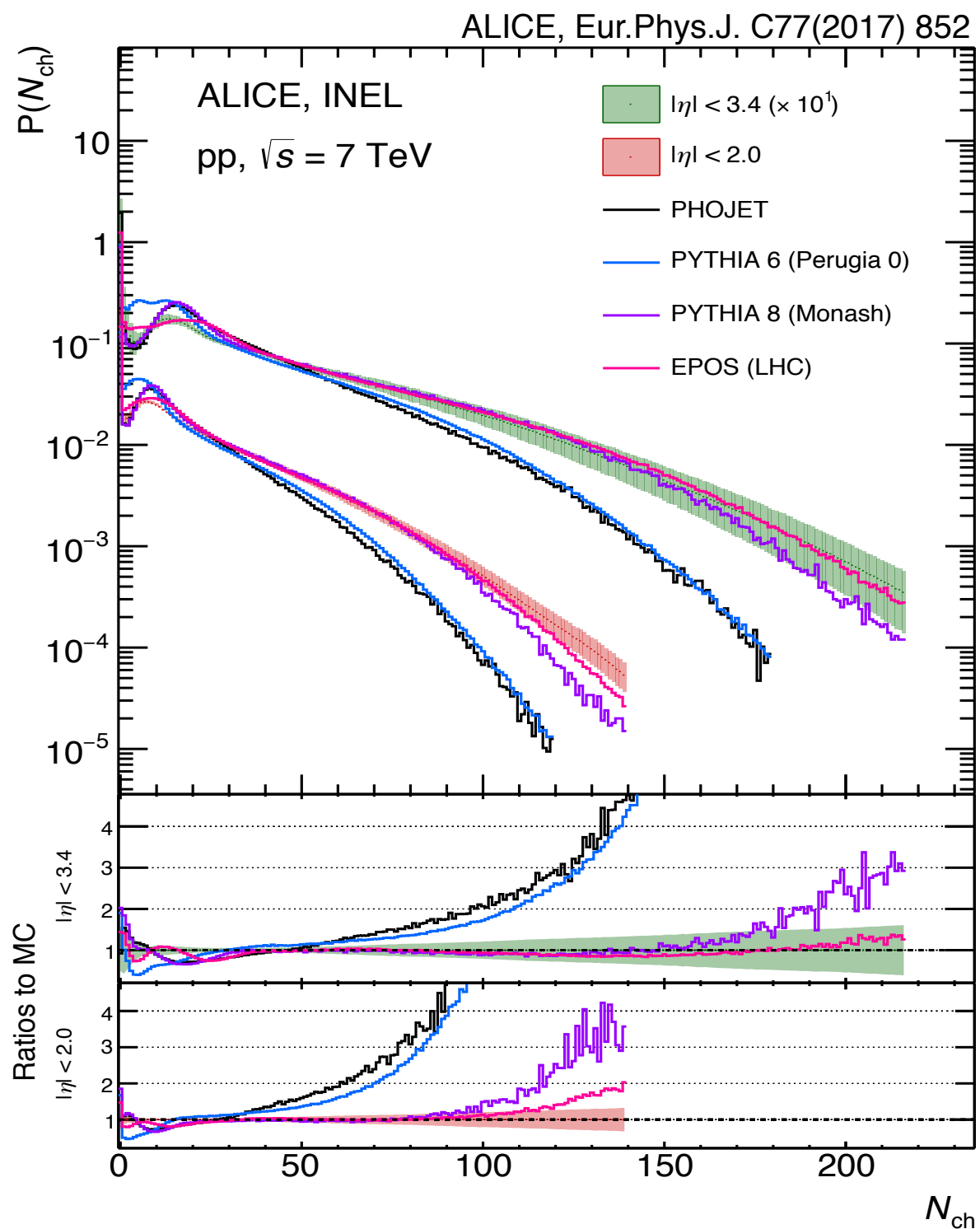
- Comparison with 4 MC models

↳ PHOJET and PYTHIA 6

↳ Underestimate the tails

↳ PYTHIA 8

→ Enhances the peak



Comparison with 4 MC models

↳ PHOJET and PYTHIA 6

↳ Underestimate the tails

↳ PYTHIA 8 → Enhances the peak

↳ EPOS LHC → Describes well the distributions

Two scenarios in p-Pb collisions

Monte Carlo Generators

EPOS LHC

Nucl.Phys.Proc.Suppl. 196 (2009) 36-43

DPMJet

Phys. Rev. D 51, 64 (1995)



ALICE

Two scenarios in p-Pb collisions

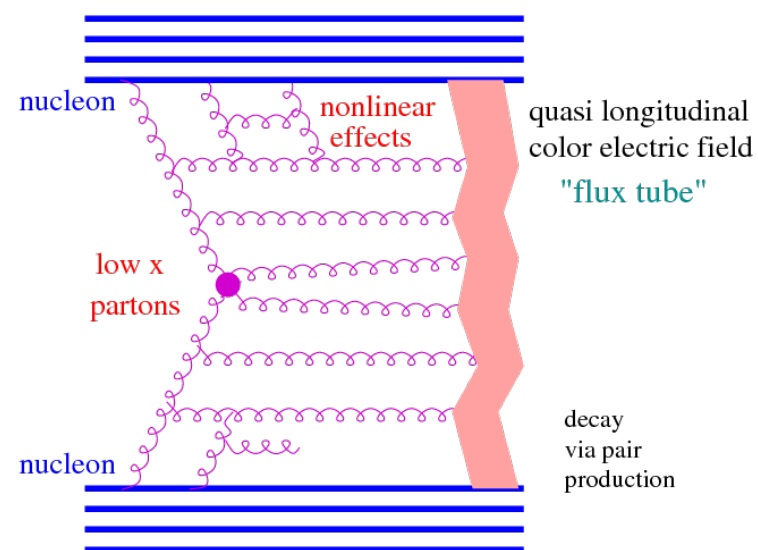


Monte Carlo Generators

EPOS LHC

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- Parton evolution → Parton ladders



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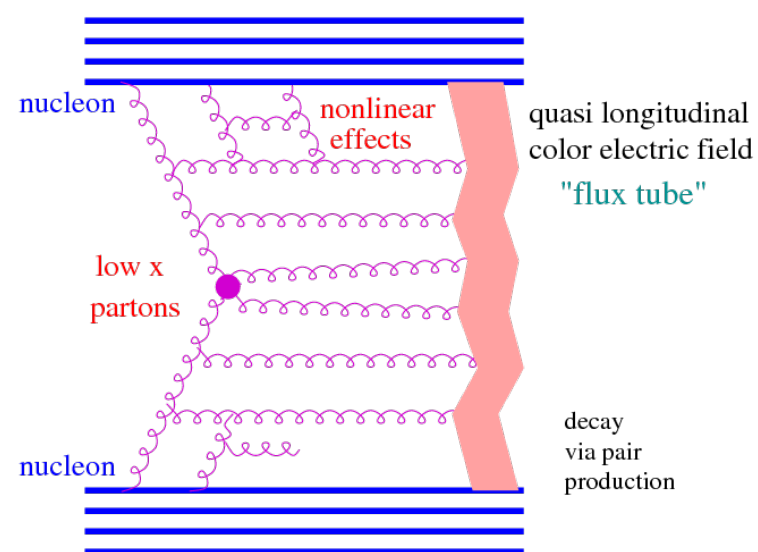


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String decays → production of quark-antiquark pairs

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Two scenarios in p-Pb collisions

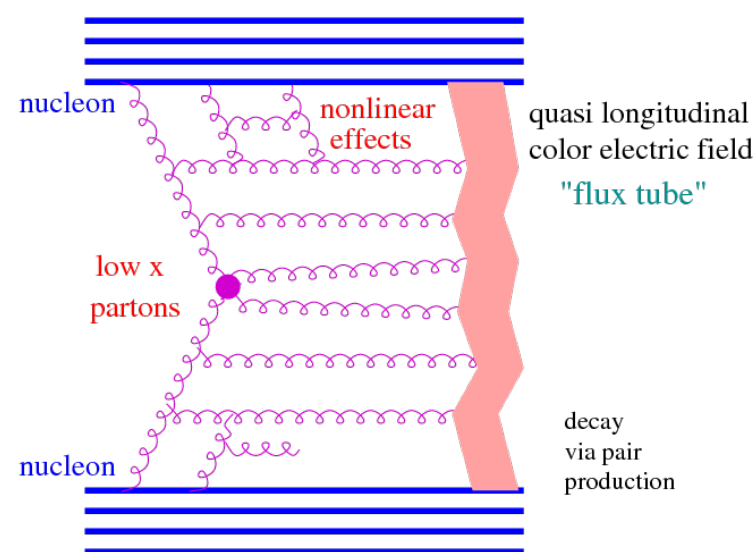


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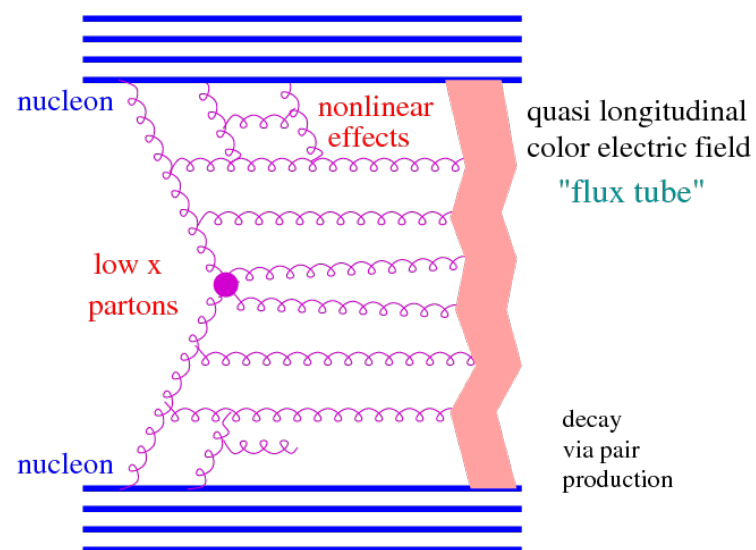


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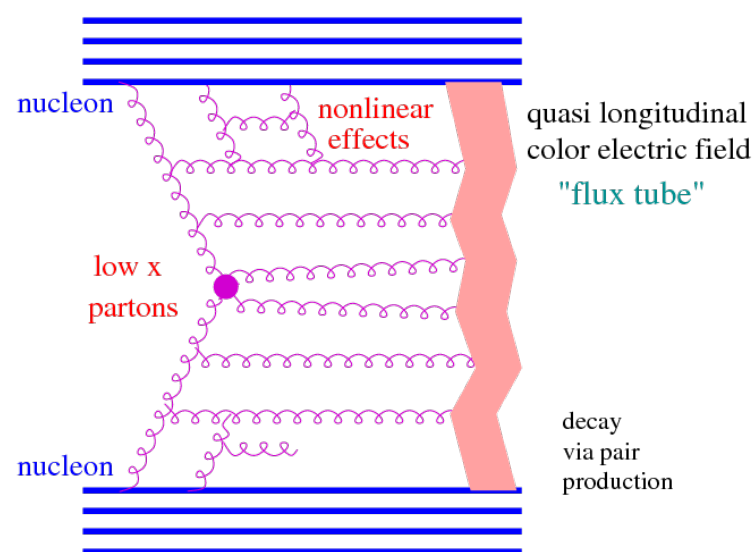


Monte Carlo Generators

EPOS LHC

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↳ String segments in dense areas - Core

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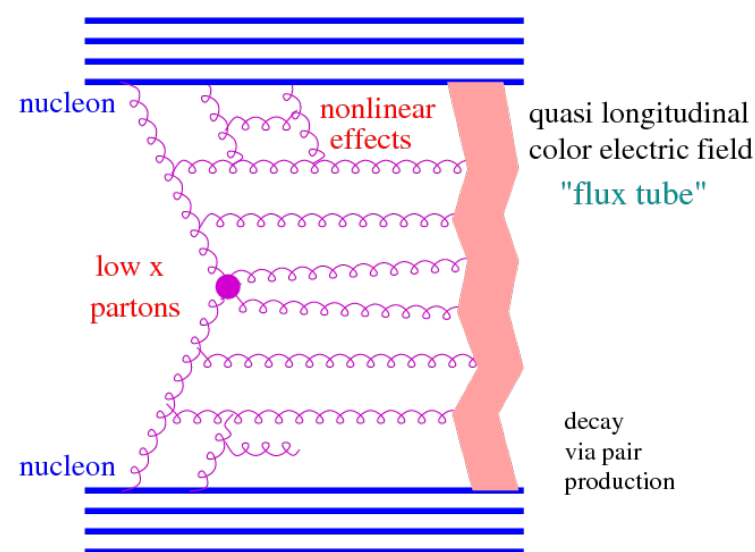


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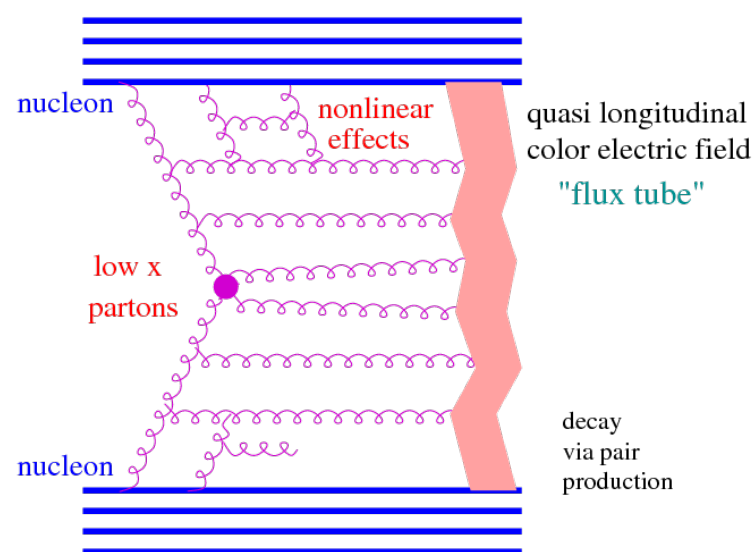


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Phys. Rev. D 51, 64 (1995)

- Based on the Dual Parton Model (DPM)



Two scenarios in p-Pb collisions

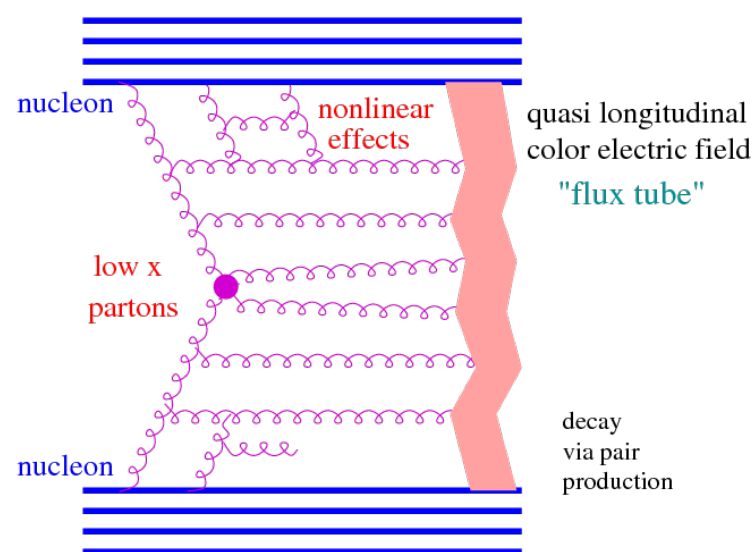


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Phys. Rev. D 51, 64 (1995)

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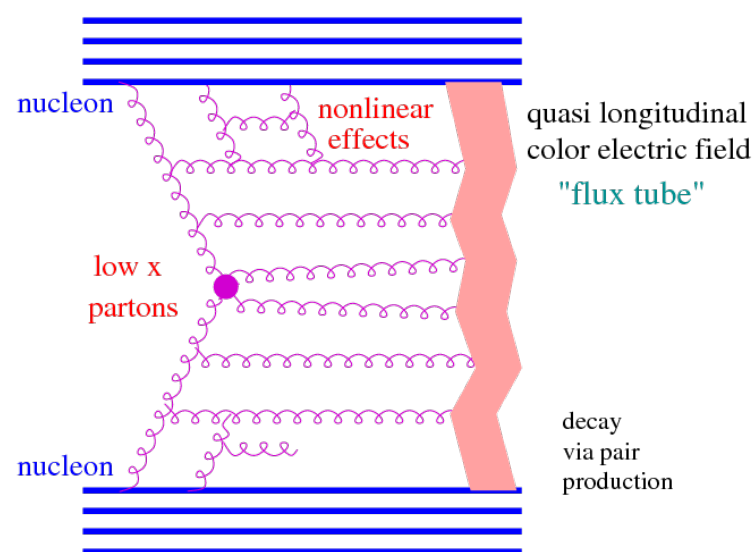


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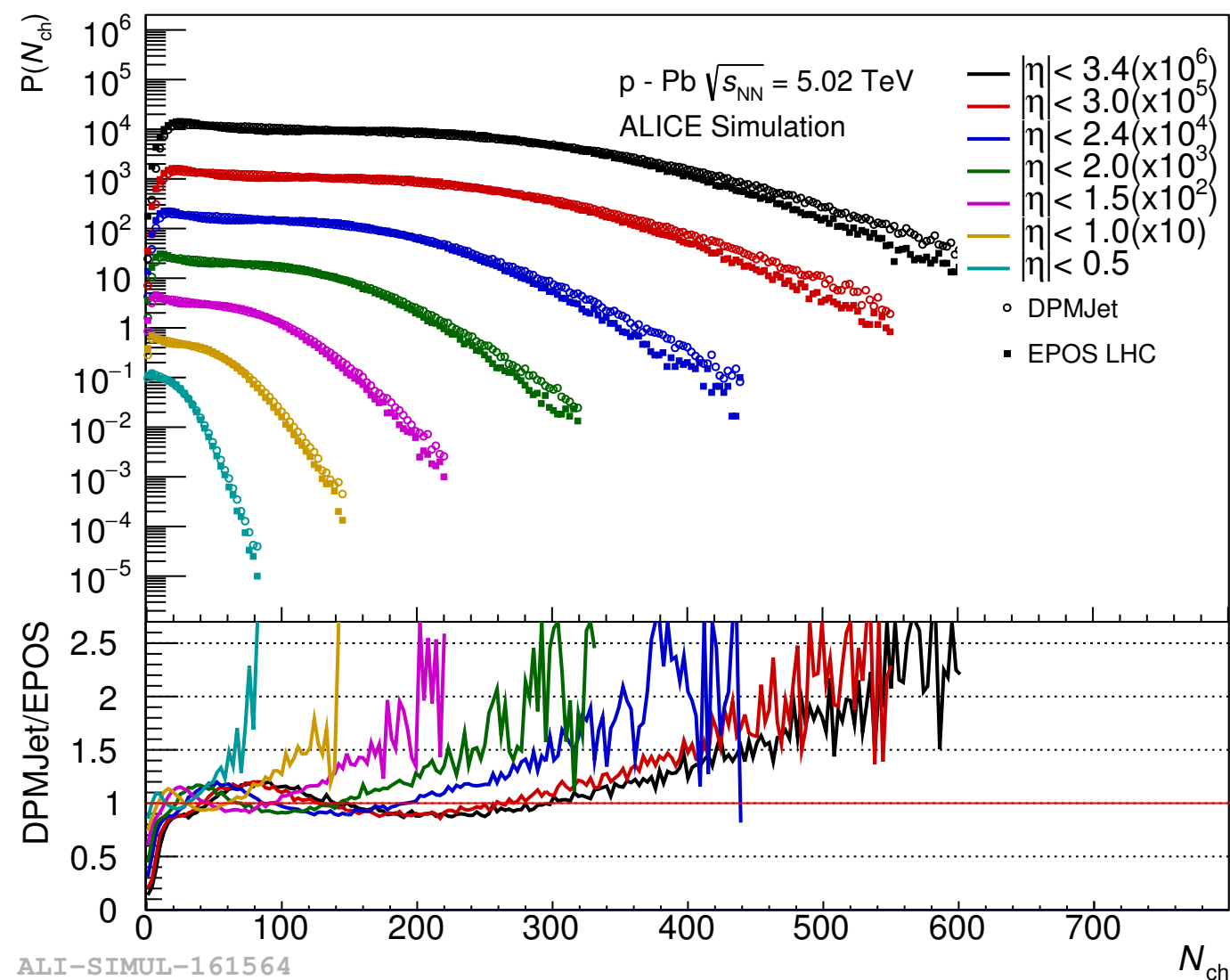
DPMJet

Phys. Rev. D 51, 64 (1995)

- Based on the Dual Parton Model (DPM)
- Individual collisions are simulated based on the PHOJET1.12
- JETSET used for the fragmentation part using string model.

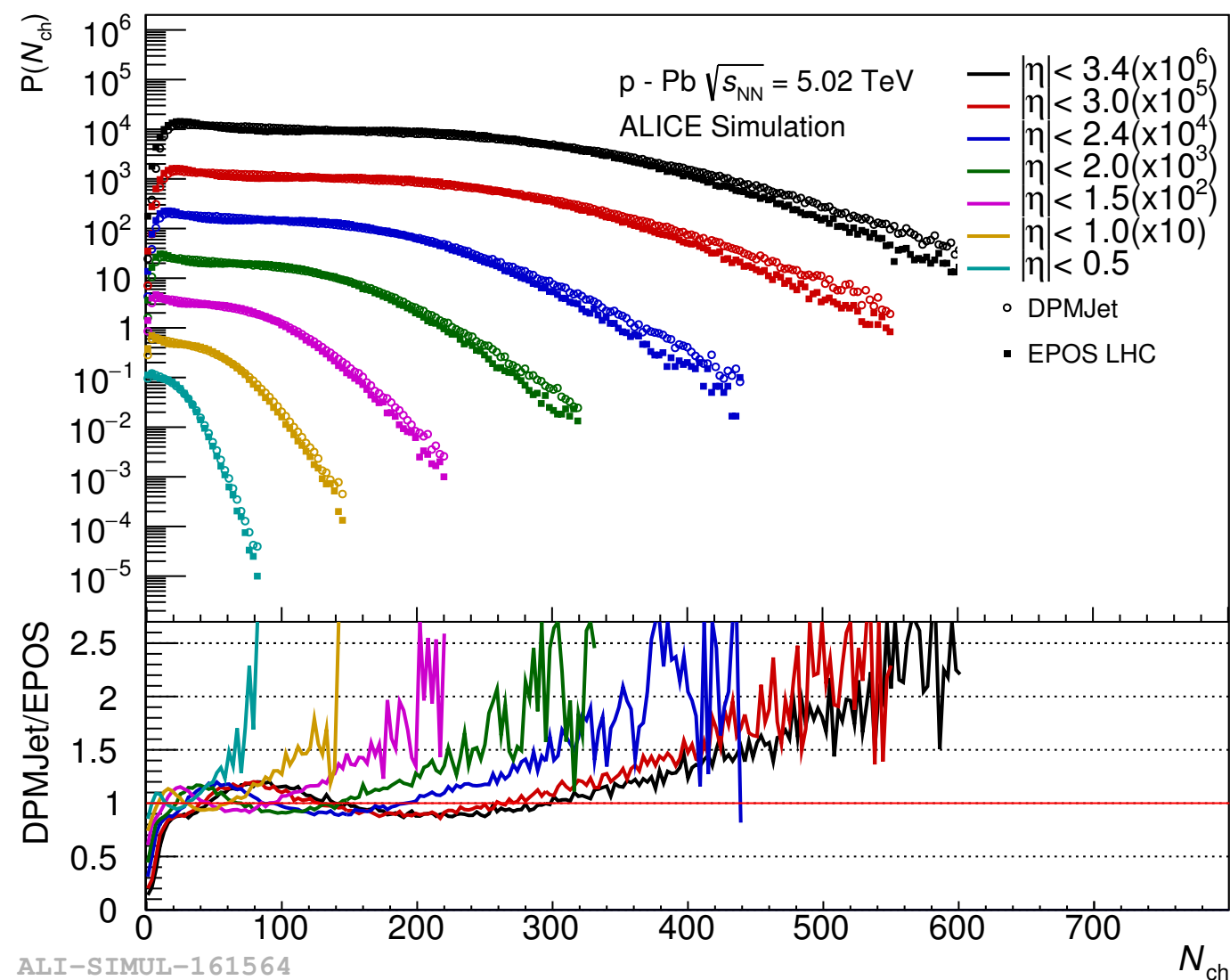


Minimum Bias



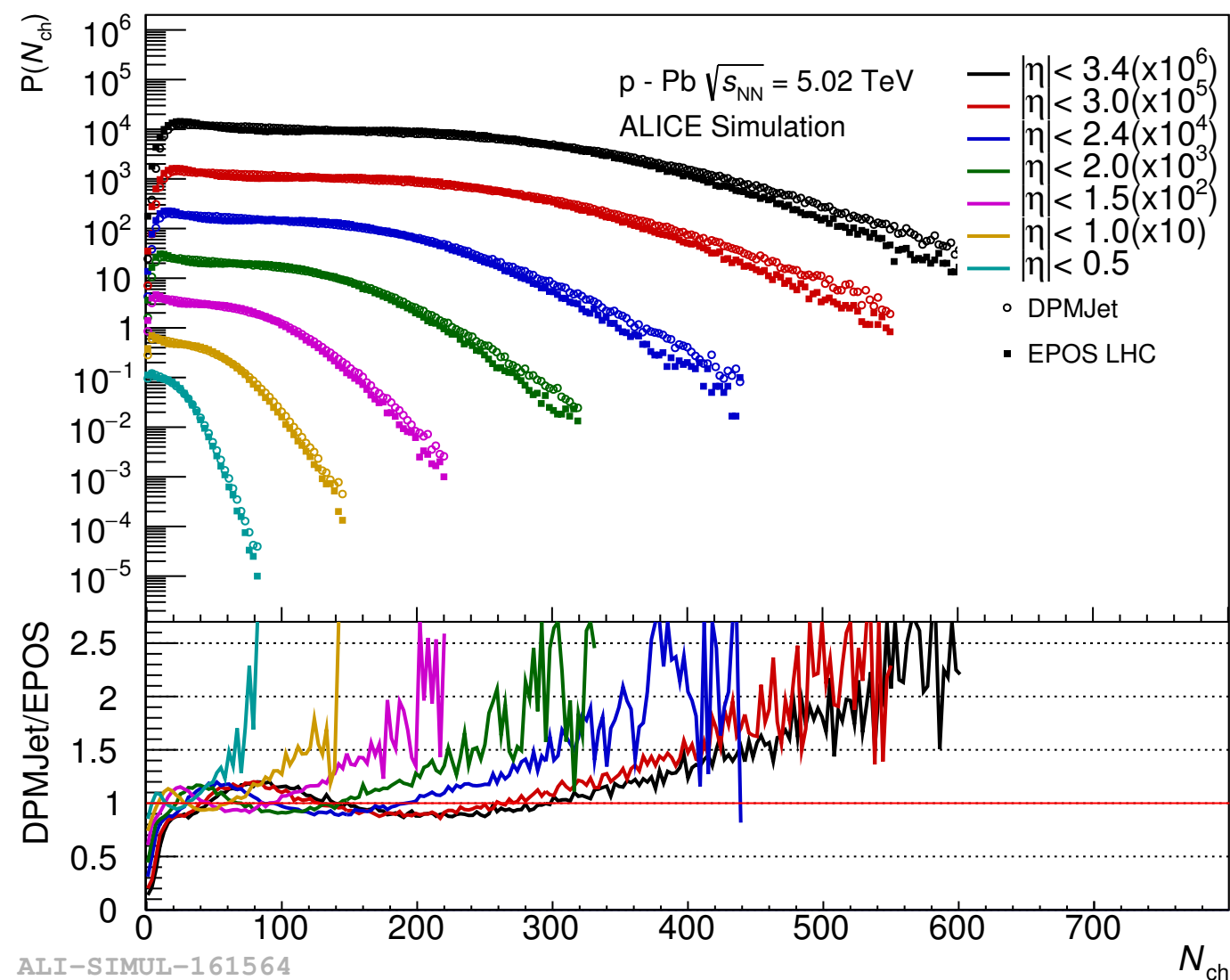
• DPMJet

Minimum Bias



- DPMJet
 - ↳ Broader tails

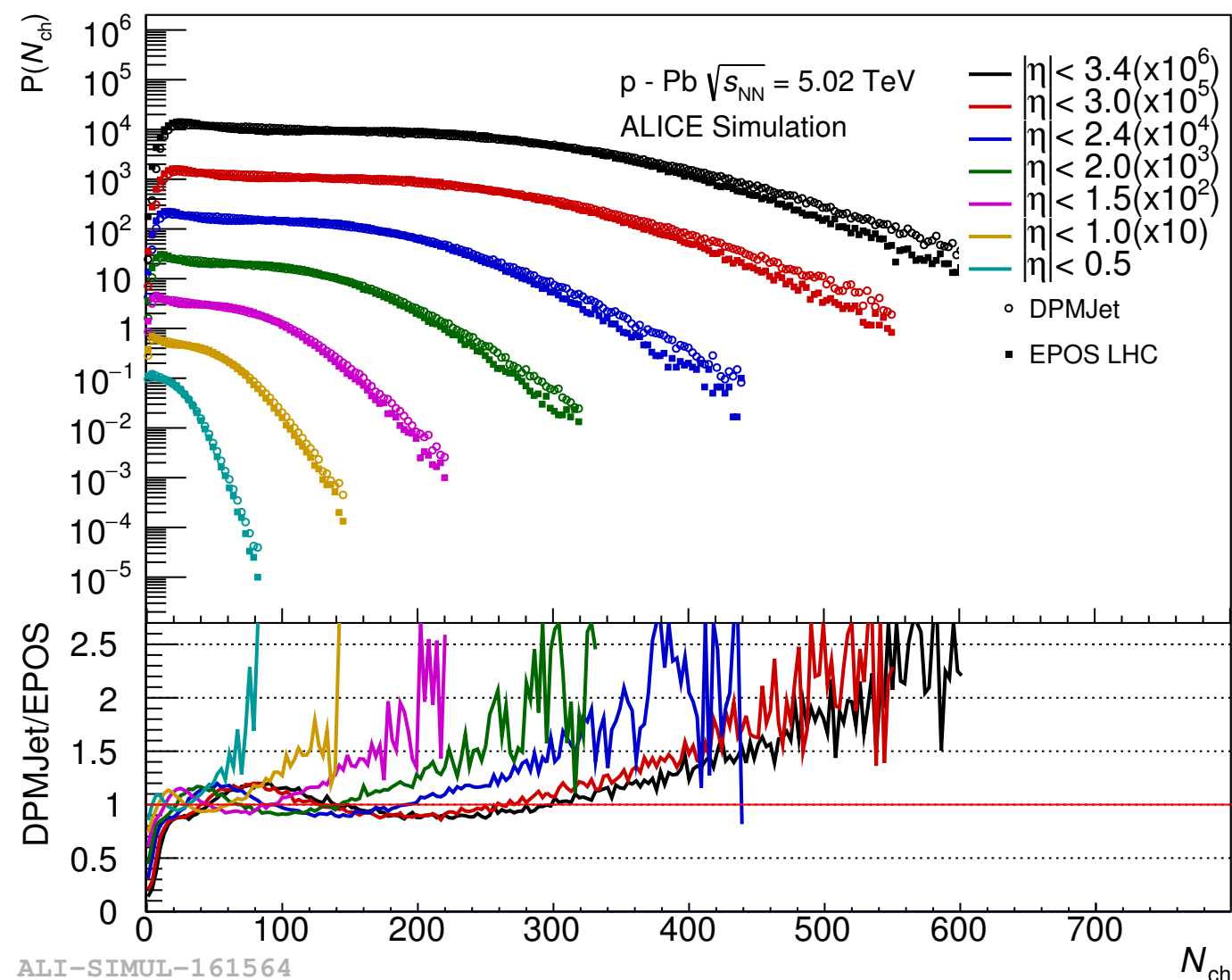
Minimum Bias



- DPMJet
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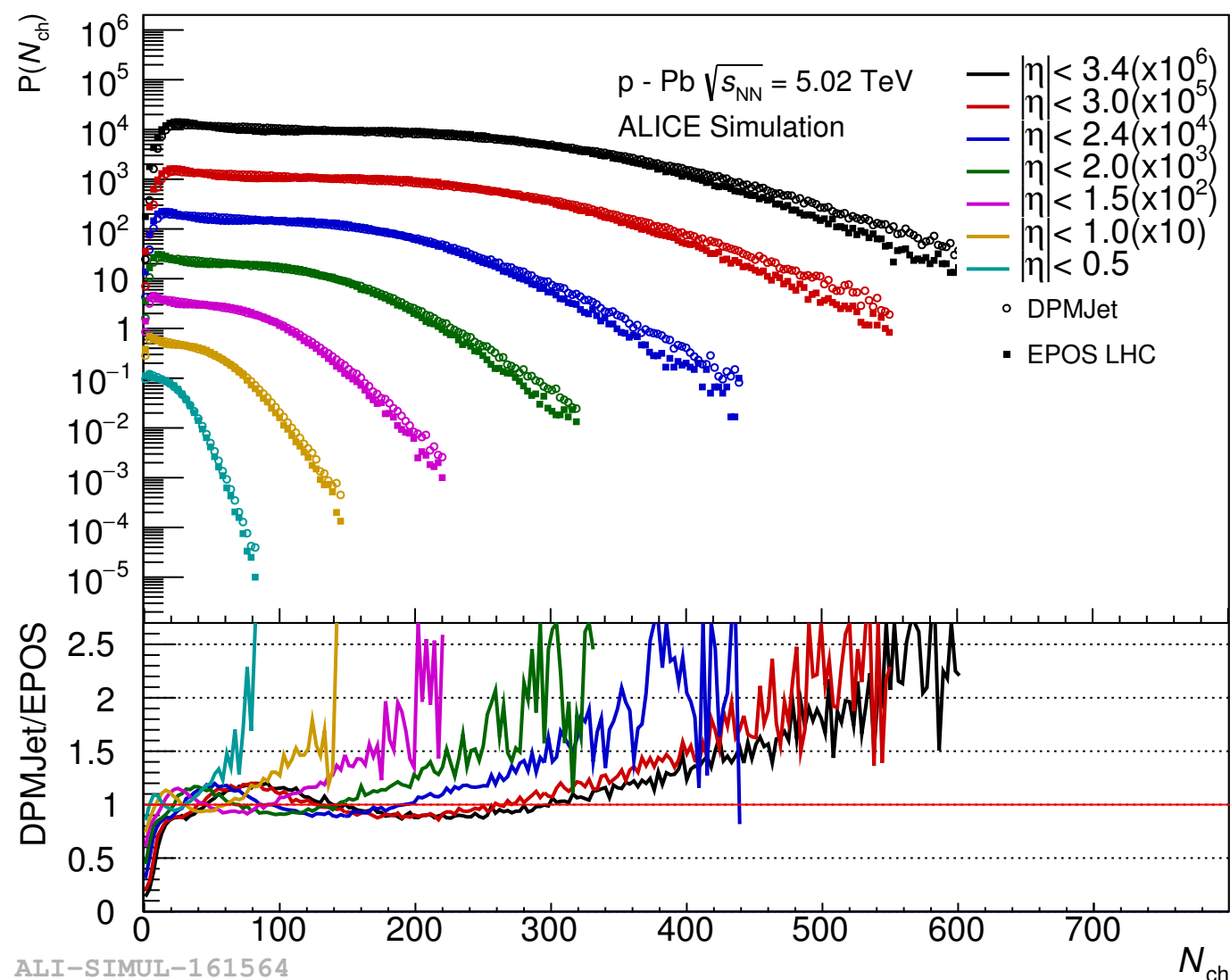
- EPOS LHC

Minimum Bias



- DPMJet
 - ↳ Broader tails
- EPOS LHC
 - ↳ Indicates two domains
 - ↳ Soft and semi-hard components (?)
 - ↳ Double NBD

Minimum Bias



- DPMJet

↳ Broader tails

- EPOS LHC

↳ Indicates two domains

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↳ Double NBD

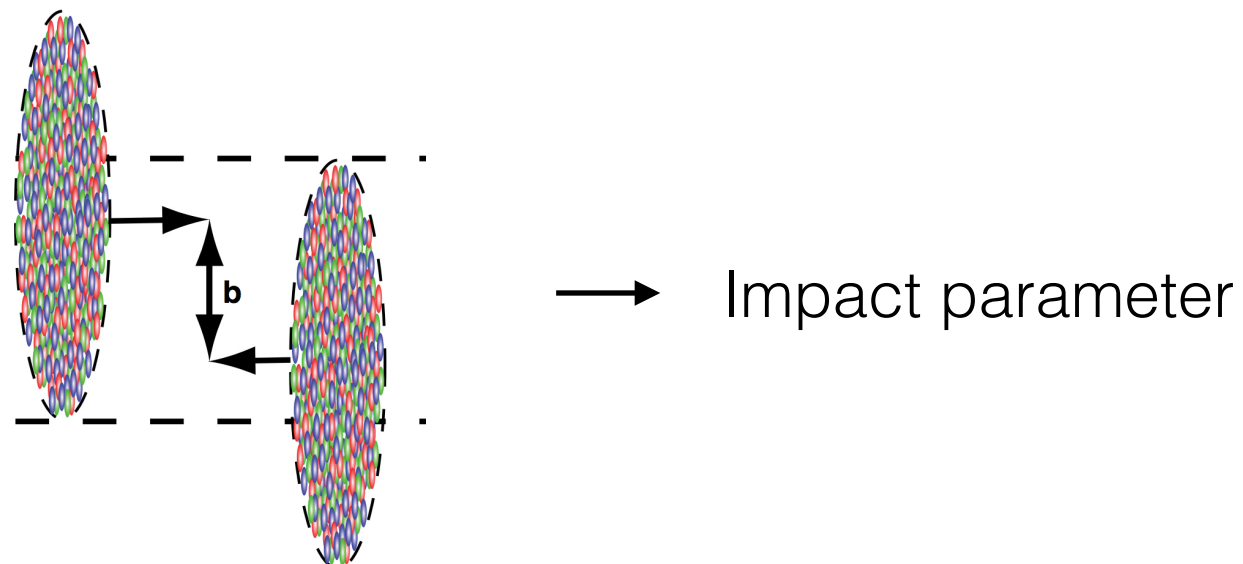
Differences in the tail

EPOS → More soft events

Centrality Slicing



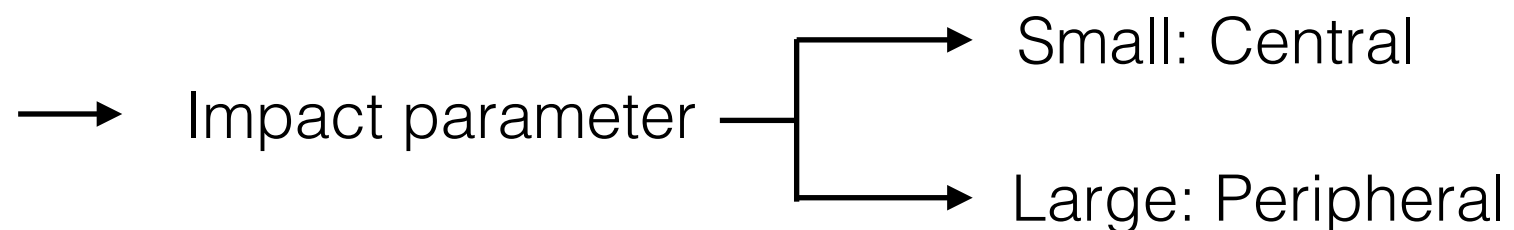
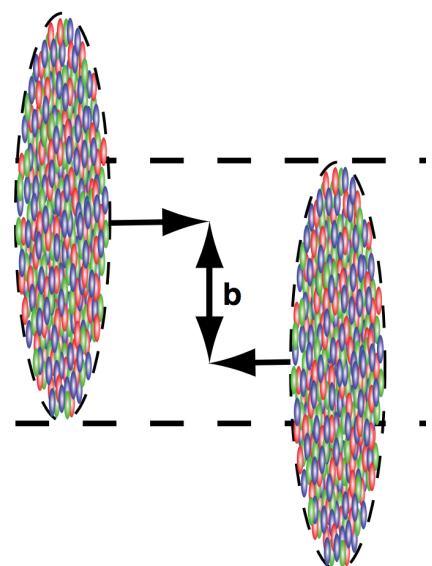
A-A Collisions



Centrality Slicing



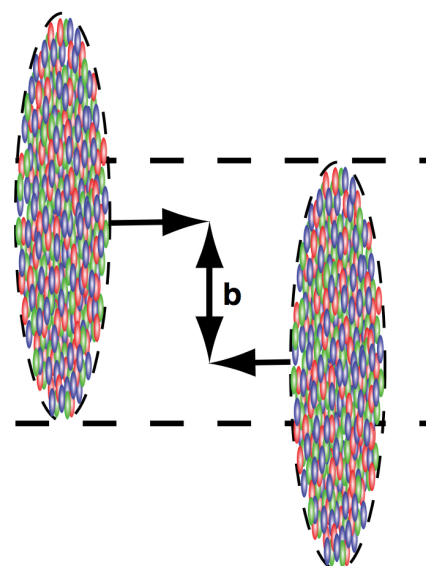
A-A Collisions



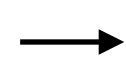
Centrality Slicing



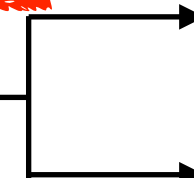
A-A Collisions



Not directly measurable



Impact parameter



Small: Central

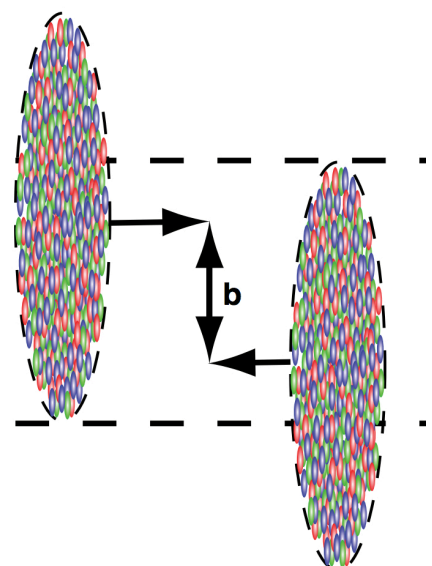
Large: Peripheral



Centrality Slicing



A-A Collisions



Not directly measurable

→ Impact parameter

Small: Central

Large: Peripheral

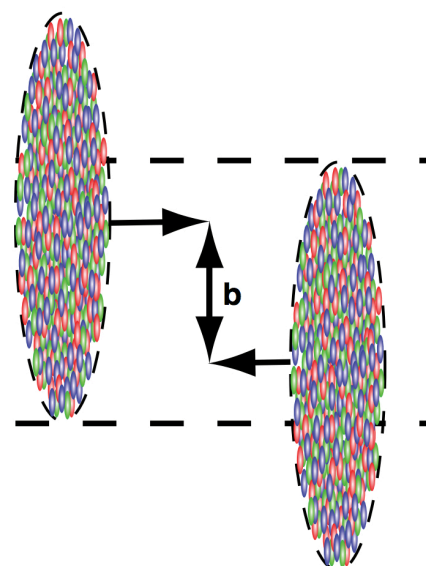
↳ theoretical models → geometrical quantities



Centrality Slicing



A-A Collisions



Not directly measurable

→ Impact parameter

Small: Central

Large: Peripheral

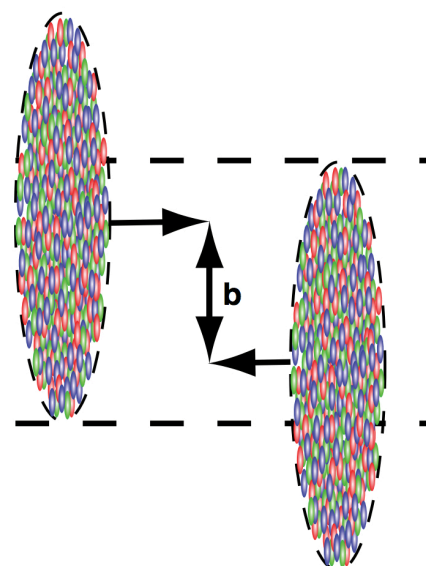
↳ theoretical models → geometrical quantities → Glauber Model

- N_{coll}
- N_{part}

Centrality Slicing



A-A Collisions

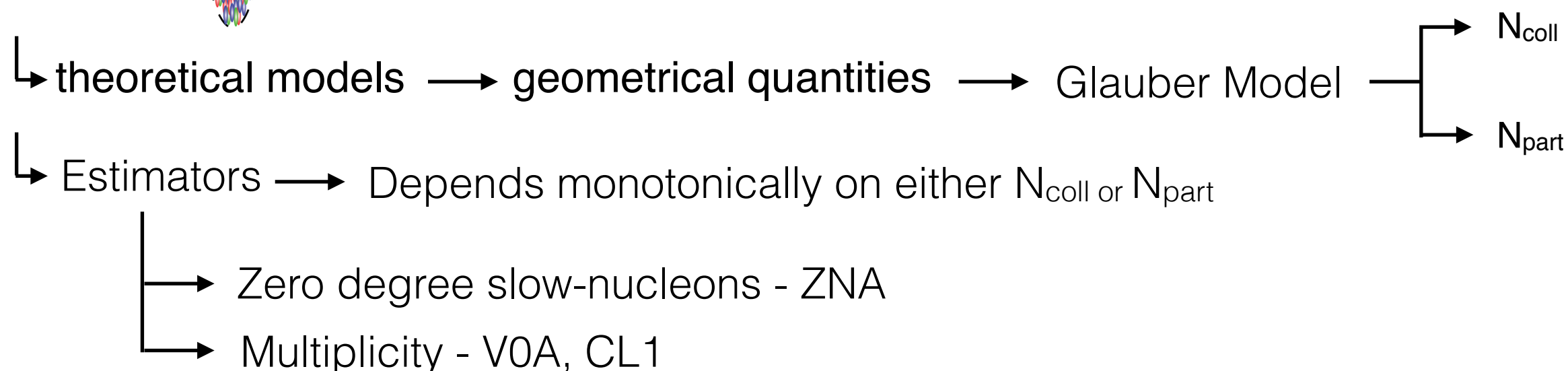


Not directly measurable

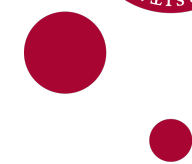
→ Impact parameter

Small: Central

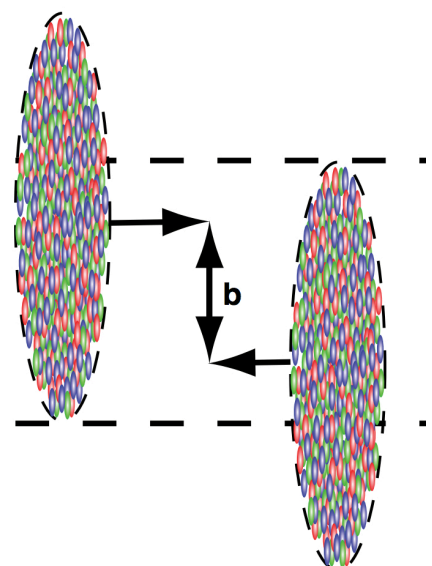
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Centrality Slicing



A-A Collisions



Not directly measurable

→ Impact parameter

Small: Central

Large: Peripheral

↳ theoretical models → geometrical quantities → Glauber Model

↳ Estimators → Depends monotonically on either N_{coll} or N_{part}

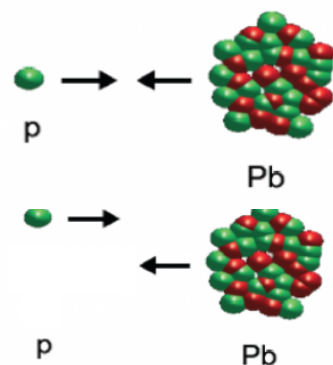
N_{coll}

N_{part}

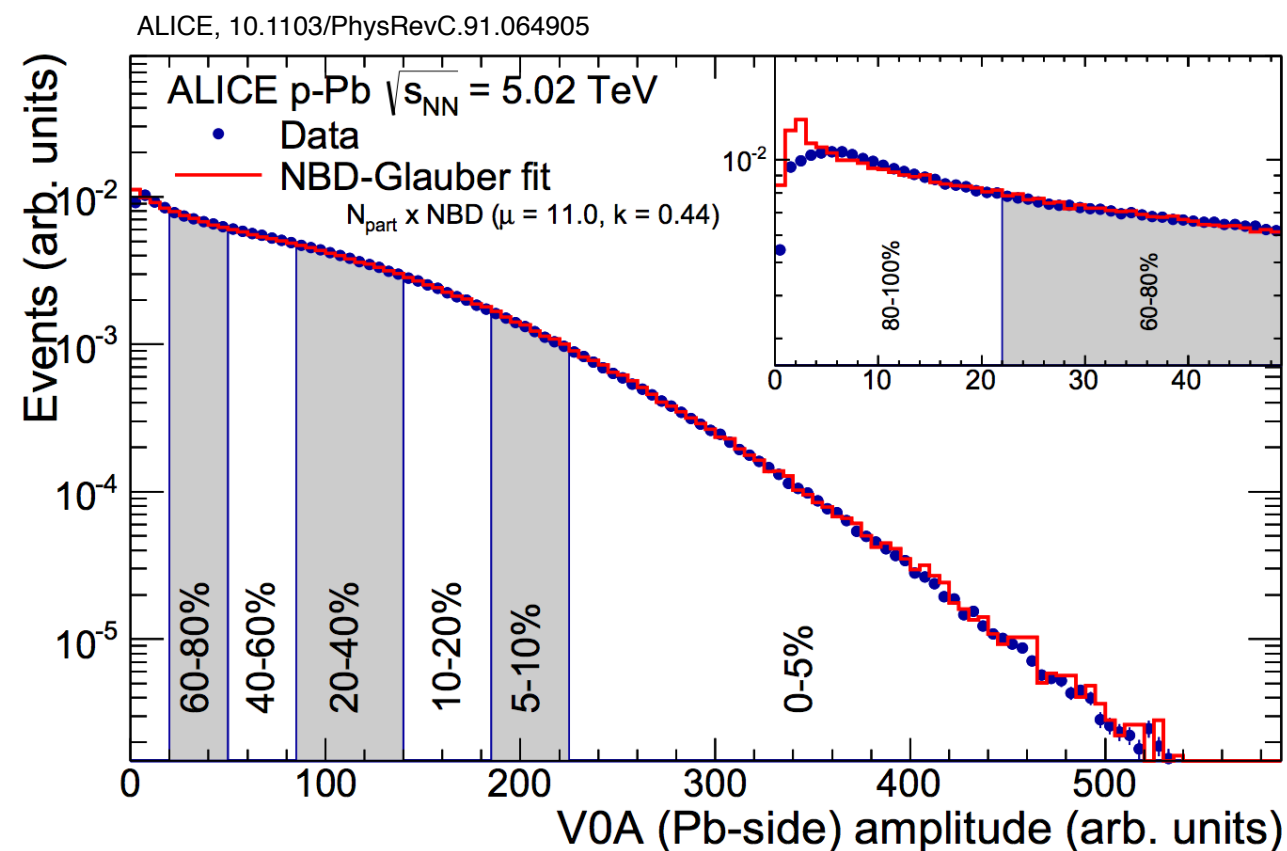
→ Zero degree slow-nucleons - ZNA

→ Multiplicity - V0A, CL1

p-A Collisions



Centrality Slicing

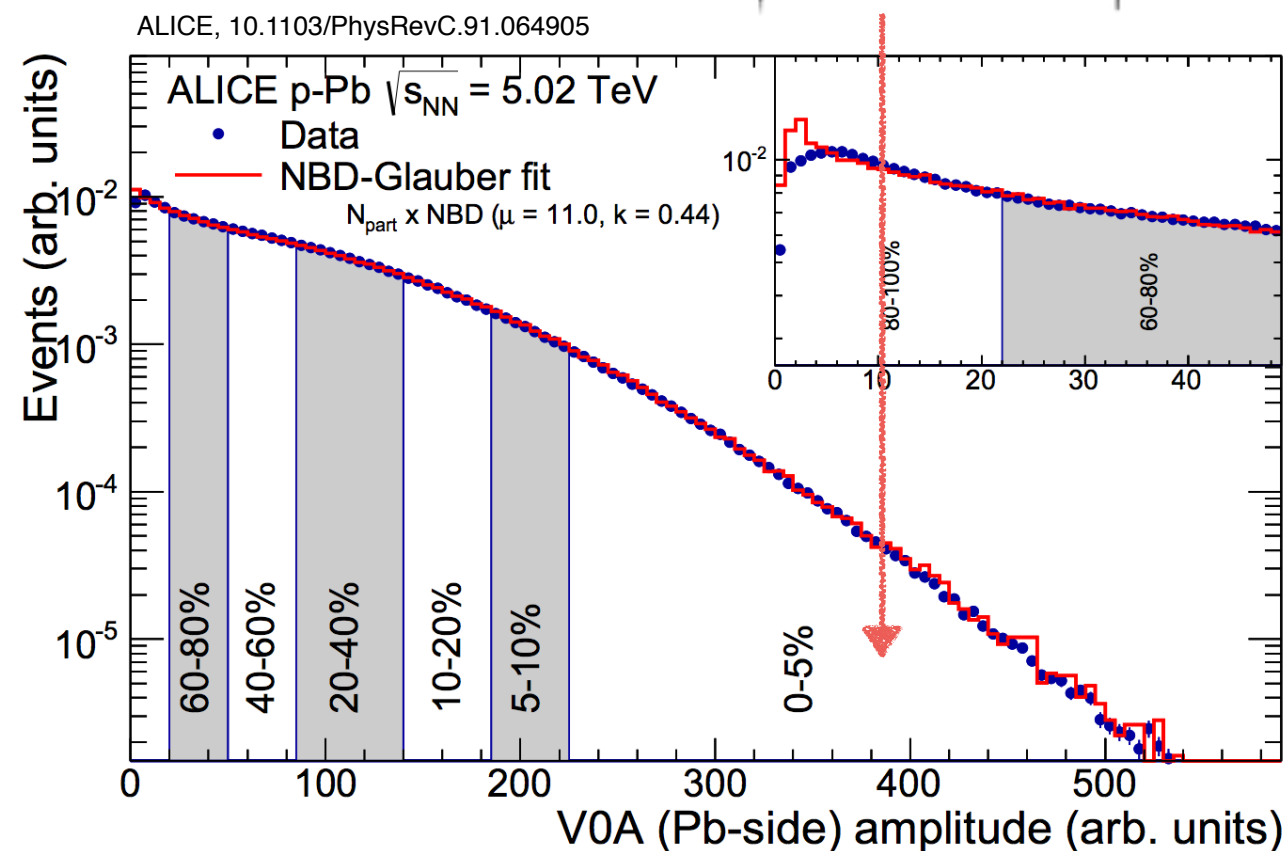


Observables related to the collision geometry N_{ch} and E_{ZDC}

Centrality classes $\rightarrow$ percentiles of the multiplicity distributions

Centrality Slicing

Central

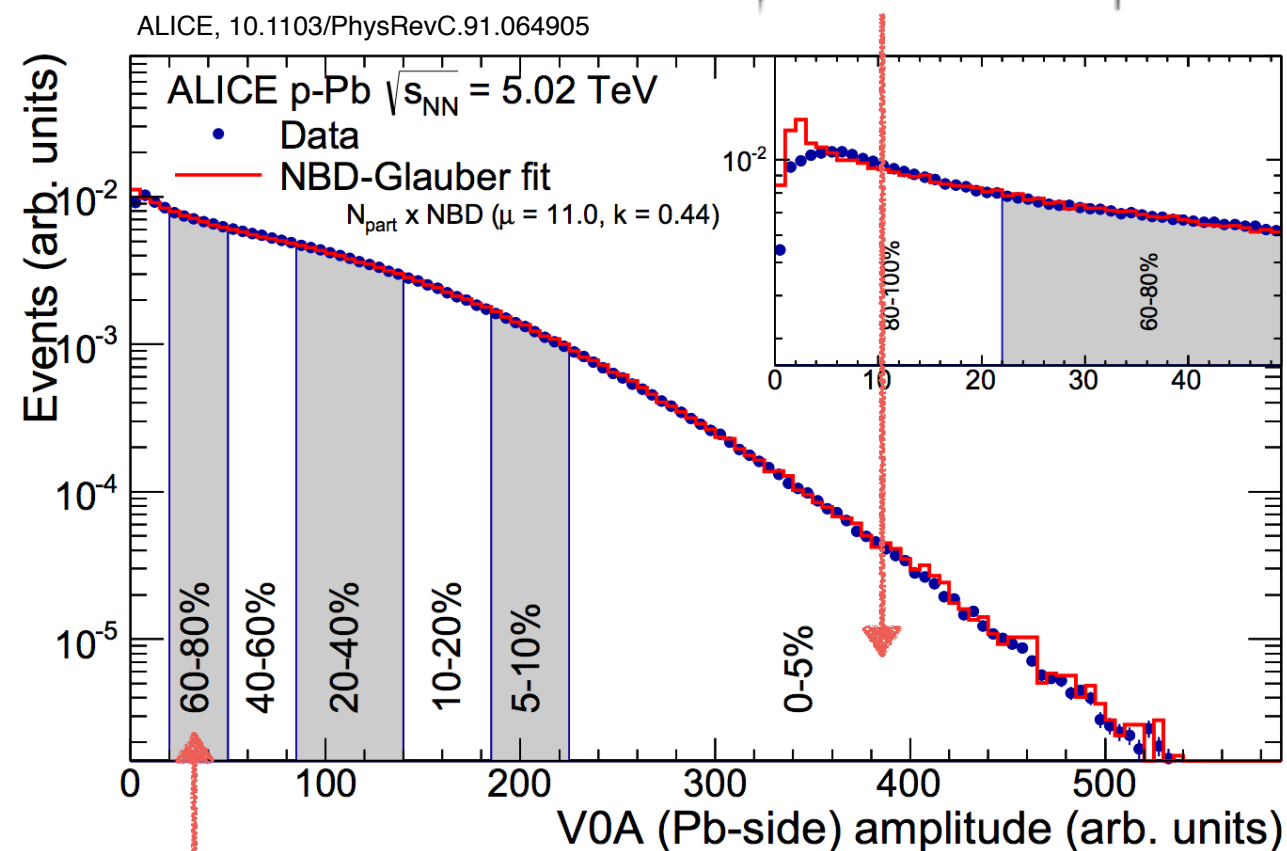


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Centrality Slicing

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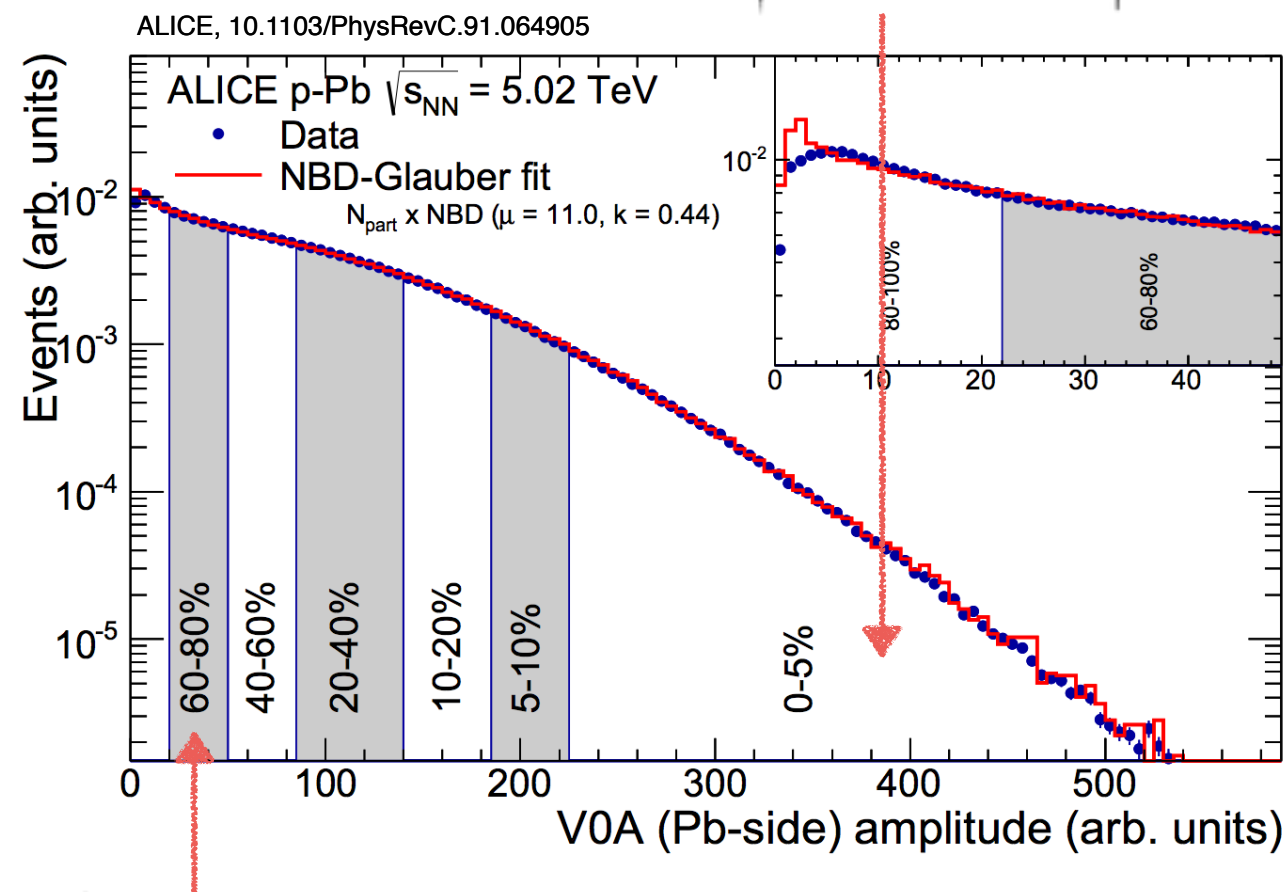
Centrality classes $\rightarrow$ percentiles of the multiplicity distributions

Peripheral



Centrality Slicing

Central

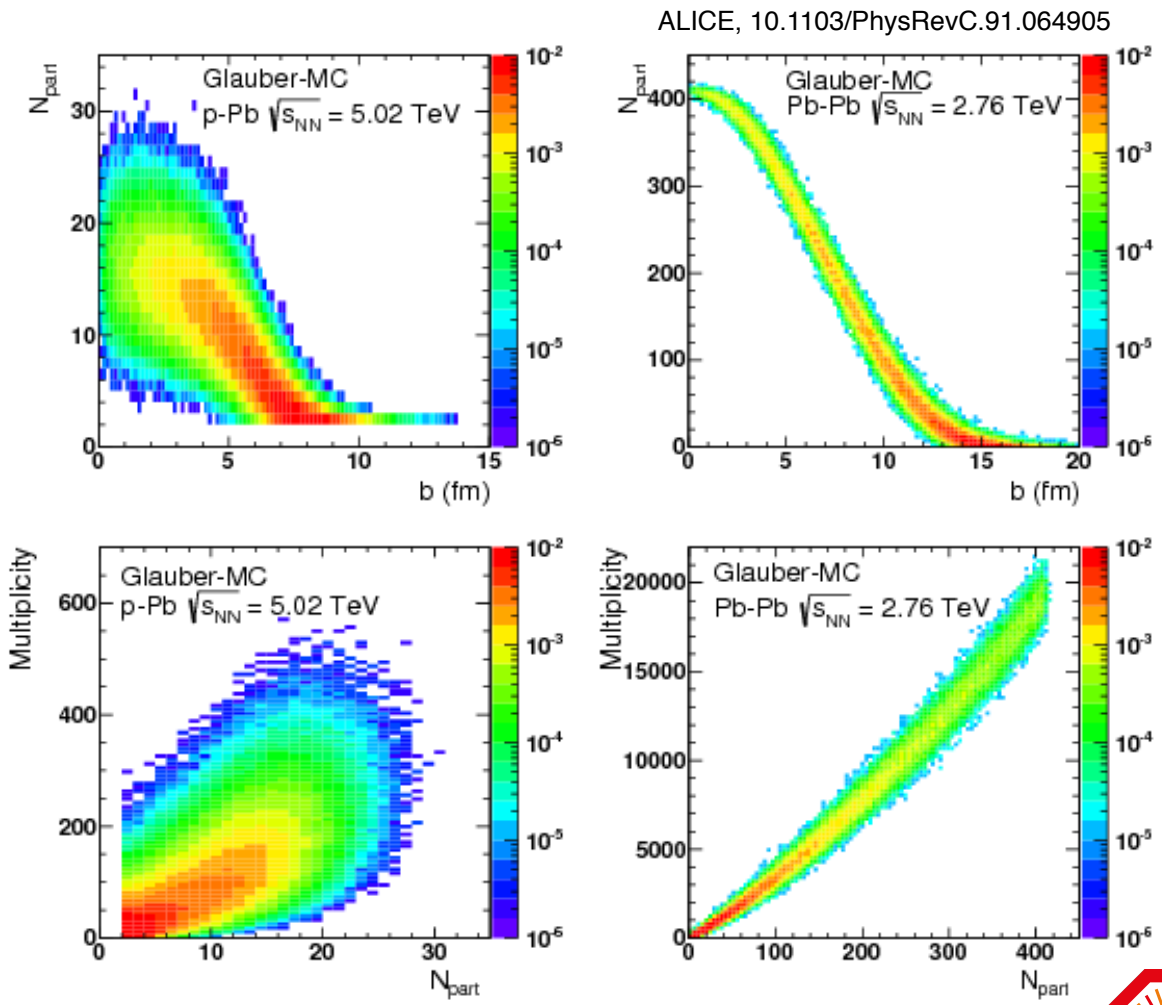


Observables related to the collision geometry N_{ch} and E_{ZDC}

Centrality classes $\rightarrow$ percentiles of the multiplicity distributions

Peripheral

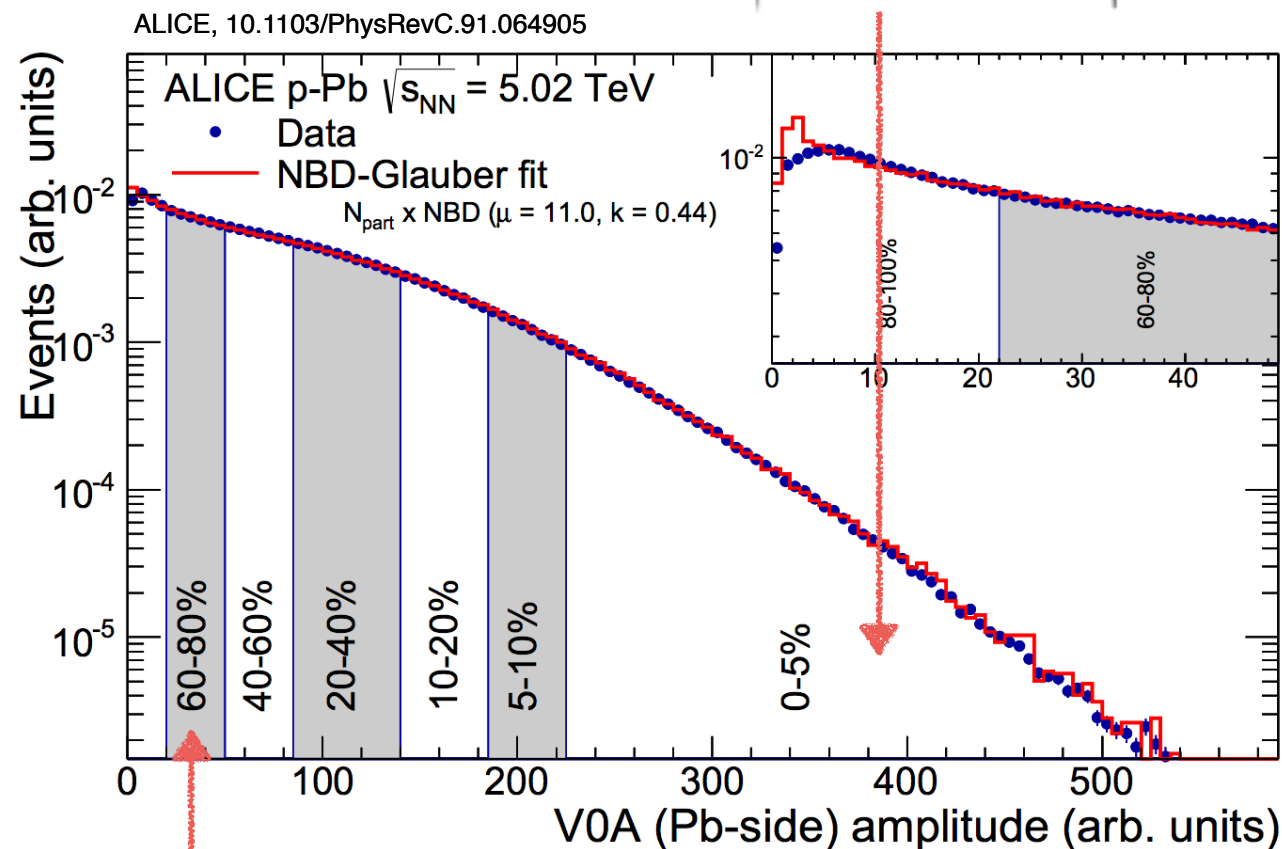
- Multiplicity fluctuations $\rightarrow$ High



p-Pb collisions - MC studies

Centrality Slicing

Central

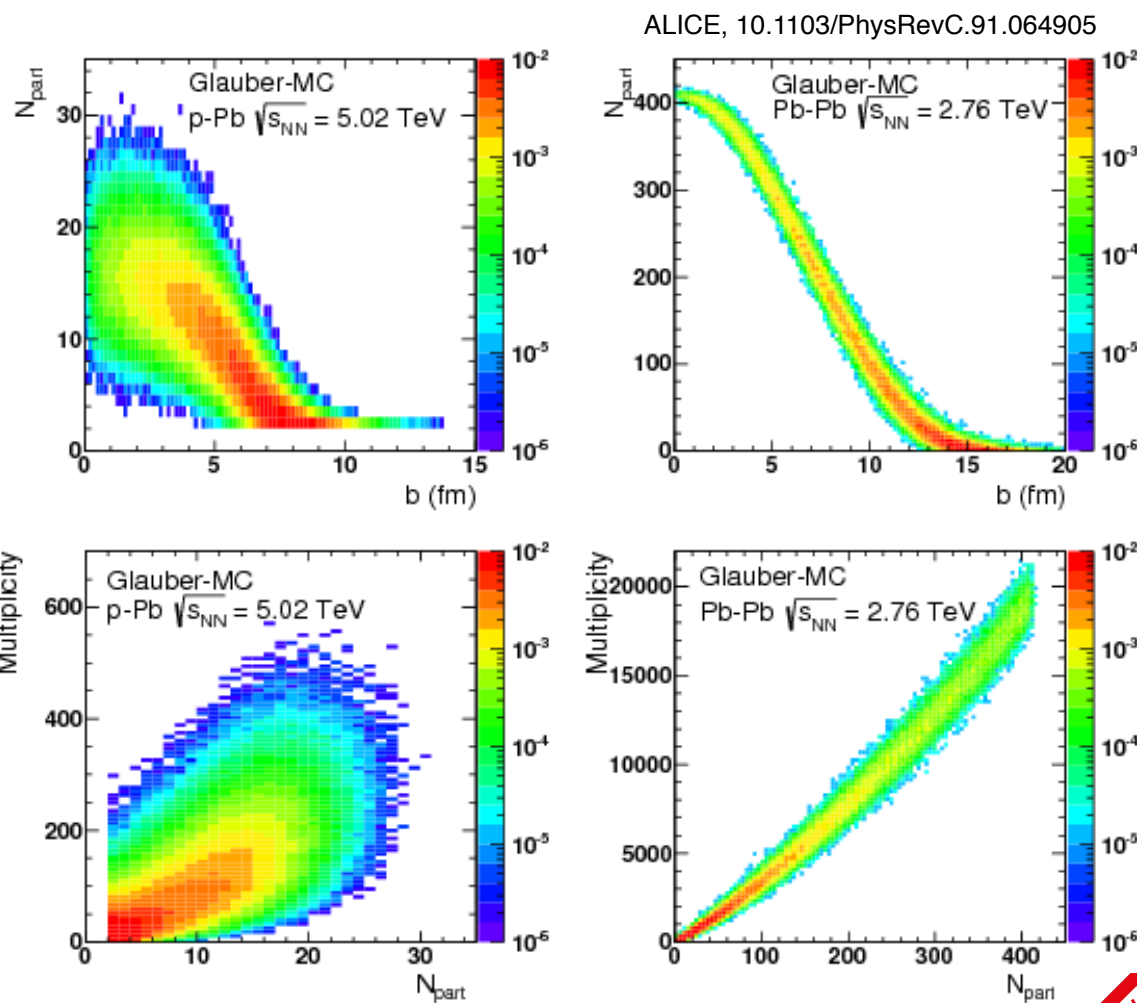


Observables related to the collision geometry N_{ch} and E_{ZDC}

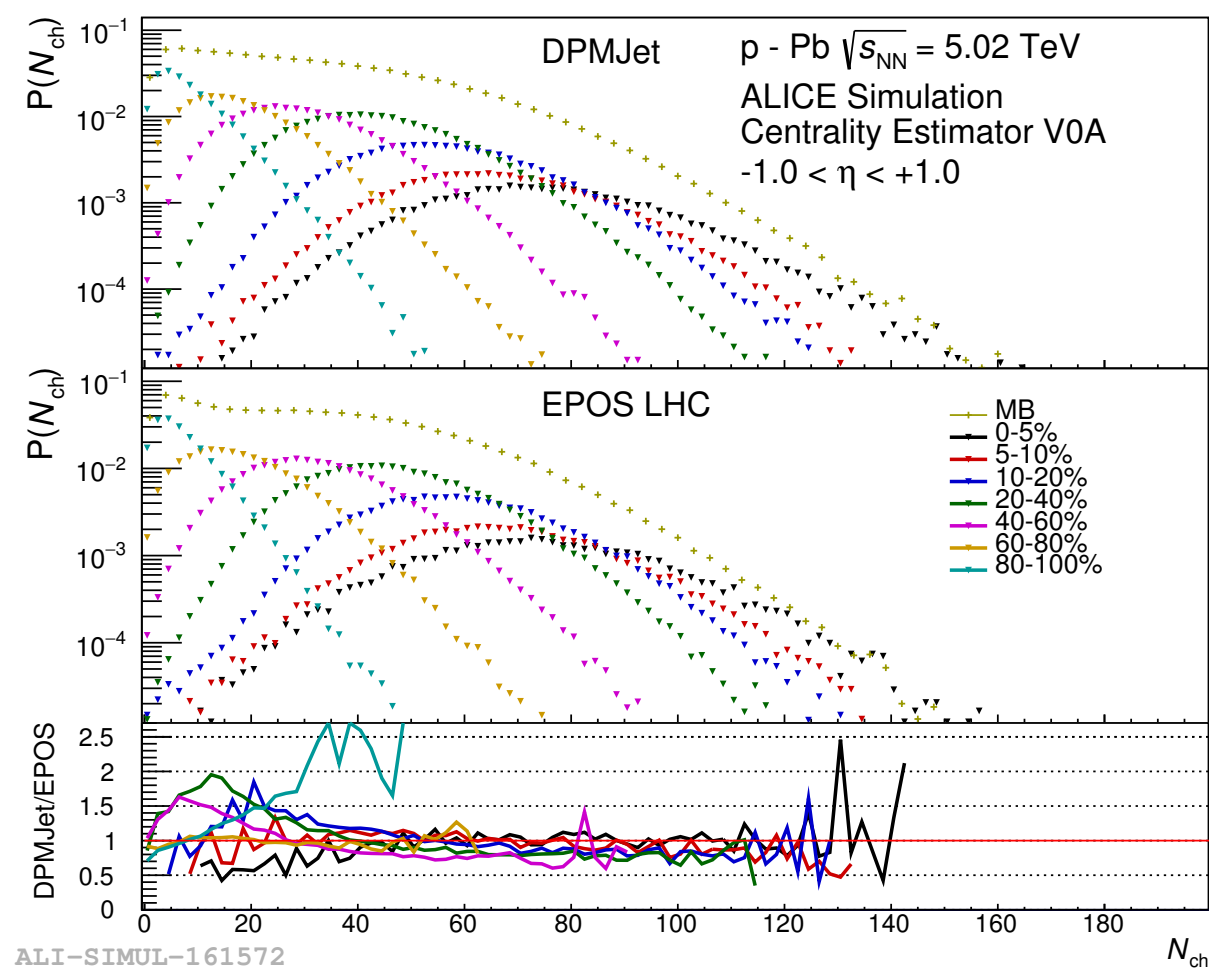
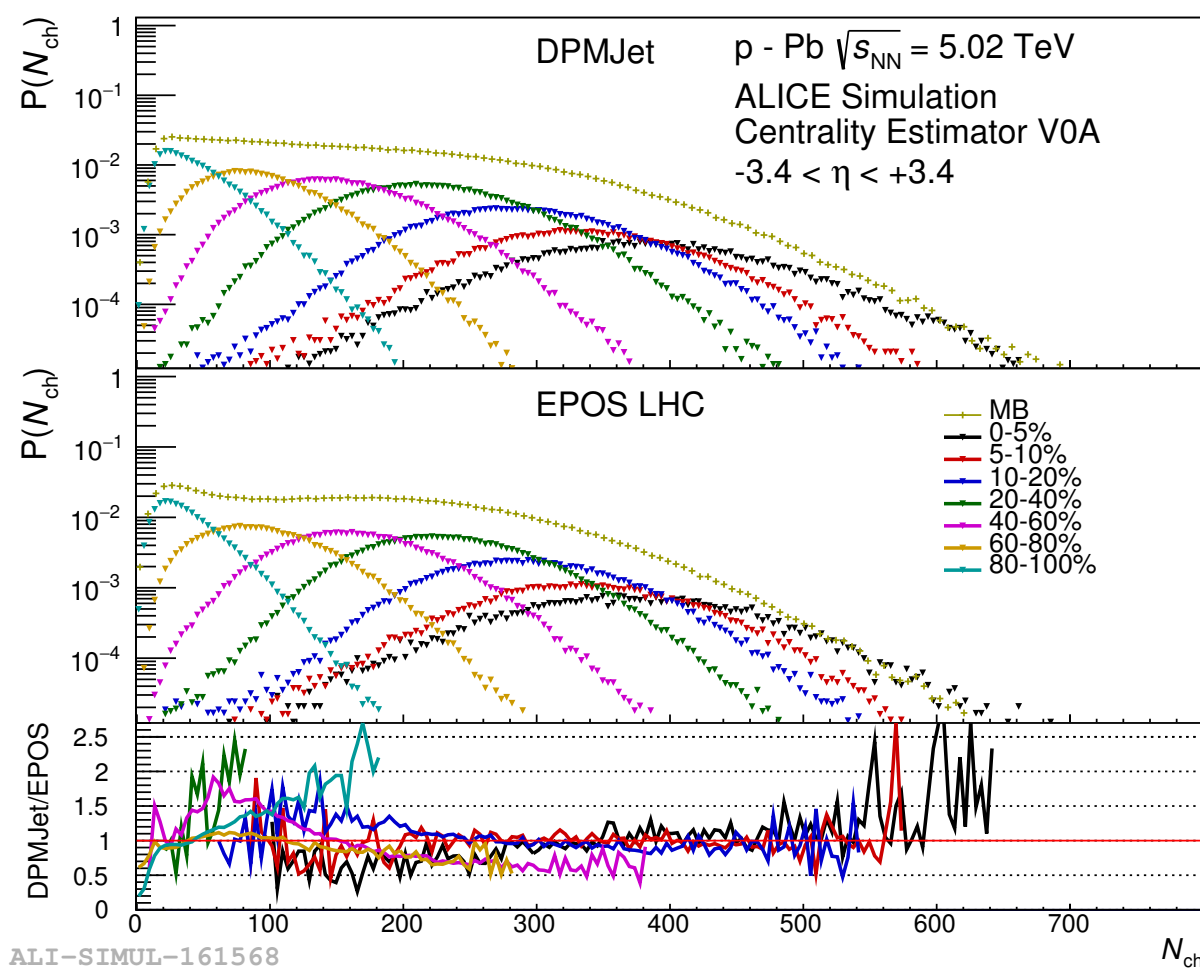
Centrality classes $\rightarrow$ percentiles of the multiplicity distributions

Peripheral

- Multiplicity fluctuations $\rightarrow$ High
- Centrality selection based on multiplicity
 $\rightarrow$ may select a biased sample

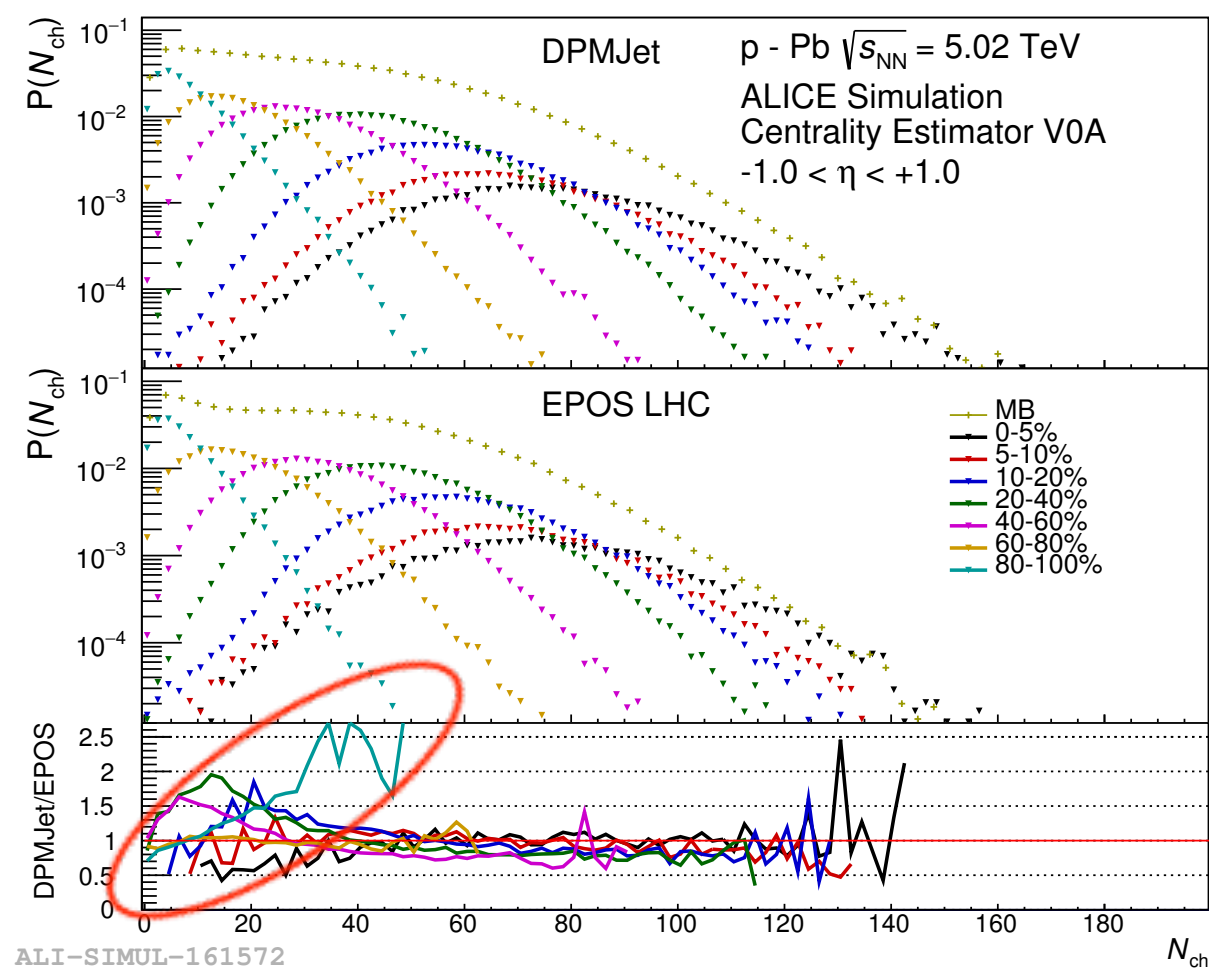
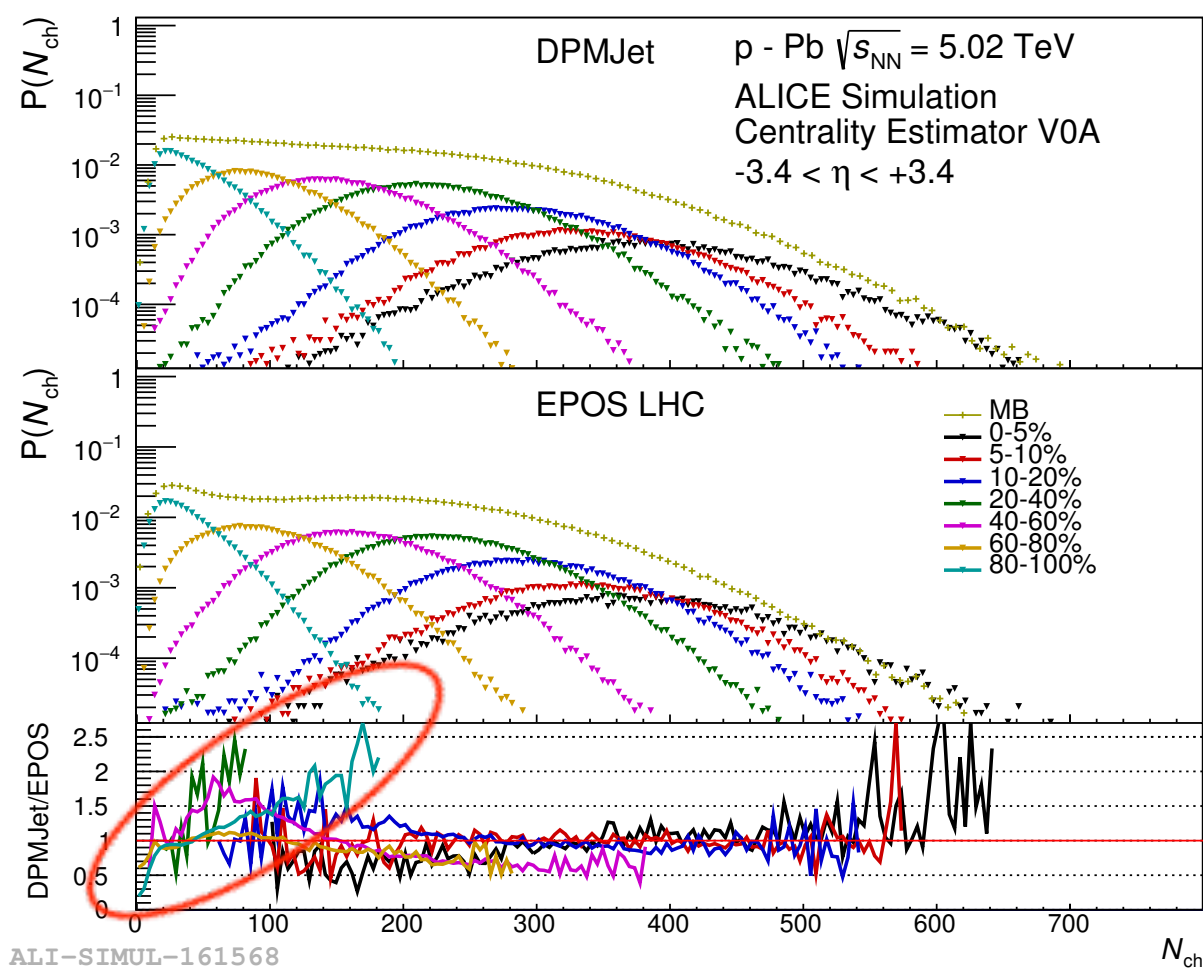


Centrality Slicing - 2 pseudorapidity ranges



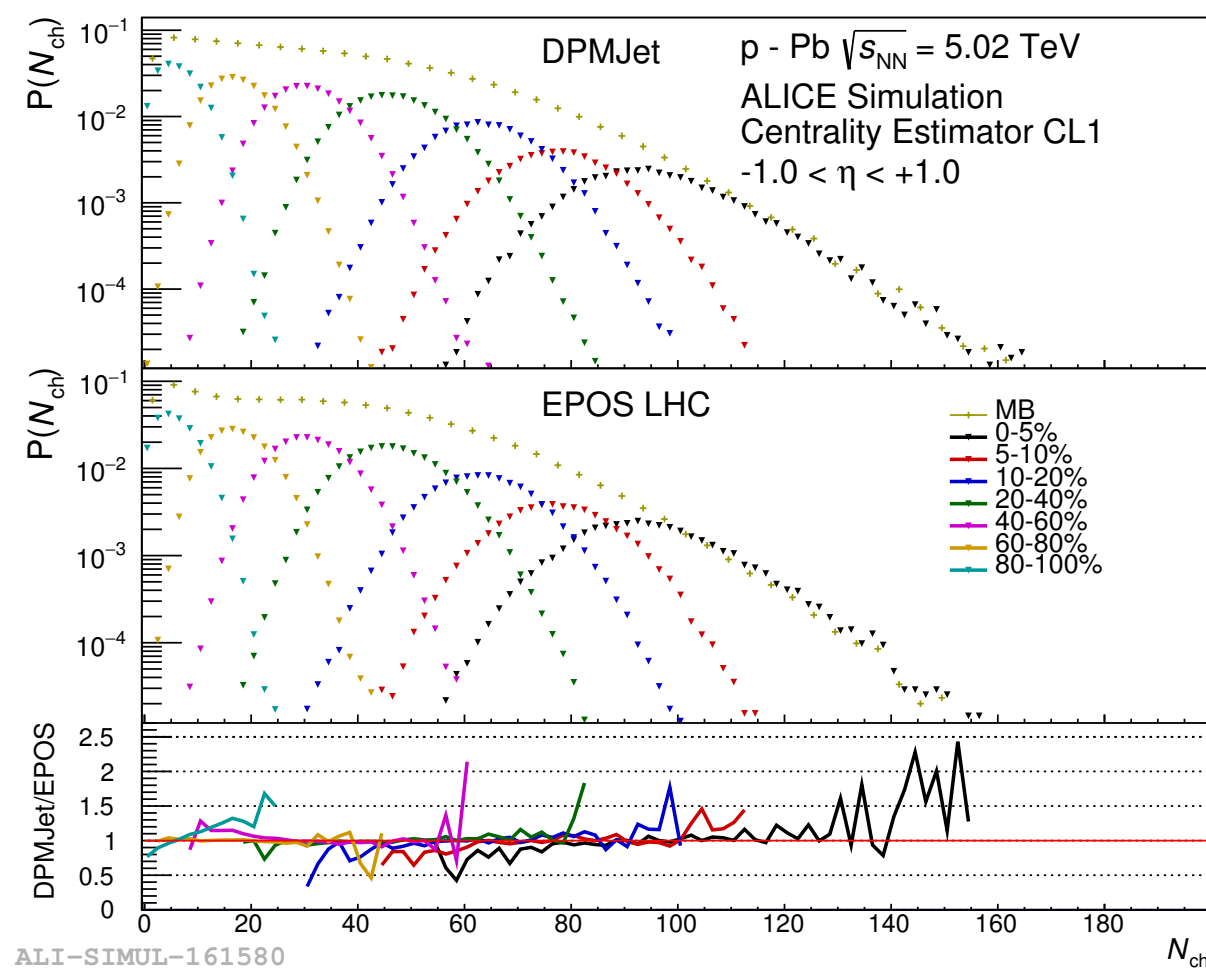
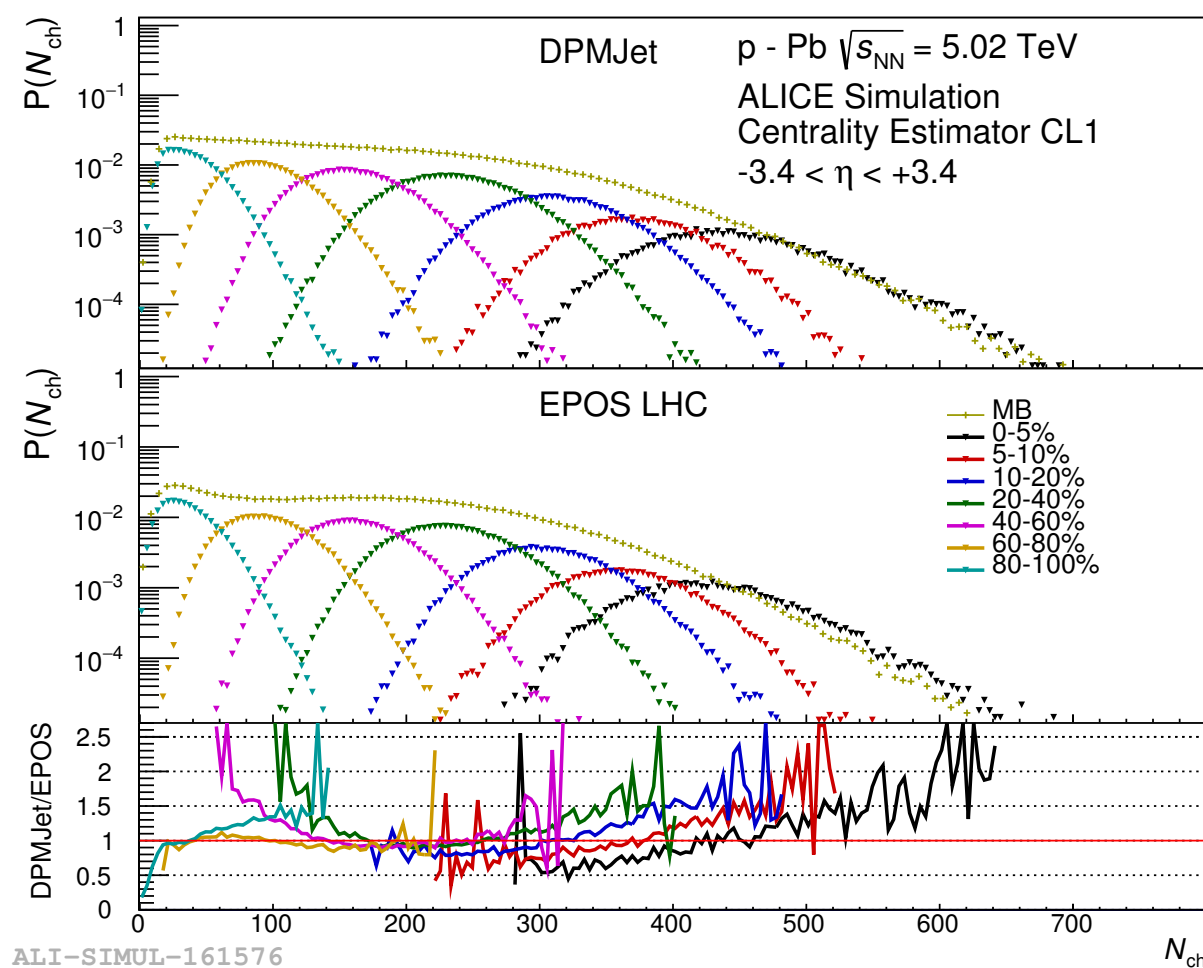
- DPMJet $\rightarrow$ Broader tails $\rightarrow$ Peripheral
- EPOS $\rightarrow$ Higher low bins $\rightarrow$ Peripheral

Centrality Slicing - 2 pseudorapidity ranges



- DPMJet $\rightarrow$ Broader tails $\rightarrow$ Peripheral
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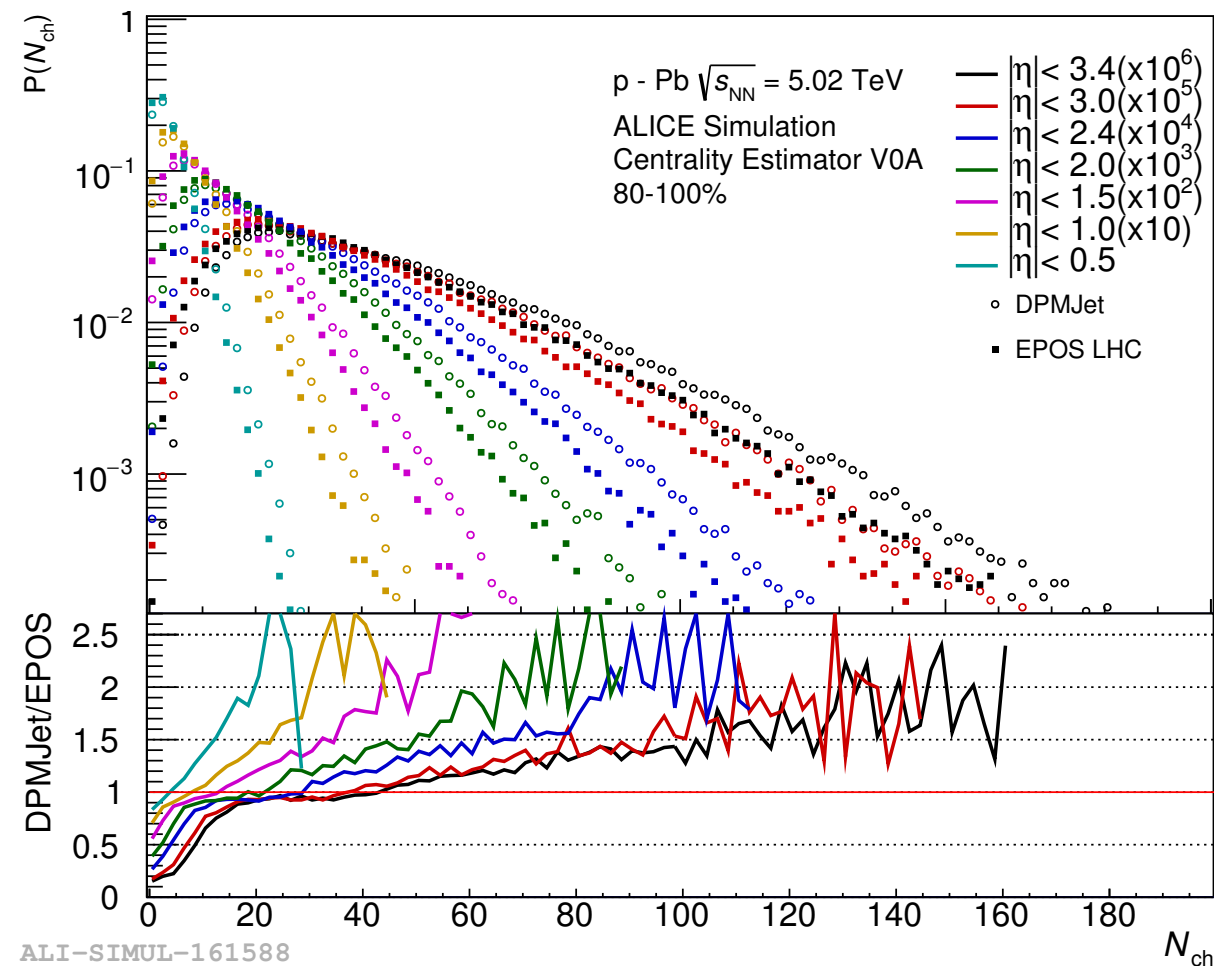
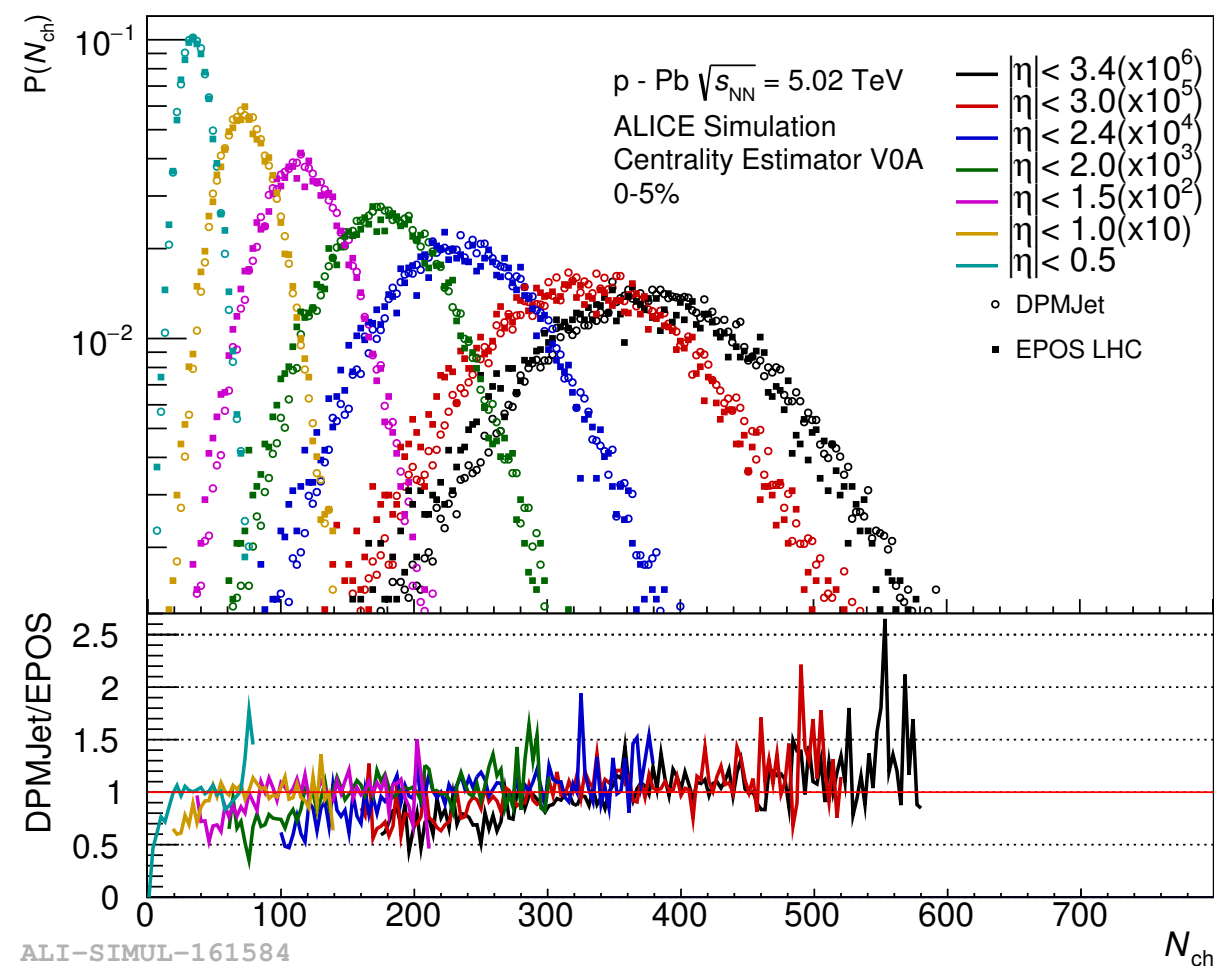
Centrality Slicing - 2 pseudorapidity ranges



- DPMJet
 - Wide pseudorapidity → Broader tails
 - Narrow pseudorapidity → Broader tails in peripheral
- EPOS → Wide and narrow pseudorapidity → Higher low bins very peripheral

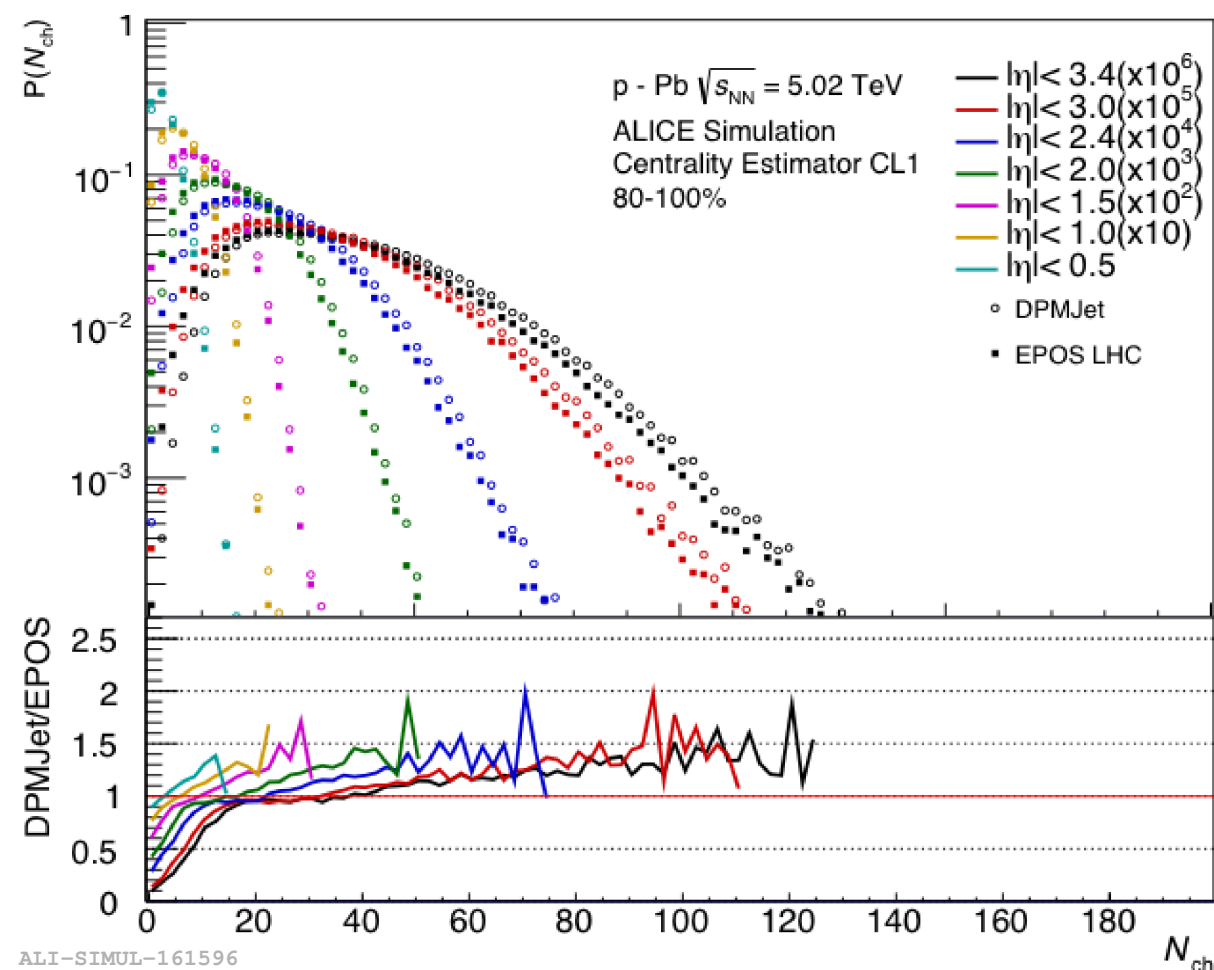
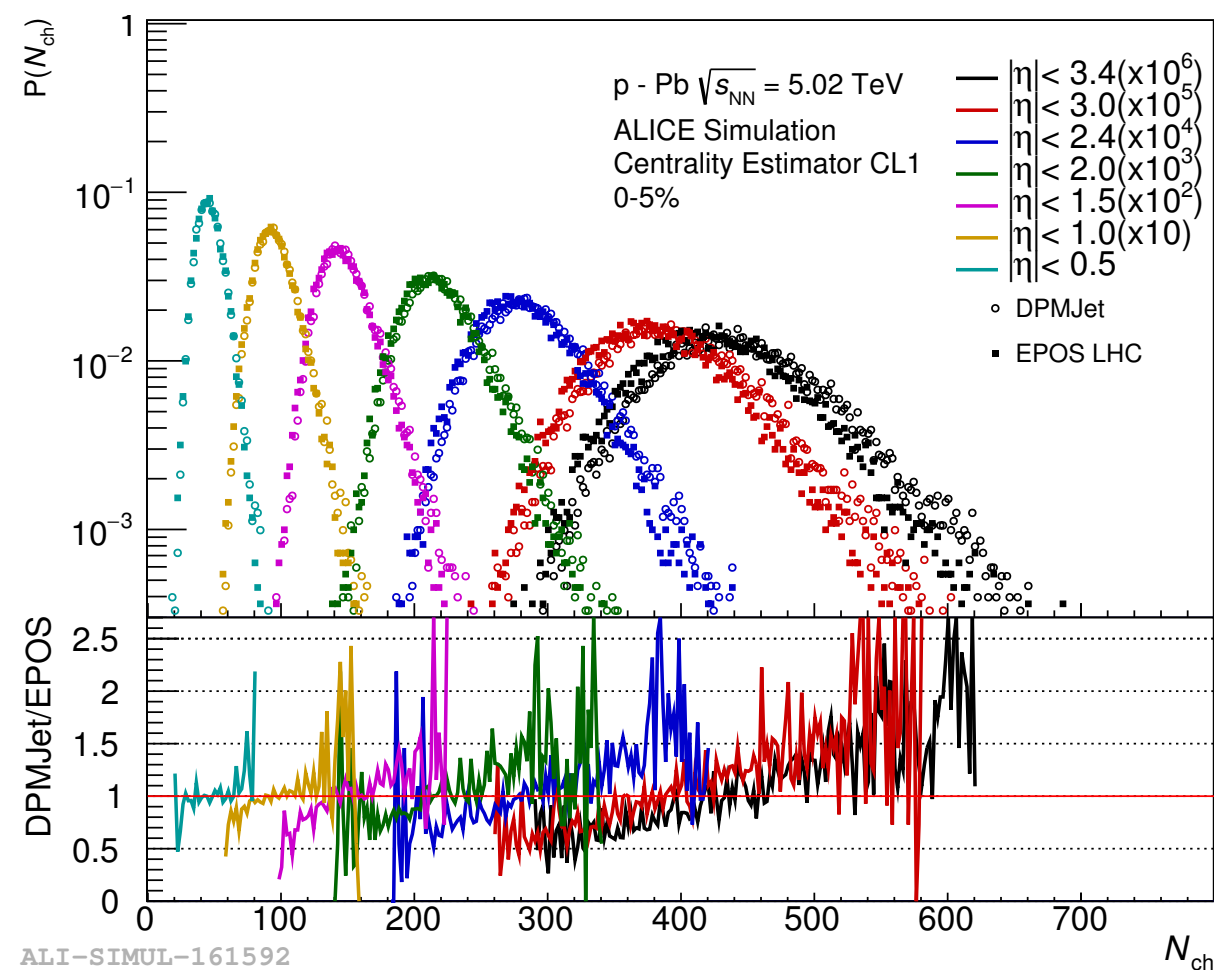


Centrality Slicing - 2 Centrality Classes



- Central $\longrightarrow$ Ratio below one in lower bins and slightly increases. Shows almost linear behavior.
- Peripheral $\longrightarrow$ Faster growth in lower bins

Centrality Slicing - 2 Centrality Classes



- Increasing ratio in both cases
- Central $\longrightarrow$ Almost linear behavior
- Peripheral $\longrightarrow$ Faster growth in lower bins

- Results for pp collisions at 7 TeV
 - ↳ Comparison with MC





- Results for pp collisions at 7 TeV

↳ Comparison with MC → EPOS LHC → Better agreement with data



- Results for pp collisions at 7 TeV
 - ↳ Comparison with MC → EPOS LHC → Better agreement with data
- p-Pb collisions at 5.02 TeV - MC studies
 - ↳ Minimum Bias
 - ↳ EPOS LHC → 2 regions → soft and semi-hard interactions
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 - ↳ Centrality Slicing
 - ↳ Holds the features observed at MB
- Next steps
 - ↳ Compare data with MC and models → Best scenario to describe p-Pb